Quantum Mechanics

Lecture Notes

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Cautions

This notes set continues from the companion PDF for Sections 1–2. This notes is based on *Lectures on Quantum Mechanics* by Steven Weinberg. This note is converted from the author's handwritten note to the LaTeX code by the use of AI assistance, therefore there may be errors or inconsistencies. Once you found the errors or inconsistencies, please notify the author.

Notations

The Heaviside-Lorentz unit is used as the convention for the electromagnetic units.

3 General Principles of Quantum Mechanics

3.1 Hilbert space and States

In this subsection, we discuss the Hilbert space and the representation of physical states as vectors within it. One of the postulates in quantum mechanics is that *every physical state can be represented as a vector in a Hilbert space.* A Hilbert space is a kind of a normed complex vector space. A (complex) vector space V is a set of vectors $\Psi, \Psi', \Psi'', \ldots$ with the following properties:

(i). If Ψ, Ψ' and $\Psi'' \in V$, and then, so is $\Psi + \Psi' \in V$.

(ii). Addition operation exists and is associative and commutative.

$$\begin{cases} \Psi + (\Psi' + \Psi'') = (\Psi + \Psi') + \Psi'' \,, \\ \Psi + \Psi' = \Psi' + \Psi \,. \end{cases} \tag{3.1}$$

(iii). Scalar product is defined: when Ψ is vector, so is $\alpha\Psi\,, (\alpha \in \mathbb{C})$ (in the case of real vector space, $\alpha \in \mathbb{R}$). The scalar multiplication is associative and distributive.

$$\begin{cases} \alpha(\alpha'\Psi) = (\alpha\alpha')\Psi \,, \\ \alpha(\Psi + \Psi') = \alpha\Psi + \alpha\Psi' \,, \\ (\alpha + \alpha')\Psi = \alpha\Psi + \alpha'\Psi \,. \end{cases} \tag{3.2}$$

(iv). There is a single zero vector (denoted as $\mathbf{0}$), and

$$\begin{cases} \mathbf{0} + \Psi = \Psi \,. \\ 0 \cdot \Psi = \mathbf{0} \,. \\ \alpha \cdot \mathbf{0} = \mathbf{0} \,. \end{cases} \tag{3.3}$$

In the following, the zero vector is just written as 0 unless it is necessary to distinguish it from the scalar zero.

A normed vector space is a vector space equipped with a norm, which is a map that assigns a non-negative length or size to each vector in the space. To define a norm, we introduce a scalar product (or inner product) $(\cdot, \cdot)$, which is a map that assigns a complex number to each pair of vectors in the space. The inner product must satisfy the following properties:

(i). Linearity: for any vectors Ψ, Ψ', Ψ'' and any scalars α, α', we have

$$(\Psi'', [\alpha\Psi + \alpha'\Psi']) = \alpha(\Psi'', \Psi) + \alpha'(\Psi'', \Psi') \,. \tag{3.4}$$

(ii). Symmetry: associated with the complex conjugate of the inner product.

$$(\Psi', \Psi)^* = (\Psi, \Psi') \,. \tag{3.5}$$

(iii). Positivity: for any vector Ψ, a norm is defined and is positive

$$(\Psi, \Psi) \geq 0\,. \tag{3.6}$$

Equality holds if and only if $\Psi = 0$.

The Dirac's bra-ket notation is widely used in quantum mechanics: for that purpose, we introduce the concept of bra vectors and ket vectors. Ket vectors are defined as vectors in the Hilbert space and are denoted by $|\Psi\rangle$. Bra vectors are defined as the dual vectors, which is linear functional mapping ket vectors in the Hilbert space to complex numbers via the inner product, and are denoted by $\langle\Psi|$. The inner product between two vectors Ψ and Φ can then be written in the bra-ket notation as $\langle\Psi|\Phi\rangle$. In the following, we do not introduce the dual vector space, and the inner product notation is used.

A vector space has a characteristic called *dimensionality*, which is the number of vectors in a basis for the space. The dimensionality is always a non-negative integer, but it does not need to be finite. Indeed, the dimensionality can be infinite in some vector spaces, such as Hilbert spaces used in quantum mechanics. A set of vectors $\Psi_1, \cdots, \Psi_n$ is independent if there is no linear combination of them vanishing, except the trivial combination where all coefficients are zero.

$$\sum_{k=1}^{n} a_k \Psi_k \neq 0\,, \tag{3.7}$$

for any choice of coefficients a_k not all zero. Here, n is the dimensionality of the vector space. For independent vectors $\Psi_1\,, \cdots, \Psi_n$, we can have

$$\sum_{k} a_k \Psi_k = 0\,, \tag{3.8}$$

if and only if all coefficients a_k are zero.

We say that a set of vectors is *orthogonal* when the inner product of each pair of distinct vectors in the set is zero:

$$(\Psi_i, \Psi_j) = 0\,, \tag{3.9}$$

for any $i \neq j$. We can prove that such a set of orthogonal vectors are independent. If the orthogonal set (not need to be independent) satisfies $\sum_i \alpha_i \Psi_i = 0$, by taking the scalar product with Ψ_j, we find

$$\left(\Psi_j, \sum_i \alpha_i \Psi_i\right) = \sum_i \alpha_i (\Psi_j, \Psi_i) = \alpha_j (\Psi_j, \Psi_j) = 0\,, \tag{3.10}$$

so $\alpha_j = 0$ for all j. Meanwhile, the converse does not always hold, namely the vectors of an independent set do not have to be orthogonal. But we can always find independent and orthogonal sets for finite-dimensional vector spaces.

A set of vectors is said to be *complete* if it spans the entire vector space. In other words, any vector Ψ can be expressed as a linear combination of the vectors in the set.

$$\Psi = \alpha_1 \Psi_1 + \cdots + \alpha_n \Psi_n \,. \tag{3.11}$$

The set of vectors is not needed to be independent, but we can always find subset that is both complete and independent. Such a subset is obtained by deleting vectors from the original set until only independent vectors remain. We can call such a subset a *basis* of the vector space. A complete set of orthogonal vectors is also called an *orthogonal basis* for the vector space. Furthermore, the orthogonal basis can be normalized to form an *orthonormal basis*, where each vector has unit norm.

A vector space is said to be *finite-dimensional* if the largest possible number of independent vectors is d. In such a case, any set of d-independent vectors Φ_i is complete because any vector Ψ in the space can be expressed as a linear combination of these d vectors:

$$\Psi = \sum_{i=1}^{d} \alpha_i \Phi_i \,. \tag{3.12}$$

If there is a vector Ψ which is not written by $\sum_{i=1}^{d} \alpha_i \Phi_i$; then Ψ and $\{\Phi_i\}$ form $d+1$ independent vectors, which contradicts the assumption that the maximum number of independent vectors is d. Also, no set of fewer than d vectors $X_i \, (i = 1, \cdots, d-1)$ could be complete. If there exists, each vector Φ_i of d independent vectors could be written as linear combination of the $d - s$ vectors (s is positive integer) X_i:

$$\Phi_i = \sum_{j=1}^{d-s} c_{ij} X_j \,. \tag{3.13}$$

If Φ_i are independent, then any linear combination should be non-zero unless all coefficients are zero.

$$\sum_{k=1}^{d} u_k \Phi_k = \sum_{k\,,j} u_k c_{kj} X_j \neq 0 \,. \tag{3.14}$$

Here, u_k are arbitrary non-zero coefficients. But, we can always take $\sum_{k=1}^{d} u_k c_{kj} = 0$ for all j, leading to a contradiction. It is natural that the d independent vectors cannot be expressed as a linear combination of fewer than d vectors. This is also ensured by the theorem in linear algebra. $(c^T)_{ji}$ is considered as a map from V^d to V^{d-s}, and the existence of a non-zero vector u_k such that $\sum_{k=1}^{d} u_k c_{kj} = 0$ for all j is guaranteed by the rank-nullity theorem:

$$\dim(\ker(c^T)) + \mathrm{rank}(c^T) = d \,. \tag{3.15}$$

Indeed, non-zero $\dim(\ker(c^T))$, namely $\ker(c^T)$ being non-trivial, ensures the existence of a non-zero vector u_k such that $\sum_{k=1}^{d} u_k c_{kj} = 0$ for all j.

Definition of Hilbert space

We have prepared several concepts of normed vector spaces. A Hilbert space is a special type of normed vector space that is complete with respect to the norm induced by an inner product. In the Quantum Mechanics context, Hilbert spaces serve as the mathematical framework for the state space of quantum systems. The dimensionality of the Hilbert spaces is up to the physical system: it can be either finite or infinite. In particular, for infinite-dimension Hilbert spaces, the set is called to be complete in the sense that any vector can be expressed as a linear combination of the infinite numbers of basis vectors.

$$\Psi = \sum_{i=1}^{\infty} \alpha_i \Phi_i \,. \tag{3.16}$$

In other words, there is a sequence of partial sums that converges to any vector Ψ in the Hilbert space: or we can equivalently define the vector $\Omega_N \equiv \Psi - \sum_{i=1}^{N} \alpha_i \Phi_i$ whose norm goes to zero as N approaches infinity, $(\Omega_N, \Omega_N) \to 0$ for $N \to \infty$.

The expansion coefficients are unique for complete orthogonal sets of basis vectors. When a vector Ψ is expanded in terms of a complete orthogonal set of basis vectors $\{\Phi_i\}$, such as Eq. (3.16), the expansion coefficients are given by

$$\alpha_i = \frac{(\Phi_i, \Psi)}{(\Phi_i, \Phi_i)} \,. \tag{3.17}$$

Therefore, any vector in the Hilbert space can be expressed as a linear combination of the complete orthogonal set of basis vectors.

$$\Psi = \sum_{i=1}^{\infty} \frac{(\Phi_i, \Psi)}{(\Phi_i, \Phi_i)} \Phi_i \,. \tag{3.18}$$

By using orthogonality of the basis vectors, we can express the inner product of any two vectors Ψ and Ψ' as

$$(\Psi, \Psi') = \sum_{i} \frac{(\Phi_i, \Psi)^* (\Phi_i, \Psi')}{(\Phi_i, \Phi_i)} \,. \tag{3.19}$$

Probability Interpretation

We can interpret the inner (scalar) products in the Hilbert space as probabilities for measurement outcomes. One can define the transition rate of a state Ψ to a state Φ_i (or measurement outcome corresponding to Φ_i from a state Ψ) as

$$P(\Psi \mapsto \Phi_i) \equiv \frac{|(\Phi_i, \Psi)|^2}{(\Phi_i, \Phi_i)(\Psi, \Psi)} \,. \tag{3.20}$$

From the expression of inner product of a vector with itself, we have

$$(\Psi, \Psi) = \sum_{i} \frac{|(\Phi_i, \Psi)|^2}{(\Phi_i, \Phi_i)} \,. \tag{3.21}$$

Therefore, summing all final states gives

$$\sum_i P(\Psi \to \Phi_i) = \sum_i \frac{|(\Phi_i, \Psi)|^2}{(\Phi_i, \Phi_i)(\Psi, \Psi)} = 1\,. \tag{3.22}$$

This shows that the total sum of all possible measurement outcomes is 1, which allows the probabilistic interpretation of quantum mechanics.

We call $P(\Psi \to \Phi_i)$ the probability of transition from the state Ψ to the state Φ_i. This probability is unchanged under the scaling transformation of the states such that

$$\Psi \to \alpha\Psi, \quad \Phi_i \to \beta_i \Phi_i\,, \tag{3.23}$$

where α and β_i are non-zero complex numbers. Therefore, we can always take any vectors to be normalized such that

$$(\Phi_i, \Phi_i) = 1\,, \qquad (\Psi, \Psi) = 1\,. \tag{3.24}$$

In this orthonormal basis, we can have $P(\Psi \mapsto \Phi_i) = |(\Phi_i, \Psi)|^2$.

There are still phase factors that do not change the probabilities.

$$\Psi \to e^{i\psi}\Psi\,, \quad \Phi_i \to e^{i\varphi_i}\Phi_i \tag{3.25}$$

Two vectors that differ only by a phase factor represent the same physical state: we identify these vectors and we call the equivalence class as *rays*. Indeed, the physical states are represented by rays in the Hilbert space rather than individual vectors.

3.2 Continuum States

Physical states are characterized not only by discrete labels $i = 1, 2, \ldots$ but also by continuum labels (such as position) ξ. Due to continuity of labels, we can define the density of labels $\rho(\xi)$ such that the number of labels in the interval ξ to $\xi + d\xi$ is given by $\rho(\xi)d\xi$. The previous discussions on the discrete labels can be generalized to the case of continuum labels, by taking $\rho(\xi)d\xi$ the approximate number of discrete labels in the interval ξ to $\xi + d\xi$. In other words, the sum over discrete labels i is replaced by an integral over the continuum labels ξ weighted by the density of labels $\rho(\xi)$:

$$\sum_i f_i \to \int f(\xi) \cdot \rho(\xi) d\xi\,. \tag{3.26}$$

The inner product of states with respect to the continuum labels is defined by

$$(\Phi_\xi, \Phi_{\xi'}) = \rho(\xi)\delta_{\xi,\xi'}\,. \tag{3.27}$$

Here, $\delta_{\xi,\xi'}$ is considered as the selection of labels, and $\rho(\xi)$ accounts for the density of labels.

Therefore, any vectors in the Hilbert space can be expanded in terms of the continuum basis $\{\Phi_\xi\}$ as follows:

$$\begin{aligned}\Psi &= \sum_i \frac{(\Phi_i, \Psi)}{(\Phi_i, \Phi_i)}\Phi_i \\ &\to \int \rho(\xi)d\xi \cdot \frac{(\Phi_\xi, \Psi)}{\rho(\xi)}\Phi_\xi = \int (\Phi_\xi, \Psi)\Phi_\xi d\xi \,. \end{aligned} \tag{3.28}$$

The inner product of two vectors Ψ and Ψ' is written in terms of the continuum basis as follows:

$$\begin{aligned}(\Psi, \Psi') &= \sum_i \frac{(\Phi_i, \Psi^*)^*(\Phi_i, \Psi')}{(\Phi_i, \Phi_i)} \\ &\to \int \rho(\xi)d\xi \cdot \frac{(\Phi_\xi, \Psi)^*(\Phi_\xi, \Psi')}{\rho(\xi)} = \int (\Phi_\xi, \Psi)^*(\Phi_\xi, \Psi')d\xi \,. \end{aligned} \tag{3.29}$$

Thus, the normalization condition for the state Ψ, $(\Psi, \Psi) = 1$, is written as

$$\int |(\Phi_\xi, \Psi)|^2 d\xi = 1 \,. \tag{3.30}$$

Compared to the probability interpretation of the measurement outcomes with the discrete basis, the differential probability for the continuum basis is given by

$$dP(\Psi \to \Phi_\xi) = \frac{|(\Phi_\xi, \Psi)|^2}{(\Phi_\xi, \Phi_\xi)} \times \rho(\xi)d\xi = |(\Phi_\xi, \Psi)|^2 \, d\xi \,. \tag{3.31}$$

From the normalization condition, it apparently follows that $\int dP\,(\Psi \to \Phi_\xi) = 1$.

Dirac Delta function

The next question is what is $\rho(\xi)\delta_{\xi',\xi}$ as a "function" of ξ. From the orthonormality condition of the continuum basis, we find the integration property of $\rho(\xi)\delta_{\xi',\xi}$:

$$\int d\xi' \rho(\xi')\delta_{\xi',\xi} = \int d\xi' (\Phi_{\xi'}, \Phi_\xi) = 1 \,, \tag{3.32}$$

as far as ξ is within the domain of integration. Furthermore, analogous to the discrete label, $\sum_{i'} f_{i'}\delta_{ii'} = f_i$, the similar property for the continuum label is imposed,

$$\int d\xi' \rho(\xi')\delta_{\xi',\xi} \cdot f(\xi') = f(\xi) \,. \tag{3.33}$$

Here, ξ is assumed to be within the domain of integration, again. This "function" picks out the special value at ξ. In particular, this function is called as Dirac's delta function and is denoted by $\delta(\xi - \xi')$.

$$\int d\xi' \delta(\xi - \xi') = 1 \,, \qquad \int d\xi' f(\xi')\delta(\xi - \xi') = f(\xi) \,. \tag{3.34}$$

Indeed, the Dirac delta function is not a conventional function in the usual sense, but rather a distribution or generalized function. There are several ways to describe the Dirac delta function as a limit of ordinary functions. The followings are examples of limit expressions.

$$f(x) = \lim_{n\to\infty} \begin{cases} n & \left(-\frac{1}{2n} \le x \le \frac{1}{2n}\right) \\ 0 & \text{(others)} \end{cases} , \tag{3.35}$$

$$f(x) = \lim_{n\to\infty} \sqrt{\frac{n}{\pi}} e^{-nx^2} , \tag{3.36}$$

$$f(x) = \lim_{\epsilon\to 0} \frac{\epsilon}{\pi\left(x^2+\epsilon^2\right)} . \tag{3.37}$$

3.3 Observables

As stated in the first subsection, the physical states are described by the vectors in a Hilbert space. The other important concepts are the observables, which correspond to measurable physical quantities in a quantum system. In quantum mechanics, the observables are represented by *linear Hermitian operators* acting on the Hilbert space of physical states. Let $A\,, B\,, \cdots$ denotes the linear operators and $\Psi, \Phi, \cdots$ denotes the physical states in the Hilbert space. There are several properties of operators acting on the physical states.

(i). The product of the operators is defined as the operator acting on the physical states from right in order, i.e., $(AB)\Psi \equiv A(B\Psi)$.

(ii). The constant scalars of operators are defined as $(\alpha A)\Psi \equiv \alpha(A\Psi)$.

(iii). The addition of operators is defined as $(A+B)\Psi \equiv A\Psi + B\Psi$.

(iv). There is a zero operator $\hat{\mathbb{O}}$ defined as $\hat{\mathbb{O}}\Psi \equiv \mathbf{0}$ for arbitrary Ψ.

Again, in the following, we write the zero operator and the zero vector as 0 for simplicity unless it is necessary to distinguish from the scalar zero. For arbitrary operators and constants, the zero operator satisfies

$$\hat{\mathbb{O}}A = \hat{\mathbb{O}}\,, \quad \hat{\mathbb{O}} + A = A\,, \quad \alpha\hat{\mathbb{O}} = \hat{\mathbb{O}}\alpha = \hat{\mathbb{O}}\,. \tag{3.38}$$

(v). There is a unit operator $\hat{\mathbb{1}}$ defined as $\hat{\mathbb{1}}\Psi \equiv \Psi$ for arbitrary states.

For arbitrary operators, the unit operator satisfies

$$\hat{\mathbb{1}}A = A\hat{\mathbb{1}} = A\,. \tag{3.39}$$

A *linear* operator A acts on states, and satisfies the following linearity conditions:

$$A(\Psi+\Psi') = A\Psi + A\Psi'\,, \quad A(\alpha\Psi) = \alpha A\Psi\,. \tag{3.40}$$

The *adjoint* $A^\dagger$ of any operator A is defined as the operator satisfying the relation

$$(\Psi', A^\dagger \Phi) \equiv (A\Psi', \Psi)\,. \tag{3.41}$$

or equivalently, by using the symmetry of the complex conjugate of the inner product,

$$(\Psi', A^\dagger \Psi) = (\Psi, A\Psi')^*\,. \tag{3.42}$$

It is easy to show that adjoint has the following general properties:

$$(AB)^\dagger = B^\dagger A^\dagger\,, \quad \left(A^\dagger\right)^\dagger = A\,, \quad (\alpha A)^\dagger = \alpha^* A^\dagger\,, \quad (A+B)^\dagger = A^\dagger + B^\dagger\,. \tag{3.43}$$

The adjoint of an operator is also called the Hermitian conjugate. When the Hermitian conjugate of an operator is equal to the operator itself, i.e.,

$$A^\dagger = A\,, \tag{3.44}$$

the operator is called *Hermitian* or *self-adjoint*. The zero operator $\hat{\mathbb{0}}$ and the unit operator $\hat{\mathbb{1}}$ are both Hermitian: $\hat{\mathbb{0}}^\dagger = \hat{\mathbb{0}}$ and $\hat{\mathbb{1}}^\dagger = \hat{\mathbb{1}}$.

Once we introduce a complete orthonormal set $\{\Phi_i\}$, we can define the matrix representation of any linear operator A with respect to this basis.

$$A_{ij} \equiv (\Phi_i, A\Phi_j) \tag{3.45}$$

Since any vectors in Hilbert space are expanded in terms of the complete orthonormal set $\{\Phi_i\}$, the matrix elements of the product of two operators A and B are given by

$$\begin{aligned}(AB)_{ij} &= (\Phi_i, AB\Phi_j) \\ &= \sum_k (\Phi_i, A\Phi_k)(\Phi_k, B\Phi_j) = \sum_k A_{ik}B_{kj}\,.\end{aligned} \tag{3.46}$$

In the intermediate step, we use the expansion of $B\Phi_j = \sum_i (\Phi_i, B\Phi_j)\Phi_i$ in terms of the complete set. We often refer this procedure as *inserting a complete set of states* (in between the operators A and B in this case). In the sense of inserting a complete set, we remind the reader that the inner product can be written as a sum over the complete set:

$$(\Psi, \Psi') = \sum_k (\Psi, \Phi_k)(\Phi_k, \Psi')\,, \tag{3.47}$$

and that the operator matrix elements satisfy $(\Phi_i, AB\Phi_j) = (A^\dagger\Phi_i, B\Phi_j)$. As you can see, the matrix element of the product of the operators has the form of a product of the matrices. The adjoint of an operator in the matrix representation is given by

$$(A^\dagger)_{ij} = A^*_{ji}\,. \tag{3.48}$$

As shown in the previous subsection, we frequently use the continuum label for the states. Let us consider the continuum labels ξ, η, and the corresponding states Φ_ξ and Φ_η, which satisfy the orthonormal condition in the continuum sense:

$$(\Phi_\xi, \Phi_\eta) = \delta(\xi - \eta)\,. \tag{3.49}$$

and we can define the matrix elements of operators in the continuum representation as follows:

$$A_{\xi\eta} \equiv (\Phi_\xi, A\Phi_\eta)\,, \tag{3.50}$$

$$(AB)_{\xi\eta} = \int d\lambda\, A_{\xi\lambda} B_{\lambda\eta}\,. \tag{3.51}$$

We have prepared to discuss the second postulate in quantum mechanics: the observables are represented by linear Hermitian operators acting on the Hilbert space of physical states. A state has a definite value a for an observable, represented by a linear Hermitian operator A, if and only if the state vector Ψ is an *eigenstate* of A.

$$A\Psi = a\Psi\,. \tag{3.52}$$

Here, a is called an *eigenvalue* of A If we also have a different eigenstate Ψ' and its eigenvalue a', namely $A\Psi' = a'\Psi'$, we have

$$\begin{aligned} a(\Psi', \Psi) &= (\Psi', A\Psi) \\ &= (A\Psi', \Psi) = a'^*(\Psi', \Psi)\,. \end{aligned} \tag{3.53}$$

Here, we use the Hermiticity of the operator A. We find the important consequences of eigenvalues and eigenstates of Hermitian operators.

(i). The allowed value of observables are real: $a^* = a$ in the case that $\Psi = \Psi' \neq 0$ and $a' = a$.

(ii). The eigenstates with the different values for any observable are orthogonal: $(\Psi', \Psi) = 0$ for $a' \neq a$

Expectation Values

Let $\{\Psi_r\}$ be a complete set (given by eigenstates for a Hermitian operator A).

$$A\Psi_r = a_r \Psi_r \tag{3.54}$$

The expectation value in a state represented by a normalized vector Ψ is given by

$$\langle A \rangle_\Psi \equiv (\Psi, A\Psi) = \sum_r (\Psi, A\Psi_r)(\Psi_r, \Psi) = \sum_r a_r |(\Psi_r, \Psi)|^2\,. \tag{3.55}$$

In the intermediate step, we insert the completeness relation to expand the state Ψ in terms of the complete set of eigenstates $\{\Psi_r\}$. For any normalized state Ψ, the expectation value of the observable A can be interpreted as the weighted average of its eigenvalues, with the weights given by the probabilities $|(\Psi_r, \Psi)|^2$.

It is easy to see that the n-th power of operator A has the eigenvalues of a_r^n corresponding to the same eigenstates Ψ_r. More generally, we can define functions $f(A)$ of the Hermitian operator A, which act on the eigenstates Ψ_r as

$$f(A)\Psi_r = f(a_r)\Psi_r\,. \tag{3.56}$$

The function of the operator $f(A)$ has the eigenvalues $f(a_r)$ corresponding to the same eigenstates Ψ_r. In other words, the expectation value of a function of the operator $f(A)$ in the eigenstates Ψ_r is simply the function of the corresponding eigenvalue:

$$\langle f(A)\rangle_{\Psi_r} = f(a_r)\,. \tag{3.57}$$

Meanwhile, the expectation value of a function of the operator $f(A)$ in a general state Ψ is not necessarily equal to the function of the expectation value of A in the same state. It is apparent to see

$$\langle f(A)\rangle_{\Psi} = \sum_r f(a_r)\,|(\Psi_r,\Psi)|^2 \neq f\left(\langle A\rangle_{\Psi}\right)\,. \tag{3.58}$$

In particular, for the square of any Hermitian operator A, we have the expectation value

$$\langle A^2\rangle_{\Psi} = \sum_r a_r^2\,|(\Psi_r,\Psi)|^2 \neq \langle A\rangle_{\Psi}^2 = \left(\sum_r a_r\,|(\Psi_r,\Psi)|^2\right)^2\,. \tag{3.59}$$

Indeed, we can show that $\langle A^2\rangle_{\Psi} \geq \langle A\rangle_{\Psi}^2$ and the equality holds when the state Ψ is the eigenstate of A. To see this, we can consider the square of Hermitian operators B^2, which is always a positive semi-definite operator.

$$\left\langle B^2\right\rangle_{\Psi} = \left(\Psi, B^2\Psi\right) = (B\Psi, B\Psi) \geq 0\,. \tag{3.60}$$

The equality holds only if the operator vanishes the state $B\Psi = 0$. Considering the Hermitian operator $B = A - \langle A\rangle_{\Psi}$, we can show that

$$0 \leq \left\langle (A - \langle A\rangle_{\Psi})^2\right\rangle_{\Psi} = \left\langle A^2\right\rangle_{\Psi} - \langle A\rangle_{\Psi}^2\,. \tag{3.61}$$

Then, we can also find that the equality holds when Ψ is an eigenstate of A.

Cauchy-Schwarz inequality

There is a general inequality possessed in any inner product spaces. Applying the statement to the Hilbert space, for any two state vectors Ψ and Ψ', we have

$$|(\Psi',\Psi)|^2 \leq (\Psi',\Psi')(\Psi,\Psi)\,. \tag{3.62}$$

The equality holds when these states are linearly dependent. It is easy to prove this inequality by considering the state

$$\Psi'' \equiv \Psi - \frac{(\Psi',\Psi)}{(\Psi',\Psi')}\Psi'\,, \tag{3.63}$$

and using the positivity of a quantity $(\Psi'',\Psi'')(\Psi',\Psi') \geq 0$. We can also see that the equality holds when the new state vector Ψ'' is zero vector, namely the state vectors Ψ and Ψ' are linearly dependent.

The Cauchy-Schwarz inequality in the Hilbert space gives a generalization of Heisenberg uncertainty principle. We now consider the root mean square deviation of an Hermitian operator A from its expectation value in a state Ψ.

$$\Delta_\Psi A \equiv \left\langle (A - \langle A \rangle_\Psi)^2 \right\rangle_\Psi^{1/2} . \tag{3.64}$$

It is convenient to use a state Ψ_A defined as

$$\Psi_A \equiv \frac{1}{\sqrt{(\Psi, \Psi)}} (A - \langle A \rangle_\Psi) \Psi , \tag{3.65}$$

then we can write $\Delta_\Psi A = (\Psi_A, \Psi_A)^{1/2}$. For any pair of Hermitian operators, A and B, we can define the similar state vectors Ψ_A and Ψ_B. The Cauchy-Schwarz inequality states that

$$\Delta_\Psi A \Delta_\Psi B \geq |(\Psi_A, \Psi_B)| . \tag{3.66}$$

Furthermore, we can rewrite the inner product of the state vectors Ψ_A and Ψ_B as follows.

$$(\Psi_A, \Psi_B) = \frac{1}{(\Psi, \Psi)} (\Psi, [AB - \langle A \rangle_\Psi \langle B \rangle_\Psi] \Psi) . \tag{3.67}$$

Remind that for any complex number c, we have $|c| = \sqrt{(\mathrm{Re}\, c)^2 + (\mathrm{Im}\, c)^2} \geq |\mathrm{Im}\, c|$. The imaginary part of the inner product (Ψ_A, Ψ_B) is given by

$$\mathrm{Im}(\Psi_A, \Psi_B) = \frac{1}{2i} \frac{1}{(\Psi, \Psi)} \left[(\Psi, AB\Psi) - (\Psi, AB\Psi)^* \right] . \tag{3.68}$$

By using the property $(\Psi, AB\Psi)^* = (\Psi, BA\Psi)$ for Hermitian operators, we can derive a generalized uncertainty relation.

$$\Delta_\Psi A \Delta_\Psi B \geq \frac{1}{2} |\langle [A, B] \rangle_\Psi| . \tag{3.69}$$

Here, we define *the commutator of two operators* as

$$[A, B] \equiv AB - BA . \tag{3.70}$$

The definition of $\Delta_\Psi A$ implies that it represents the standard deviation of the observable A in the state Ψ. Therefore, this leads to an important consequence in quantum mechanics: any pair of observables represented by non-commuting Hermitian operators cannot be simultaneously measured with arbitrary precision.

For example, the operators representing position and momentum in quantum mechanics, X and P, satisfy the canonical commutation relation $[X, P] = i\hbar$. Then, in any state Ψ, the uncertainty in position and momentum operators leads to

$$\Delta_\Phi X \Delta_\Phi P \geq \frac{\hbar}{2} . \tag{3.71}$$

This is known as the Heisenberg uncertainty principle.

Meanwhile, when there are two commuting Hermitian operators, A and B, any eigenstate of A can also be chosen to be an eigenstate of B. Such a state is called a *simultaneous eigenstate* of A and B. Indeed, the simultaneous eigenstate labeled by a and b, eigenstates of A and B respectively, satisfies

$$A\Psi_{a,b} = a\Psi_{a,b}\,, \quad B\Psi_{a,b} = b\Psi_{a,b}\,. \tag{3.72}$$

Hence, the operation of the commutator on the simultaneous eigenstate does not lead to any inconsistency, as expected for commuting operators: $[A, B]\Psi_{a,b} = 0$.

Similarly to the trace of matrices, the trace of an operator in a Hilbert space is defined by introducing a complete orthonormal set of state vectors. Let $\{\Psi_i\}$ be a complete orthonormal set of state vectors in the Hilbert space. The trace of an operator A is then defined as

$$\mathrm{Tr}A \equiv \sum_i (\Psi_i, A\Psi_i)\,. \tag{3.73}$$

This definition is quite useful because it allows us to define the trace of an operator independently of the choice of the complete orthonormal set of state vectors, as far as trace exists. Using another orthonormal set $\{\Phi_j\}$, the state vectors Ψ_i can be expressed as a linear combination of Φ_j's. Then, the trace of the operator A can be expressed in terms of the new orthonormal set as follows:

$$\begin{aligned}\mathrm{Tr}A &= \sum_i (\Psi_i, A\Psi_i) \\ &= \sum_{ij} (\Phi_j, A\Psi_i)(\Psi_i, \Phi_j) = \sum_j (\Phi_j, A\Phi_j)\,.\end{aligned} \tag{3.74}$$

In the intermediate step, we use the completeness relation for the orthonormal set $\{\Phi_j\}$. Similarly to the trace of matrices, the trace of an operator has also obvious properties:

(i). Linearity: $\mathrm{Tr}(\alpha A + \beta B) = \alpha \mathrm{Tr}A + \beta \mathrm{Tr}B$.

(ii). Hermitian conjugate: $\mathrm{Tr}A^\dagger = (\mathrm{Tr}A)^*$.

(iii). Cyclic property: $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

These properties are derived from the definition of trace in Hilbert space, Eq. (3.73). Meanwhile, the trace is not always defined for all operators, especially in infinite-dimensional Hilbert spaces. For example, the trace of the unit operator $\mathbb{1}$ in a d-dimensional Hilbert space is $\mathrm{Tr}\mathbb{1} = d$. In an infinite-dimensional Hilbert space, this trace is ill-defined.

Related to the uncertainty principle, the trace of the commutator of two operators provides important insights. For instance, consider the position and momentum operators X and P satisfying the canonical commutation relation $[X, P] = i\hbar\mathbb{1}$. In a finite-dimensional Hilbert space, taking the trace of both sides leads to a contradiction:

$$\mathrm{Tr}([X, P]) = \mathrm{Tr}(i\hbar\mathbb{1}) = i\hbar d\,, \tag{3.75}$$

where d is the dimension of the Hilbert space. However, the cyclic property of the trace implies $\mathrm{Tr}([X, P]) = 0$, which contradicts the previous result. Therefore, the uncertainty relation can only be realized in an infinite-dimensional Hilbert space.

3.4 Symmetries

One of important concepts in physics is symmetry. Even in quantum mechanics, symmetries play a crucial role in determining the behavior of physical systems.

Symmetry Principle

When we change our point of view in certain ways, the physics does not change. As an example, the change of the laboratory frame of measurements does not change the physical outcomes. A symmetry transformation is an operation that maps states to different states while preserving the physical properties of the system. Thus, a symmetry transformation can be described by operators mapping rays into different rays. Let Φ be a state vector, and a symmetry transformation is represented by a linear operator U acting on Φ as follows:

$$\Phi \mapsto U\Phi \,. \tag{3.76}$$

The meaning that the physical outcomes do not change is that the probability between any two states remains invariant under the symmetry transformation:

$$P(\Psi \to \Phi) = |(\Phi, \Psi)|^2 \,, \tag{3.77}$$

Here, Φ and Ψ are normalized state vectors. Under the symmetry transformation, the inner product between any two states transforms as follows:

$$(\Phi, \Psi) \mapsto (U\Phi, U\Psi) = (\Phi, U^\dagger U\Psi) \,. \tag{3.78}$$

Here, in the last equality, the adjoint of an operator U is used. One way to make the inner product invariant under the symmetry transformation is to require that U be a *unitary* operator.

$$U^\dagger U = \mathbb{1} \,. \tag{3.79}$$

The existence of an inverse transformation may not be guaranteed for a general operator. For example, rotation and translation have inverses, which undo the original transformation. The inverse of the unitary operator U exists and U^{-1} maps $U\Phi$ back to Φ for any state vector Φ: $U^{-1}U = UU^{-1} = \mathbb{1}$. It is also true that inverse operator U^{-1} also has its inverse $(U^{-1})^{-1}$, which is equal to the original operator U: by multiplying $(U^{-1})^{-1}$ to both sides of $U^{-1}U = \mathbb{1}$, we get $(U^{-1})^{-1} = U$. Applying the inverse transformation U^{-1} to both sides of $U^\dagger U = \mathbb{1}$, we have $U^\dagger = U^{-1}$

There is a question that is this the only way to represent symmetry transformations in quantum mechanics? Indeed, there is an alternative way to represent symmetry

transformations in quantum mechanics, which is given by an operator U being *antiunitary* and *antilinear*.

$$U(\alpha\Psi + \alpha'\Psi') = \alpha^* U\Psi + \alpha'^* U\Psi' \,, \tag{3.80}$$

$$(U\Phi, U\Psi) = (\Phi, \Psi)^* \,, \tag{3.81}$$

respectively. Under the symmetry transformation represented by an antiunitary operator U, the inner product between any two states transforms as the complex conjugate of the original inner product:

$$(\Phi, \Psi) \to (U\Phi, U\Psi) = (\Phi, \Psi)^* \,. \tag{3.82}$$

We remark that the antiunitary operator cannot be a linear operator. If the antiunitary operator were linear, it would contradict the definition of antiunitarity, as shown in the following argument. Let U be antiunitary and linear operator, then

$$\alpha(U\Phi, U\Psi) = (U\Phi, U\alpha\Psi) = (\alpha\Psi, \Phi)^* = \alpha^*(\Psi, \Phi)^* = \alpha^*(U\Phi, U\Psi) \,. \tag{3.83}$$

This is contradict for $\alpha \in \mathbb{C}$. But, most symmetries are represented by linear unitary operators. There is a known theorem that any symmetry transformation in quantum mechanics can be represented either by a unitary or an antiunitary operator: *Wigner's theorem*.

The unitary operators satisfy all the group axioms, and therefore they form a group:

(i). Existence of an identity element: $\mathbb{1}$

(ii). Existence of inverse elements: U^{-1} such that $UU^{-1} = U^{-1}U = \mathbb{1}$

(iii). Closure: If U_1 and U_2 are unitary, then U_1U_2 is also unitary

(iv). Associativity: $(U_1U_2)U_3 = U_1(U_2U_3)$

This fact tells us about the importance of learning *group theory* and *representation theory*. The group theory provides the mathematical framework to study the structure and properties of symmetry groups. Meanwhile, the representation theory becomes important, as it allows us to understand how the symmetry group acts on the Hilbert space of quantum states.

There are two types of symmetries: one is discrete (such as parity or time-reversal), and the other is continuous (such as spatial translations or rotations). As for the continuous symmetries, there is a class of symmetry transformations that can be connected smoothly to the identity transformation. The linear operators arbitrary close to a unit operator $\mathbb{1}$ can be written as $U_\epsilon = \mathbb{1} + i\epsilon T + \mathcal{O}(\epsilon^2)$ with an infinitesimal parameter ϵ. T is called as a *generator* of the continuous symmetry transformation. Such a unitary operator also satisfies the unitarity condition, $U_\epsilon^\dagger U_\epsilon = \mathbb{1}$, and hence the generator T must be Hermitian, $T = T^\dagger$.

So far, we discuss the symmetry transformation of state vectors $\Psi \mapsto U\Psi$. Under the symmetry transformation, the expectation value of an observable A transforms as

$$(\Psi, A\Psi) \mapsto (\Psi, U^{-1}AU\Psi) \,. \tag{3.84}$$

Thus, instead of the symmetry transformation of the state vectors, we can equivalently consider the transformation of the operators.

$$A \to U^{-1}AU\,. \tag{3.85}$$

Transformations of this type are known as a *similarity transformation* in Mathematics. We remark that similarity transformations preserve algebraic properties of operators:

$$U^{-1}AU \times U^{-1}BU = U^{-1}(AB)U\,, \tag{3.86}$$

$$U^{-1}AU + U^{-1}BU = U^{-1}(A+B)U\,, \tag{3.87}$$

As for the infinitesimal unitary transformation, we have

$$A \mapsto A - i\epsilon[T, A] + \mathcal{O}\left(\epsilon^2\right)\,, \tag{3.88}$$

Thus, the infinitesimal unitary transformation of an operator is determined by the commutator with the generator. Since the generator is itself an operator, it can also undergo transformations under the unitary symmetry operations.

There are two equivalent ways to view transformations: one is the active view, and the other is the passive view. In the active view, we consider the transformation as acting directly on the state vectors, while in the passive view, we interpret the transformation as a change of the basis. From the active/passive view of transformation of the basis vectors, the transformation $(\Psi \mapsto U\Psi\,, A \mapsto A)$ is considered to be active, while $(\Psi \mapsto \Psi\,, A \mapsto U^{-1}AU)$ is considered to be passive.

3.5 Space translation

As an example of a symmetry transformation, let us consider space translation: the space translation is a fundamental symmetry in physics, which corresponds to shifting the coordinates of *all* particles by a constant vector. The law of physics should remain invariant even if we change the origin of the coordinate system. Under the space translation by a vector $\boldsymbol{a}$, any particle at position $\boldsymbol{X}_n$ is shifted to $\boldsymbol{X}_n + \boldsymbol{a}$ (n denotes the particle label). In terms of operators acting on the Hilbert space, the symmetry transformation is represented by the unitary operator $U(\boldsymbol{a})$, this unitary transformation gives the similarity transformation of the position operator,

$$U^{-1}(\boldsymbol{a})\boldsymbol{X}_n U(\boldsymbol{a}) = \boldsymbol{X}_n + \boldsymbol{a}\,, \tag{3.89}$$

In the infinitesimal limit, the unitary operator can be expanded to first order in $\boldsymbol{a}$, giving

$$U(\boldsymbol{a}) = 1 - i\frac{\boldsymbol{P}}{\hbar}\cdot\boldsymbol{a} + \mathcal{O}\left(\boldsymbol{a}^2\right)\,. \tag{3.90}$$

Here, $\boldsymbol{P}$ is the generator of space translations. The $\hbar$ in the denominator ensures that the generator $\boldsymbol{P}$ has the dimension of momentum. Then, the commutation relation between the position operator and the generator of translations can be derived.

$$U^{-1}(\boldsymbol{a})\boldsymbol{X}_n U(\boldsymbol{a}) = \boldsymbol{X}_n + \frac{i}{\hbar}\left[\boldsymbol{P}\cdot\boldsymbol{a}, \boldsymbol{X}_n\right] + \mathcal{O}(\boldsymbol{a}^2)\,, \tag{3.91}$$

and this should coincide with the finite translation result $\boldsymbol{X}_n + \boldsymbol{a}$, leading to the commutation relation

$$\frac{i}{\hbar}\left[\boldsymbol{P}\cdot\boldsymbol{a}, \boldsymbol{X}_n\right] = \boldsymbol{a}\,, \tag{3.92}$$

or equivalently,

$$[X_{ni}, P_j] = i\hbar\delta_{ij}\,. \tag{3.93}$$

Here, this commutation relation coincides with the canonical commutation relation between position and momentum operators in quantum mechanics. The question is then: what is the physical meaning of the momentum operator $\boldsymbol{P}$? The space translation makes the coordinates of all particles shift by the same amount, and therefore the generator of translations $\boldsymbol{P}$ corresponds to the total momentum of the system. The total momentum is given by the sum of the individual momenta of all particles, which is represented by the operator $\boldsymbol{P} = \sum_n \boldsymbol{P}_n$, with $\boldsymbol{P}_n$ being the momentum operator of the n-th particle. This ensures that the commutation relations between the total momentum and the position operators of individual particles are consistent with the canonical commutation relations:

$$[X_{ni}, P_{mj}] = i\hbar\delta_{ij}\delta_{nm}\,. \tag{3.94}$$

Here, n, m label the particles, and i, j label the spatial components of the position and momentum operators.

The order of translations does not matter for space translations: a translation by $\boldsymbol{a}$ followed by a translation by $\boldsymbol{b}$ is equivalent to a translation by $\boldsymbol{b}$ followed by a translation by $\boldsymbol{a}$. In terms of the translation operators, this means that the operators corresponding to translations by different vectors commute with each other:

$$U(\boldsymbol{a})U(\boldsymbol{b}) = U(\boldsymbol{b})U(\boldsymbol{a})\,, \tag{3.95}$$

The infinitesimal version of this commutation relation can be derived by expanding the translation operators to the order of $a_i b_j$:

$$U(\boldsymbol{a})U(\boldsymbol{b}) = 1 - \frac{i}{\hbar}\boldsymbol{P}\cdot(\boldsymbol{a}+\boldsymbol{b}) - \frac{1}{\hbar^2}(\boldsymbol{P}\cdot\boldsymbol{a})(\boldsymbol{P}\cdot\boldsymbol{b}) + O(a^2, b^2)\,, \tag{3.96}$$

$$U(\boldsymbol{b})U(\boldsymbol{a}) = 1 - \frac{i}{\hbar}\boldsymbol{P}\cdot(\boldsymbol{a}+\boldsymbol{b}) - \frac{1}{\hbar^2}(\boldsymbol{P}\cdot\boldsymbol{b})(\boldsymbol{P}\cdot\boldsymbol{a}) + O(a^2, b^2)\,. \tag{3.97}$$

It leads to the commutation relation between the components of the momentum operator:

$$[P_i, P_j] = 0\,. \tag{3.98}$$

We can generalize this idea to finite translations by considering the limit of many infinitesimal translations. A translation vector is denoted by $\boldsymbol{r}_N = N\boldsymbol{a}$, where N is a large integer and $\boldsymbol{a}$ is an infinitesimal translation vector. The finite translation vector is a limit of many infinitesimal translations: $\boldsymbol{r} = \lim_{N\to\infty}\boldsymbol{r}_N$. Therefore, the finite

translation operator can be expressed as the limit of many infinitesimal translation operators:

$$U(\boldsymbol{r}) = \lim_{N\to\infty} [U(\boldsymbol{a})]^N = \lim_{N\to\infty} \left[1 - \frac{i}{\hbar}\boldsymbol{P}\cdot\boldsymbol{a}\right]^N = \exp\left(-\frac{i}{\hbar}\boldsymbol{P}\cdot\boldsymbol{r}\right) . \tag{3.99}$$

Here, we can find the s-th power term of the expansion of the N-th power of the infinitesimal translation operator:

$$\left[1 - \frac{i}{\hbar}\boldsymbol{P}\cdot\boldsymbol{a}\right]^N = \sum_{s=0}^{N} \binom{N}{s} \left(-\frac{i}{\hbar}\boldsymbol{P}\cdot\boldsymbol{a}\right)^s , \tag{3.100}$$

and use the definition of exponentiation of operators in the limit $N \to \infty$.

$$e^{iA} = \sum_{n=0}^{\infty} \frac{i^n}{n!} A^n , \tag{3.101}$$

with A denoting an operator.

Let us define Φ_0 as one-particle state with a definite position at the origin. It means that the operation of $\boldsymbol{X}$ on Φ_0 gives zero, $\boldsymbol{X}\Phi_0 = 0$. We can construct a state with a definite position $\boldsymbol{x}$ by applying the translation operator $U(\boldsymbol{x})$ to Φ_0.

$$\Phi_x \equiv U(\boldsymbol{x})\Phi_0 , \quad \boldsymbol{X}\Phi_x = \boldsymbol{x}\Phi_x . \tag{3.102}$$

Now, we can define a momentum-representation state Ψ_p, which is an eigenstate of the momentum operator with eigenvalue $\boldsymbol{p}$: $\boldsymbol{P}\Psi_p = \boldsymbol{p}\Psi_p$. Then, we can consider a scalar product of the momentum-representation state vector Ψ_p and the position-representation state vector Φ_x: (Ψ_p, Φ_x). The scalar product (Ψ_p, Φ_x) can be evaluated using the translation operator.

$$(\Psi_p, \Phi_x) = (\Psi_p, U(\boldsymbol{x})\,\Phi_0) = e^{-i\boldsymbol{p}\cdot\boldsymbol{x}/\hbar}(\Psi_p, \Phi_0) . \tag{3.103}$$

In the last equality, we used the fact that Ψ_p is an eigenstate of the momentum operator with eigenvalue $\boldsymbol{p}$. It is convenient to choose the normalization of the state as

$$(\Psi_p, \Phi_x) \equiv \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{-i\boldsymbol{p}\cdot\boldsymbol{x}/\hbar} . \tag{3.104}$$

The complex conjugate of this scalar product gives the coordinate wave function in the position representation:

$$\psi_p(x) \equiv (\Phi_x, \Psi_p) = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{i\boldsymbol{p}\cdot\boldsymbol{x}/\hbar} \tag{3.105}$$

This is the plane wave solution representing a particle with definite momentum $\boldsymbol{p}$ in the position representation. The normalization chosen here ensures that the position and momentum eigenstates are properly normalized according to the Dirac delta function

conventions. Here, the scalar product of state vectors in the position representation is given by

$$(\Phi_{x'}, \Phi_x) = \delta^3(\boldsymbol{x}' - \boldsymbol{x})\,, \tag{3.106}$$

which ensures that the position eigenstates are orthonormal according to the Dirac delta function conventions. Then, the scalar product of state vectors in the momentum representation is

$$\begin{aligned}(\Psi_{p'}, \Psi_p) &= \int d^3x (\Psi_{p'}, \Phi_x)(\Phi_x, \Psi_p) \\ &= \frac{1}{(2\pi\hbar)^3} \int d^3x e^{i(\boldsymbol{p}-\boldsymbol{p}')\cdot\boldsymbol{x}/\hbar} = \delta^3(\boldsymbol{p} - \boldsymbol{p}')\,. \end{aligned} \tag{3.107}$$

3.6 Time translation

The second example for the symmetry transformations is the *time translation*. For a state vector $\Psi(t)$ at time t, the time translation operator $U(\tau)$, which induces the time shift $t \to t + \tau$, acts as

$$U(\tau)\Psi(t) = \Psi(t + \tau)\,, \tag{3.108}$$

In a similar way to the previous subsection, we introduce an Hermitian operator H to generate the time translation.

$$U(\tau) \equiv e^{-i\frac{1}{\hbar}H\tau}\,. \tag{3.109}$$

Analogous to the momentum operator generating spatial translations, the Hermitian operator H generates time translations. This can be taken as a definition of the *Hamiltonian* H. Thus, the time dependence of any physical state can be factored out as a unitary evolution generated by the Hamiltonian:

$$\Psi(t) = e^{-i\frac{1}{\hbar}Ht}\Psi(0)\,, \tag{3.110}$$

and this time evolution leaves the scalar product invariant.

$$(\Psi(t), \Psi(t)) = (\Psi(0), \Psi(0))\,, \tag{3.111}$$

Since we have factored out the time dependence as a unitary evolution, the state vector $\Psi(t)$ satisfies the following time evolution equation:

$$i\hbar \cdot \frac{d}{dt}\Psi(t) = H\Psi(t)\,, \tag{3.112}$$

This is nothing but the *time-dependent Schrödinger equation*. One remark is that this is *the time evolution equation for the state vector*, not for the wavefunction in a particular representation.

There are two equivalent formalisms in quantum mechanics: the *Schrödinger picture* and the *Heisenberg picture*. These are related to the choice of whether the time

dependence is assigned to the state vectors (Schrödinger picture: state vectors evolve in time $\Psi(t)$, operators are fixed $A = A(0)$) or to the operators (Heisenberg picture: state vectors are fixed $\Psi = \Psi(0)$, operators evolve in time $A(t)$).

Operators in two different pictures are related by the following unitary transformation (similarity transformation of the operator A):

$$A_H(t) = e^{iHt/\hbar} A_S e^{-iHt/\hbar} , \tag{3.113}$$

Here, A_S denotes the operator A in the Schrödinger picture, and $A_H(t)$ denotes the operator A in the Heisenberg picture at time t. The Hamiltonian is also an operator, so we can define the Hamiltonian operators in both the Schrödinger and Heisenberg pictures. However, the Hamiltonian commutes with itself at all times, and the Hamiltonian in two pictures is exactly same.

Symmetry and Conserved Quantity

The time dependence of an operator A in the Heisenberg picture leads to so called *the Heisenberg equation of motion*: the time evolution is governed by the commutator with the Hamiltonian.

$$\frac{d}{dt} A_H(t) = \frac{i}{\hbar} \left[H, A_H(t) \right] , \tag{3.114}$$

It is easy to prove the time evolution equation. Since the Hamiltonian H commutes with itself, we can easily compute the derivative of the unitary operator:

$$\frac{d}{dt} \left(e^{-iHt/\hbar} \right) = -\frac{i}{\hbar} H e^{-iHt/\hbar} = -\frac{i}{\hbar} e^{-iHt/\hbar} H . \tag{3.115}$$

The Hamiltonian determines the time evolution of the operators in the Heisenberg picture. From the Heisenberg equation of motion, we can see that if an operator commutes with the Hamiltonian, its time derivative vanishes, indicating that $A_H(t)$ is a conserved quantity. Such a case, the operator $A_H(t)$ is time independent, and hence there is no difference between the Schrödinger and Heisenberg pictures for this operator.

Moving on to the discussion of symmetries and conserved quantities in Schrödinger picture. When a symmetry is present in the Schrödinger picture, there is a symmetry transformation connecting the state vectors Ψ and $U\Psi$. Since these state vectors represent the same physical state from different viewpoints (or observers), their time evolutions must be consistent.

$$\Psi(t) = e^{-iHt/\hbar} \Psi(0) , \qquad (U\Psi)(t) = e^{-iHt/\hbar} (U\Psi)(0) . \tag{3.116}$$

On the other hand, the time evolution of the transformed state is also written as $(U\Psi)(t) = U e^{-iHt/\hbar} \Psi(0)$. Therefore, to be consistent, we must have the relation between the time translation operator and Hamiltonian as follows.

$$e^{-iHt/\hbar} U = U e^{-iHt/\hbar} . \tag{3.117}$$

From this relation, we can deduce that the Hamiltonian commutes with the symmetry operator:

$$U^{-1}HU = H\,. \tag{3.118}$$

Once the infinitesimal transformation is considered, we can write $U = e^{ixT}$ for some Hermitian operator T and small parameter x. The commutation relation with the Hamiltonian then becomes

$$[H, T] = 0\,, \tag{3.119}$$

indicating that T is a conserved quantity associated with the symmetry. We can see that the presence of a symmetry leads to a conserved quantity in the system. For example, if the system is invariant under spatial translations, the corresponding conserved quantity is the momentum. Similarly, if the system is invariant under time translations, the conserved quantity is the energy.

So far, we consider the case where the symmetry operator associated with time translation U is unitary and linear. In this case, the Hamiltonian commutes with U, leading to conserved quantities associated with the symmetry.

We may consider the case where the symmetry operator associated with time translation U is antilinear. In this case, the commutation relation with the Hamiltonian may take a different form, and the analysis of conserved quantities needs to be adjusted accordingly. Once we start from Eq. (3.117), we obtain the similarity transformation of the Hamiltonian under the symmetry operator U as $U^{-1}HU = -H$. The Hamiltonian changes sign under the antilinear symmetry transformation, and it implies that a positive energy state is mapped to a negative energy state under this symmetry. If such a symmetry exists, it would imply the existence of negative energy states, which is in conflict with the observed stability of matter. To avoid such a situation, the antilinear transformation associated with time must be accompanied by the time reversal. Once the time reversal is included, the time evolution of the transformed state is

$$(U\Psi)(t) = e^{\frac{i}{\hbar}Ht}(U\Psi)(0)\,, \tag{3.120}$$

and the corresponding commutation relation between the time translation operator and the symmetry operator is modified accordingly.

$$e^{+\frac{i}{\hbar}Ht}U = Ue^{-\frac{i}{\hbar}Ht}\,. \tag{3.121}$$

As a result, we obtain the commutation relation between the Hamiltonian and the symmetry operator, $U^{-1}HU = H$, and no negative energy state is not required after the symmetry transformation.

4 Particle States in Central Potential

It is one of central programs of quantum mechanics to understand the behavior of a state. In this section, we first consider the meaning of Schrödinger equation for a single particle moving in three-dimensional space. In particular, we focus on the case where the particle is subject to a central potential, which depends only on the distance from the origin of coordinates. In the usual textbooks, the treatment often starts from one-dimensional problems. However, for a central potential, the problem can be effectively reduced to a one-dimensional problem along the radial direction. There is no fundamental difference between one-dimensional and three-dimensional treatments in this context.

4.1 Schrödinger Equation for a Single Particle

Let us consider a single particle of mass μ moving in a central potential. The central potential depends only on the distance from the origin, i.e., $V = V(r)$ with $r = \sqrt{x_1^2 + x_2^2 + x_3^2}$. The Hamiltonian for this system is given by

$$H = \frac{1}{2\mu}\boldsymbol{p}^2 + V(r)\,, \tag{4.1}$$

where the first and the second terms represent the kinetic term and the potential term, respectively. The momentum operator is defined as $\boldsymbol{p} = -i\hbar\boldsymbol{\nabla}$.

The time-independent Schrödinger equation for a state with a definite energy E is given by

$$H\psi(\boldsymbol{x}) = \left[-\frac{\hbar^2}{2\mu}\nabla^2 + V(r)\right]\psi(\boldsymbol{x}) = E\psi(\boldsymbol{x})\,, \tag{4.2}$$

$$\nabla^2 \equiv \partial_1^2 + \partial_2^2 + \partial_3^2\,, \quad \partial_i \equiv \frac{\partial}{\partial x_i}\,. \tag{4.3}$$

Here, $\boldsymbol{x} = (x_1, x_2, x_3)$ represents the position vector of the particle in the three-dimensional space, in particular in Cartesian coordinates. For a state with a definite energy E, the time dependence of the wave function is factored out as $e^{-iEt/\hbar}$. To find a solution to the Schrödinger equation under a central potential, it is convenient to switch to spherical coordinates.

$$(x_1, x_2, x_3) \quad \longrightarrow \quad (r, \theta, \varphi)\,. \tag{4.4}$$

Here, r is the radial distance, θ is the polar angle, and φ is the azimuthal angle. In the spherical coordinate system, the Laplacian operator takes the form

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\,. \tag{4.5}$$

In an analogy to the classical mechanics, under a spherical potential, the rotational motion of the particle is conserved. It means that the angular momentum of the

particle is conserved. This situation is same even in quantum mechanics, and in the terminology of quantum mechanics, it means that the angular momentum operator commutes with the Hamiltonian. Now, let us define the angular momentum operator in quantum mechanics as

$$\boldsymbol{L} \equiv \boldsymbol{x} \times \boldsymbol{p} = -i\hbar \boldsymbol{x} \times \boldsymbol{\nabla} \,. \tag{4.6}$$

The components of the angular momentum operator in Cartesian coordinates are given by

$$L_i = -i\hbar \sum_{j,k} \epsilon_{ijk} x_j \partial_k \,, \tag{4.7}$$

for i, j, k running over three Cartesian coordinates $1, 2, 3$. ϵ_{ijk} denotes the *Levi-Civita symbol*, which is totally antisymmetric in its indices.

$$\epsilon_{ijk} \equiv \begin{cases} +1 & \text{even permutation} \\ -1 & \text{odd permutation} \\ 0 & \text{otherwise} \end{cases} \,, \tag{4.8}$$

with the convention $\epsilon_{123} = +1$.

To show commutation of the angular momentum operator with Hamiltonian, we first need to establish the commutation relations between the angular momentum operator and the position and momentum operators. Using the commutation relation of the position and momentum operators, $[x_i, \partial_j] = \delta_{ij}$, we can derive the commutation relations between the angular momentum operator and the position and momentum operators as follows.

$$[L_i, x_j] = i\hbar \sum_k \epsilon_{ijk} x_k \,, \qquad [L_i, \partial_j] = i\hbar \sum_k \epsilon_{ijk} \partial_k \,, \tag{4.9}$$

$$[L_i, x^2] = 0 \,, \qquad [L_i, \partial^2] = 0 \,. \tag{4.10}$$

Here, $x^2 \equiv \sum_i x_i^2$ and $\partial^2 \equiv \sum_i \partial_i^2$. The commutation relation with the scalar quantities like x^2 and ∂^2 is zero, as expected for rotationally invariant quantities. For any vectors v_i constructed from x_i and ∂_i, it is possible to show that the commutation relation with the angular momentum operator takes the same form as for the position and momentum operators:

$$[L_i, v_j] = i\hbar \sum_k \epsilon_{ijk} v_k \,. \tag{4.11}$$

It is straightforward to verify the commutation relation with the three-dimensional vectors, which is given by $\boldsymbol{v} = f(x^2, p^2)\boldsymbol{x}$ or $\boldsymbol{v} = f(x^2, p^2)\boldsymbol{p}$. Another possible vector is proportional to the angular momentum operator itself, $\boldsymbol{L} = \boldsymbol{x} \times \boldsymbol{p}$. Indeed, the commutation relation of the angular momentum operator with itself is given by

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k \,. \tag{4.12}$$

To show the commutation relation of the angular momentum operator with itself, we can use a relation for the Levi-Civita symbol, which is known as the *Jacobi-type identity* for the Levi-Civita symbol.

$$\sum_k (\epsilon_{ijk}\epsilon_{lkm} + \epsilon_{ikl}\epsilon_{jkm} + \epsilon_{ikm}\epsilon_{jlk}) = 0\,. \tag{4.13}$$

For the vector $\boldsymbol{v}$ satisfying Eq. (4.11), the commutation relation of v^2 with the angular momentum operator is given by

$$[L_i, v^2] = \sum_j ([L_i, v_j]v_j + v_j[L_i, v_j]) = 0\,. \tag{4.14}$$

In particular, L^2 commutes with each component of the angular momentum operator, $[L^2, L_i] = 0$. Since the angular momentum operator $\boldsymbol{L}$ commutes with the scalar quantities like x^2 and ∂^2, it commutes with the Hamiltonian for a central potential, and L^2 does as well. It is important to note that while L^2 commutes with each component of the angular momentum operator, the individual components L_i do not commute with each other. Thus, the *simultaneous eigenstates* of the Hamiltonian and the angular momentum operator are typically chosen to be eigenstates of L^2 and one component of $\boldsymbol{L}$, usually L_3. In addition, since the angular momentum operator $\boldsymbol{L}$ commutes with r^2, the operation of $\boldsymbol{L}$ on the wave function actually acts on the angular coordinates (θ, φ).

Now, let us find the expression of the angular momentum operators in terms of spherical coordinates. The Cartesian coordinates are

$$x_1 = r\sin\theta\cos\varphi\,, \quad x_2 = r\sin\theta\sin\varphi\,, \quad x_3 = r\cos\theta\,. \tag{4.15}$$

We can find each component of the angular momentum in the spherical coordinates as follows.

$$L_1 = i\hbar\left(\sin\varphi\frac{\partial}{\partial\theta} + \cot\theta\cos\varphi\frac{\partial}{\partial\varphi}\right)\,, \tag{4.16}$$

$$L_2 = i\hbar\left(-\cos\varphi\frac{\partial}{\partial\theta} + \cot\theta\sin\varphi\frac{\partial}{\partial\varphi}\right)\,, \tag{4.17}$$

$$L_3 = -i\hbar\frac{\partial}{\partial\varphi}\,. \tag{4.18}$$

From the expressions of the angular momentum operators in spherical coordinates, we can find that it is convenient to label the states by the eigenvalue of L_3: the expression of L_3 in the spherical coordinate is rather simple. The square of angular momentum operator L^2 in spherical coordinates is computed by taking the sum of its components squared:

$$L^2 = \sum_{i=1}^{3} L_i^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]\,. \tag{4.19}$$

Then, the Laplacian operator is written in terms of the radial derivative and the square of the angular momentum operator as follows.

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{L^2}{r^2\hbar^2}\,. \tag{4.20}$$

Therefore, the time-independent Schrödinger equation in spherical coordinates is written as follows.

$$E\psi(\boldsymbol{x}) = \left[-\frac{\hbar^2}{2\mu r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{L^2}{2\mu r^2} + V(r)\right]\psi(\boldsymbol{x})\,. \tag{4.21}$$

Now, let us consider the spectrum of the angular momentum operator L^2. As long as the potential $V(r)$ is not extremely singular at the origin, the wave function behave smoothly near $r \to 0$. In particular, the presence of the angular momentum term in the Hamiltonian implies that the wave function smoothly connected zero near the origin. Thus, introducing an integer ℓ, the wave function behaves as

$$\psi(\boldsymbol{x}) \to r^\ell Y(\theta,\varphi)\,. \tag{4.22}$$

Here, $Y(\theta,\varphi)$ denotes the function depending only on the angular coordinates θ and φ. We assume that the potential $V(r)$ is less singular than $1/r^2$ to allow the potential term in the Schrödinger equation to be negligible compared to the angular momentum term near the origin. Under the assumption, the Schrödinger equation near the origin is dominated by the kinetic term and the angular momentum term, leading to the following equation.

$$0 = -\frac{\hbar^2}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial}{\partial r}\left(r^\ell Y(\theta,\varphi)\right)\right] + \frac{L^2}{r^2}r^\ell Y(\theta,\varphi)\,. \tag{4.23}$$

Therefore, the function $Y(\theta,\varphi)$ must satisfy the following equation for the angular momentum operator L^2 near the origin.

$$L^2 Y(\theta,\varphi) - \hbar^2\ell(\ell+1)Y(\theta,\varphi) = 0\,. \tag{4.24}$$

So far, we consider the behavior of the wave function near the origin. However, once we chose $\psi(\boldsymbol{x})$ as an eigenfunction of L^2 (commute with H), the eigenvalue of L^2 is fixed for the entire wave function.

$$L^2\psi = \hbar^2\ell(\ell+1)\psi(\boldsymbol{x})\,. \tag{4.25}$$

Since the operator L^2 acts only on the angular part of the wave function, the wave function should be proportional to a product of a radial function and an angular function, i.e., we can separate variables as $\psi(\boldsymbol{x}) = R_{n,\ell}(r)Y_\ell^m(\theta,\varphi)$. The functions $Y_\ell^m(\theta,\varphi)$ are labeled by the quantum numbers ℓ and m, corresponding to the eigenvalues of L^2 and L_3, respectively. ℓ is called the *azimuthal quantum number*, while m is called the *magnetic quantum number*. The functions $Y_\ell^m(\theta,\varphi)$ are called the *spherical harmonics*.

$$L^2 Y_\ell^m(\theta,\varphi) = \hbar^2\ell(\ell+1)Y_\ell^m(\theta,\varphi)\,, \tag{4.26}$$

$$L_3 Y_\ell^m(\theta,\varphi) = \hbar m Y_\ell^m(\theta,\varphi)\,. \tag{4.27}$$

Meanwhile, the functions $R_{n,\ell}(r)$ are labeled by the quantum numbers n and ℓ, corresponding to the energy level and azimuthal quantum numbers, respectively. After separating the variables, we obtain the radial Schrödinger equation,

$$\left[-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right)+\frac{\hbar^2\ell(\ell+1)}{2\mu r^2}+V(r)\right]R_{n,\ell}(r)=ER_{n,\ell}(r)\,. \tag{4.28}$$

By separating the variables, we reduce the three-dimensional Schrödinger equation to a one-dimensional radial equation for $R_{n,\ell}(r)$, in particular ordinary differential equation (ODE). There is a unitary (normalization) condition for the wave function to be normalizable, which gives us the probability interpretation of the wave function. The integration of the wave function square over the three dimensions should be finite:

$$\int d^3x\,|\psi(\boldsymbol{x})|^2<\infty\,, \tag{4.29}$$

and the finiteness of the radial integral leads to the requirement on the radial wave function $R_{n,\ell}(r)$:

$$\int_0^\infty dr r^2|R_{n,\ell}(r)|^2<\infty\,. \tag{4.30}$$

We may also define the radial wave function as $u(r)\equiv rR(r)$, which gives the radial Schrödinger equation in a form similar to the one-dimensional Schrödinger equation.

$$-\frac{\hbar^2}{2\mu}\frac{d^2u(r)}{dr^2}+\left[V(r)+\frac{\ell(\ell+1)\hbar^2}{2\mu r^2}\right]u(r)=Eu(r)\,, \tag{4.31}$$

and the normalization condition for $u(r)$ becomes

$$\int_0^\infty dr\,|u(r)|^2<\infty\,. \tag{4.32}$$

This is quite similar to the one-dimensional Schrödinger equation, but there are several differences. One of the main differences is the presence of the centrifugal term in the potential term: this term behaves like a repulsive potential for states with nonzero angular momentum, and the wave function approaches zero as $r\to 0$ in such a case. The second difference is the boundary condition for the radial wave function: in one dimension, the wave function is defined over the entire real line, whereas in the radial case, it is defined only for $r\in[0,\infty)$.

4.2 Spherical Harmonics

As discussed in the previous subsection, the angular part of the wave function is described by the spherical harmonics, $Y_\ell^m(\theta,\varphi)$. This subsection is devoted to the properties and mathematical structure of the spherical harmonics. The spherical harmonics are the eigenfunctions of the angular momentum operators L^2 and L_3, satisfying the eigenvalue equations, Eqs. (4.26) and (4.27). We recall that Eq. (4.26) is rewritten as

$\nabla^2\left(r^\ell Y_\ell^m\right) = 0$, and $r^\ell Y_\ell^m$ is the homogeneous polynomial of degree ℓ in $x_\pm \equiv x_1 \pm i x_2$ and x_3. Here, the homogeneous polynomial of degree ℓ contains only terms that are of total degree ℓ in $x_\pm$ and x_3. For example, a homogeneous polynomial of degree one is $\sum_i a_i x_i$, and a homogeneous polynomial of degree two is $\sum_{ij} a_{ij} x_i x_j$. It is apparent because the function $\Phi(r,\theta,\varphi) \equiv r^\ell Y_\ell^m(\theta,\varphi)$ scales as λ^ℓ when r is scaled by λ, which is the defining property of a homogeneous polynomial of degree ℓ.

The eigenvalue m of $L_3 = -i\hbar\partial_\varphi$ just counts the difference between the number of x_+ and x_- factors in the homogeneous polynomial: $m \equiv n_+ - n_-$. Indeed, in the spherical coordinate system, remind that $x_\pm = r\sin\theta e^{\pm i\varphi}$ and $x_3 = r\cos\theta$, and φ dependence of the homogeneous polynomial is entirely contained in the factors of $x_\pm$. For example, $L_3(x_+)^n = \hbar n (x_+)^n$ and $L_3(x_+x_-) = 0$. The function $r^\ell Y_\ell^m$ may $(x_+)^\ell$ or $(x_-)^\ell$, namely $m_{\max} = \ell$ and $m_{\min} = -\ell$. Since m is an integer and m runs from $m = -\ell\,, -\ell+1\,, \cdots \ell-1\,, \ell$, there are $2\ell+1$ possible values of m for a given ℓ.

Let us consider the uniqueness of the spherical function for given labels ℓ and m. Any homogeneous polynomials of order ℓ is written as

$$\sum_{\nu_+,\nu_-} C_{\nu_+\nu_-}\,(x_+)^{\nu_+}\,(x_-)^{\nu_-}\,(x_3)^{\ell-\nu_+-\nu_-}\,. \tag{4.33}$$

The homogeneous polynomial $r^\ell Y_\ell^m$ can contain any terms with $\nu_+ - \nu_- = m$. If there are more than two terms for given m, it is not apparent at first glance that only single combination of these terms remain. Otherwise, there are multiple linearly independent homogeneous polynomials for the same m, and the uniqueness of Y_ℓ^m is not guaranteed. For given ℓ, the powers of orders of x_+ and x_- (and also x_3) is constrained as, $0 \le \nu_+ \le \ell$ and $0 \le \nu_- \le \ell - \nu_+$. Hence, the total number of the homogeneous polynomial of order ℓ is computed as

$$N_\ell = \sum_{\nu_+=0}^{\ell}\sum_{\nu_-=0}^{\ell-\nu_+} 1 = \frac{1}{2}(\ell+1)(\ell+2)\,. \tag{4.34}$$

Meanwhile, the Laplacian of homogeneous polynomial of order ℓ is a homogeneous polynomial of order $\ell-2$: therefore, there are $N_{\ell-2}$ independent constraints on the coefficients of the homogeneous polynomial of order ℓ. Hence, the number of independent homogeneous polynomials of order ℓ that satisfy the Laplace equation is

$$N_\ell - N_{\ell-2} = \frac{1}{2}(\ell+1)(\ell+2) - \frac{1}{2}(\ell-1)\ell = 2\ell+1\,, \tag{4.35}$$

which matches the number of possible values of m for a given ℓ. In other words, the number of independent homogeneous polynomials of order ℓ that satisfy the Laplacian constraints is exactly equal to the number of spherical harmonics Y_ℓ^m for a given ℓ, which is $2\ell+1$: this demonstrates the uniqueness of the spherical harmonics for given ℓ and m.

Because we choose the spherical harmonics Y_ℓ^m to be simultaneous eigenfunctions of L^2 and L_3. The application of L_3 gives the eigenvalue $\hbar m$, and hence the spherical harmonics Y_ℓ^m must have the form $e^{im\varphi}$ in the azimuthal angle φ.

$$Y_\ell^m(\theta,\phi) \propto P_\ell^{|m|}(\cos\theta)e^{im\phi}\,. \tag{4.36}$$

The function $P_\ell^{|m|}(\cos\theta)$ is known as the *associated Legendre function*, and $P_\ell^{|m|}(\cos\theta)$ satisfies the associated Legendre differential equation:

$$\left[-\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d}{d\theta}\right)+\frac{m^2}{\sin^2\theta}\right]P_\ell^{|m|}(\cos\theta)=\ell(\ell+1)P_\ell^{|m|}(\cos\theta)\,. \tag{4.37}$$

We remark that the differential equation for $P_\ell^{|m|}(\cos\theta)$ only depend on m^2, and hence the function $P_\ell^{|m|}(\cos\theta)$ does not have the dependence on sign of m.

Now, let us see several examples of how to construct Y_ℓ^m. We remind the reader that $x_\pm = r\sin\theta e^{\pm i\varphi}$ and $x_3 = r\cos\theta$. There are several requirements for constructing the spherical harmonics Y_ℓ^m.

(i). $r^\ell Y_\ell^m(\theta,\varphi)$ are homogeneous polynomials.

(ii). $r^\ell Y_\ell^m(\theta,\varphi)$ satisfy the Laplacian constraint, i.e., $\nabla^2(r^\ell Y_\ell^m)=0$.

(iii). The spherical harmonics is normalized, i.e., $\int d\Omega|Y_\ell^m(\theta,\varphi)|^2=1$, where $d\Omega=\sin\theta d\theta d\varphi$.

First, for $\ell=0$ case, there is only one spherical harmonic, which is the homogeneous polynomial of order zero, i.e., a constant function. The corresponding spherical harmonic is

$$Y_0^0(\theta,\varphi)=\frac{1}{\sqrt{4\pi}}\,. \tag{4.38}$$

Next, for $\ell=1$ case, there are three spherical harmonics corresponding to $m=0,\pm1$. There are three different homogeneous polynomials of order one corresponding to the three spherical harmonics. Indeed,

$$Y_1^{\pm1}(\theta,\varphi)\propto\hat{x}_\pm\,,\quad Y_1^0(\theta,\varphi)\propto\hat{x}_3\,. \tag{4.39}$$

Here, $\hat{x}_\pm$ and $\hat{x}_3$ are the normalized Cartesian coordinates on the unit sphere, i.e.,

$$\hat{x}_\pm=\sin\theta e^{\pm i\varphi},\quad \hat{x}_3=\cos\theta\,. \tag{4.40}$$

Thus, once the normalization is fixed, the spherical harmonics for $\ell=1$ are given by

$$Y_1^{\pm1}(\theta,\varphi)=\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\varphi}\,, \tag{4.41}$$

$$Y_1^0(\theta,\varphi)=\sqrt{\frac{3}{4\pi}}\cos\theta\,. \tag{4.42}$$

The sign convention for Y_ℓ^m is chosen $(-1)^m$ for $m>0$, which is known as the Condon-Shortley phase: it is convenient to obtain the all spherical harmonics using the ladder operators $L_\pm=L_1\pm iL_2$.

Finally, we consider the case with $\ell = 2$. There are six homogeneous polynomials, but they are not all independent due to the Laplace equation constraint. Indeed,

$$Y_2^2 \propto (\hat{x}_+)^2 \quad Y_2^{-2} \propto (\hat{x}_-)^2 \,, \tag{4.43}$$

$$Y_2^1 \propto \hat{x}_+\hat{x}_3 \,, \quad Y_2^{-1} \propto \hat{x}_-\hat{x}_3 \,, \tag{4.44}$$

$$Y_2^0 \propto A\hat{x}_+\hat{x}_- + B\hat{x}_3^2 \,. \tag{4.45}$$

As for the spherical harmonics $m = \pm 1\,, \pm_2$, there is single polynomial for each m, but there are two for $m = 0$. A and B are undetermined constants, but there is a constraint from Laplacian. Once we solve $\nabla^2(r^2 Y_2^0) = 0$, we obtain the relation between A and B as $2A + B = 0$. Therefore, we find $Y_2^0 \propto \hat{x}_+\hat{x}_- - 2\hat{x}_3^2$. Once taking a proper normalization into account, we obtain the explicit forms of Y_2^m as follows.

$$Y_2^{\pm 2}(\theta, \varphi) = \sqrt{\frac{15}{32\pi}} \sin^2\theta e^{\pm 2i\varphi} \,, \tag{4.46}$$

$$Y_2^{\pm 1}(\theta, \varphi) = \mp\sqrt{\frac{8}{32\pi}} \sin\theta\cos\theta e^{\pm i\varphi} \,, \tag{4.47}$$

$$Y_2^0(\theta, \varphi) = \sqrt{\frac{5}{16\pi}} \left(3\cos^2\theta - 1\right) \,. \tag{4.48}$$

For later reference, we need to summarize several properties. One is the orthonormal relation of the spherical harmonics:

$$\int d\Omega \left[Y_\ell^m(\theta, \varphi)\right]^* Y_{\ell'}^{m'}(\theta, \varphi) = \delta_{\ell\ell'}\delta_{mm'} \,. \tag{4.49}$$

It is easy to prove the orthonormality of the spherical harmonics with the normalization above. Once we ignore the overall normalization factor, the spherical harmonics can be expressed as $Y_\ell^m(\theta, \varphi) \propto P_\ell^{|m|}(\theta)e^{im\varphi}$. Thus, the integration over φ gives the orthonormal condition for the state vectors with different m values.

$$\int_0^{2\pi} d\varphi e^{-i(m-m')\varphi} = 2\pi\delta_{mm'} \,. \tag{4.50}$$

Then, we can proceed to the integration over θ, which gives the orthonormal condition for the state vectors with different ℓ values, as follows. Since the φ integration gives the orthonormal condition for the m quantum numbers, we only need to consider the θ integration to establish the orthonormality for the ℓ quantum numbers, so let us set $m' = m$ for the θ integration. By considering the following combinations and using the associated Legendre differential equation, we can find

$$[\ell(\ell+1) - \ell'(\ell'+1)] \int_{-1}^{1} d\cos\theta P_\ell^{|m|}(\cos\theta) P_{\ell'}^{|m|}(\cos\theta) \tag{4.51}$$

$$= -\left[\sin\theta \frac{dP_\ell^{|m|}(\cos\theta)}{d\theta} P_{\ell'}^{|m|}(\cos\theta) - \sin\theta P_\ell^{|m|}(\cos\theta) \frac{dP_{\ell'}^{|m|}(\cos\theta)}{d\theta}\right]_0^\pi = 0 \tag{4.52}$$

In the last equation, the boundary terms vanish because the associated Legendre functions $P_\ell^{|m|}(\cos\theta)$ are regular at the boundaries $\theta = 0$ and $\theta = \pi$ and the sine function $\sin\theta$ goes to zero at these boundaries. Thus, the integral vanishes for $\ell \neq \ell'$, establishing the orthonormality of the associated Legendre functions for different ℓ values.

$$\int_{-1}^{1} d\cos\theta P_\ell^{|m|}(\cos\theta) P_{\ell'}^{|m|}(\cos\theta) = 0 \quad (\ell' \neq \ell)\,, \\ \neq 0 \quad (\ell' = \ell)\,. \tag{4.53}$$

Therefore, taking the proper normalization into account, we can write the orthonormality condition for the associated Legendre functions as

$$\int_{-1}^{1} d\cos\theta P_\ell^{|m|}(\cos\theta) P_{\ell'}^{|m|}(\cos\theta) = \delta_{\ell',\ell}\,. \tag{4.54}$$

In the last of this subsection, we consider the parity property of the spherical harmonics. In the Cartesian coordinates, the parity transformation corresponds to $\hat{\boldsymbol{x}} \to -\hat{\boldsymbol{x}}$. In terms of the spherical coordinates, the parity transformation corresponds to $(r, \theta, \varphi) \to (r, \pi - \theta, \pi + \varphi)$. Since Y_ℓ^m are homogeneous polynomials with rank ℓ of unit vector $\hat{\boldsymbol{x}}$, under the parity transformation $\hat{\boldsymbol{x}} \to -\hat{\boldsymbol{x}}$, they acquire a factor of $(-1)^\ell$.

$$Y_\ell^m(\pi - \theta, \pi + \varphi) = (-1)^\ell Y_\ell^m(\theta, \varphi)\,. \tag{4.55}$$

4.3 Radial Schrödinger Equation

From the succeeding discussion, the radial part of the Schrödinger equation for a particle is considered. Before going into specific potentials, there is a remark on states. The radial Schrödinger equation under a generic potential is written as

$$\left[-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right)+V(r)+\frac{\ell(\ell+1)\hbar^2}{2\mu r^2}\right]R_{n,\ell}(r)=ER_{n,\ell}(r)\,. \tag{4.56}$$

Equivalently, using $u_{n,\ell}(r)=rR_{n,\ell}(r)$, the equation is given by

$$-\frac{\hbar^2}{2\mu}\frac{d^2u_{n,\ell}(r)}{dr^2}+\left[V(r)+\frac{\ell(\ell+1)\hbar^2}{2\mu r^2}\right]u_{n,\ell}(r)=Eu_{n,\ell}(r)\,. \tag{4.57}$$

The effective potential $V_{\text{eff}}(r)$ is defined as the sum of the potential and the centrifugal potential:

$$V_{\text{eff}}(r)\equiv V(r)+\frac{\ell(\ell+1)\hbar^2}{2\mu r^2}\,. \tag{4.58}$$

There are two kinds of solutions for the radial Schrödinger equation: *bound states* and *scattering states*. The bound states are the solution for energy $E<\lim_{r\to\infty}V_{\text{eff}}(r)$. The bound states correspond to localized wave functions that decay to zero as $r\to\infty$, and hence the bound state wave function is normalized. One of the characteristics of the bound state is that the energy eigenvalues are discrete.

Meanwhile, the scattering states are the solution for energy $E\geq\lim_{r\to\infty}V_{\text{eff}}(r)$. The scattering states correspond to delocalized wave functions that oscillate as $r\to\infty$, and hence the scattering state wave function spreads over all space and is not normalizable in the usual sense. In this sense, the position of the particle is not measurable with arbitrary precision for scattering states, and hence in turn the momentum is measurable with an arbitrary precision. Someone may concern the probability interpretation for scattering states, but one can define a *wave packet* that is a superposition of scattering states, which is normalizable and allows for a probabilistic interpretation.

In the following, we will discuss specific examples of central potentials and solve the bound state solutions for the radial Schrödinger equation.

4.4 Spherical well potential

As the first example, we consider the spherical well potential defined by

$$V(r)=\begin{cases}-V_0 & (r\leq a)\\ 0 & (r>a)\end{cases}\,. \tag{4.59}$$

Here, $V_0>0$ is the depth of the spherical well, and a determines the radius of the well. As far as the energy $E<0$, the particle is bound within the spherical well.

Now, we solve the radial Schrödinger equation for the spherical well potential, given by Eq. (4.56), for each region: inside the well ($r\leq a$) and outside the well ($r>a$).

The inside region corresponds to $r \leq a$, where the potential is constant $-V_0$, and the radial Schrödinger equation has the form of:

$$-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{dR_{n,\ell}}{dr}\right)+\frac{\hbar^2\ell(\ell+1)}{2\mu r^2}R_{n,\ell}-(E+V_0)\,R_{n,\ell}(r)=0\,, \tag{4.60}$$

We consider the general solution of the radial Schrödinger equation inside the well for $-V_0 \leq E < 0$. We introduce the parameters to make the radial Schrödinger equation dimensionless,

$$\rho \equiv kr\,, \quad k \equiv \sqrt{\frac{2\mu}{\hbar^2}\,(E+V_0)}\,, \tag{4.61}$$

and then the radial Schrödinger equation can be rewritten in terms of the dimensionless variable ρ.

$$-\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{dR_{n,\ell}}{d\rho}\right)+\frac{\ell(\ell+1)}{\rho^2}R_{n,\ell}-R_{n,\ell}=0\,. \tag{4.62}$$

This equation is known as the *spherical Bessel differential equation*, and the solution is known as *spherical Bessel functions*. Before going to the spherical Bessel functions, let's review the general Bessel differential equation and its solutions. Indeed, the replacement of the function and parameter as

$$R_{n,\ell}=\sqrt{\frac{\pi}{2\rho}}F_{n,p}(\rho)\,, \quad p=\ell+\frac{1}{2} \tag{4.63}$$

the radial Schrödinger equation can be transformed into the standard Bessel differential equation for $F_{n,p}(\rho)$.

$$\frac{d^2F_{n,p}}{d\rho^2}+\frac{1}{\rho}\frac{dF_{n,p}}{d\rho}+\left(1-\frac{p^2}{\rho^2}\right)F_{n,p}=0\,. \tag{4.64}$$

The solution of the Bessel differential equation is given by the Bessel functions. The Bessel function of the first kind is defined as

$$J_p(z)\equiv\sum_{k=0}^{\infty}\frac{(-1)^k}{\Gamma(k+1)\Gamma(k+p+1)}\left(\frac{z}{2}\right)^{2k+p}\,, \tag{4.65}$$

If $p \in \mathbb{Z}$, the Bessel function of the first kind satisfies the relation $J_{-p}(z) = (-1)^p J_p(z)$, and then J_{-p} is not independent of J_p. In general, if $p \notin \mathbb{Z}$, J_p and J_{-p} are independent solutions of the Bessel differential equation. In other way, the Neumann function (or Bessel function of the second kind) is defined as

$$N_p(z)=\frac{\cos(p\pi)J_p(z)-J_{-p}(z)}{\sin(p\pi)}\,, \tag{4.66}$$

which provides an independent solution to the Bessel differential equation when $p \notin \mathbb{Z}$. We may sometimes use the different combinations of Bessel functions, such as the Hankel functions.

$$\begin{cases} H_p^{(1)}(z) \equiv J_p(z) + iN_p(z)\,, \\ H_p^{(2)}(z) \equiv J_p(z) - iN_p(z)\,, \end{cases} \tag{4.67}$$

The solution of Eq. (4.60) is known as the spherical Bessel functions. Indeed, the spherical Bessel functions of the first and second kinds are related to the ordinary Bessel functions of half-integer order as follows.

$$j_\ell(\rho) := \sqrt{\frac{\pi}{2\rho}} J_{\ell+\frac{1}{2}}(\rho)\,, \tag{4.68}$$

$$n_\ell(\rho) := \sqrt{\frac{\pi}{2\rho}} N_{\ell+\frac{1}{2}}(\rho)\,. \tag{4.69}$$

In a similar way, the spherical Hankel functions are also defined

$$h_\ell^{(\alpha)}(\rho) := \sqrt{\frac{\pi}{2\rho}} H_{\ell+\frac{1}{2}}^{(\alpha)}(\rho)\,, \quad (\alpha = 1, 2)\,. \tag{4.70}$$

To find a proper solution of the radial Schrödinger equation, we need to know asymptotic property of the spherical Bessel functions at $\rho \to 0$. Indeed, the asymptotic behavior of the spherical Bessel functions at $\rho \to 0$ is given by

$$j_\ell(\rho) \sim \frac{\rho^\ell}{(2\ell+1)!!}\,, \tag{4.71}$$

$$n_\ell(\rho) \sim -\frac{(2\ell-1)!!}{\rho^{\ell+1}}\,. \tag{4.72}$$

Therefore, even the general solution for $r \le a$ has the following form $R_{n,\ell}(r) = Aj_\ell(kr) + Bn_\ell(kr)$ in order to ensure the regularity of the solution at the origin, we must have $B = 0$. Then, the solution for $r \le a$ is

$$R_{n,\ell}(r) = Aj_\ell(kr)\,, \tag{4.73}$$

with a constant A.

Now, let us think about the solution outside the well $r > a$. The particle motion is not classically allowed in this region, but there is non-zero solution can be found in quantum mechanics. The radial Schrödinger equation for $r > a$ is given by

$$-\frac{\hbar^2}{2\mu r^2}\frac{d}{dr}\left(r^2\frac{dR_{n,\ell}(r)}{dr}\right) + \frac{\hbar^2\ell(\ell+1)}{2\mu r^2}R_{n,\ell}(r) - ER_{n,\ell}(r) = 0\,. \tag{4.74}$$

For $-V_0 \le E < 0$, there exists a non-zero solution outside the well. In a similar way to the previous case, we can define a parameter as

$$\rho \equiv \kappa r\,, \quad \kappa \equiv \sqrt{\frac{-2\mu E}{\hbar^2}}\,, \tag{4.75}$$

and the ODE is simplified as follows.

$$\frac{d^2R_{n,\ell}}{d\rho^2} + \frac{2}{\rho}\frac{dR_{n,\ell}}{d\rho} - \left(1 + \frac{\ell(\ell+1)}{\rho^2}\right) R_{n,\ell} = 0\,. \tag{4.76}$$

We remark that the sign of the last term is opposite fro the previous case. The solution of the ODE is known as the modified spherical Bessel function. There are two independent functions, denoted by $i_\ell(z)$ and $k_\ell(z)$, which are named as the modified spherical Bessel functions of the first and second kinds, respectively. Thus, a general solution for the differential equation is given by $R_{n,\ell}(r) = Ci_\ell(\kappa r) + Dk_\ell(\kappa r)$.

To find a proper asymptotic behavior of the solution at infinity, we consider the limits of the modified spherical Bessel functions as $\rho \to \infty$.

$$i_\ell(\rho) \sim \frac{1}{2\rho}\rho^\rho\,, \qquad k_\ell(\rho) \sim \frac{\pi}{2\rho}e^{-\rho}\,. \tag{4.77}$$

The asymptotic behavior of i_ℓ is exponentially growing, while that of k_ℓ is exponentially dumping. Therefore, to obtain the regular solution at $\rho \to \infty$, we have $C = 0$ and the solution is

$$R_{n,\ell}(r) = Dk_\ell(\kappa r)\,. \tag{4.78}$$

Therefore, even for $E < 0$, there is finite wave function in the classically-forbidden region. Physically, the solution is kind of penetrating into the classically-forbidden region, which is a manifestation of the *quantum tunneling* effect.

Since the radial wave function must be continuous and smooth at the boundary $r = a$, we impose the matching conditions for the wave function and its derivative at $r = a$. The matching conditions at $r = a$ are given by

$$Aj_\ell(ka) = Dk_\ell(\kappa a)\,, \qquad Akj'_\ell(ka) = D\kappa k'_\ell(\kappa a)\,, \tag{4.79}$$

and by taking the ratio, we can omit the coefficients A and D and find the condition for the allowed energy levels.

$$\frac{kaj'_\ell(ka)}{j_\ell(ka)} = \frac{\kappa ak'_\ell(\kappa a)}{k_\ell(\kappa a)}\,. \tag{4.80}$$

Here, prime denotes the derivative of the (modified) spherical Bessel functions with respect to their arguments. We remark that, there is a relation between k and κ given by

$$k^2 + \kappa^2 = \frac{2\mu V_0}{\hbar^2} \equiv \xi^2/a^2\,, \qquad \kappa a = \sqrt{\xi^2 - x^2}\,, \quad x \equiv ka\,. \tag{4.81}$$

Then, from Eq. (4.80), there exist solutions only for the specific values of k, which correspond to the discrete energy levels of the particle in the spherical well. In other words, the wave functions in the inner region and outer region are connected smoothly at the boundary only for specific values of k. In Fig. 1, the discrete energy levels of

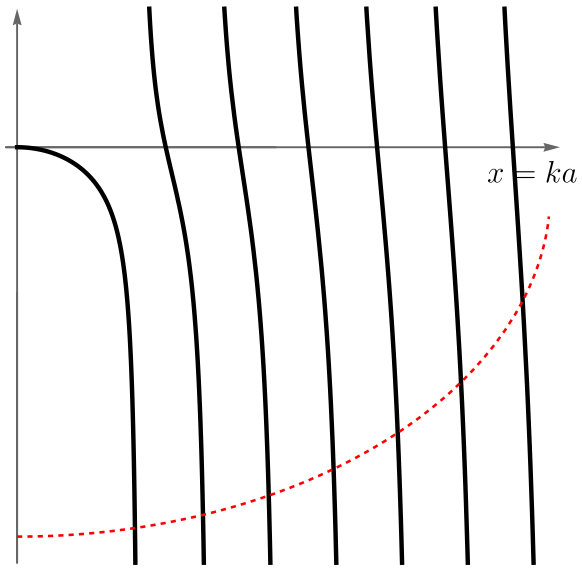


Figure 1: Possible solutions for Eq. (4.80): the wave function exists only for discrete values of k, corresponding to the intersections of the black-solid lines (the left-hand side of Eq. (4.80)) and the red-dashed lines (the right-hand side of Eq. (4.80)). In this figure, ℓ and ξ are fixed.

a particle in a spherical well potential are illustrated. The black thick lines show the left-hand side of the matching condition as a function of $x \equiv ka$, while the red dashed lines show the right-hand side. Therefore, the intersections of the black and red lines indicate the allowed values for k for which the wave function satisfies the matching condition. It leads to the discrete energy levels for the particle in the spherical well.

Let us now consider the explicit form of the matching conditions for the $\ell = 0$ case. The spherical Bessel function for $\ell = 0$ and the modified spherical Bessel function for $\ell = 0$ are given by

$$j_0(kr) = \frac{\sin(kr)}{kr}\,, \qquad k_0(\kappa r) = \frac{e^{-\kappa r}}{\kappa r}\,. \tag{4.82}$$

Therefore, the matching condition for $\ell = 0$ can be explicitly written as follows.

$$x \cot x = -\sqrt{\xi^2 - x^2}\,. \tag{4.83}$$

Here, we use $x \equiv ka$ and $\kappa a = \sqrt{\xi^2 - x^2}$ to simplify the matching condition. As seen, the cotangent function is periodic and there exist multiple solutions for x, corresponding to the discrete energy levels of the particle in the spherical well.

4.5 Hydrogen-like Atoms

This subsection is devoted to discussing hydrogen-like atoms, which are systems consisting of a nucleus with charge $+Ze$ and a single electron bound to it by the Coulomb

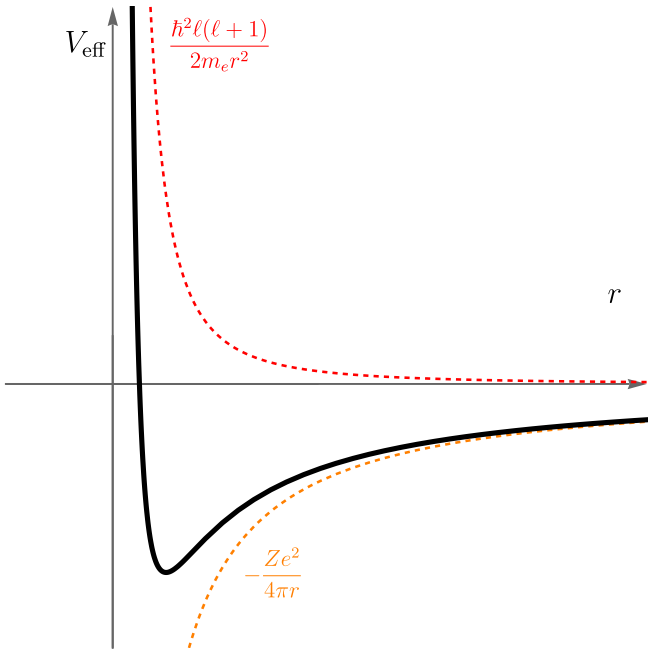


Figure 2: The effective potential for the electron in a hydrogen-like atom, including both the Coulomb potential and the centrifugal potential.

potential. The purpose is to find the electron wave function and the corresponding energy levels in the Coulomb potential. The Coulomb potential for the electron in the field of the nucleus is given by

$$V(r) = -\frac{Ze^2}{4\pi}\frac{1}{r}\,, \tag{4.84}$$

Here, Z denotes the atomic number of the nucleus, and e is the elementary charge in the Heaviside-Lorentz unit system. The potential is attractive and decreases with the distance from the nucleus. The radial Schrödinger equation is given by

$$-\frac{\hbar^2}{2m_e r^2}\frac{d}{dr}\left(r^2\frac{dR_{n,\ell}(r)}{dr}\right) + \frac{\hbar^2\ell(\ell+1)}{2m_e r^2}R_{n,\ell}(r) + V(r)R_{n,\ell}(r) = ER_{n,\ell}(r)\,. \tag{4.85}$$

Here, m_e is the electron mass. There are bound state solutions for $E < 0$ since both the Coulomb potential and the centrifugal potential goes to zero as $r \to \infty$. As discussed above, we use $R_{n,\ell}(r) \equiv \frac{u_{n,\ell}(r)}{r}$ Hence, the radial Schrödinger equation can be rewritten in terms of $u_{n,\ell}(r)$ as follows.

$$-\frac{\hbar^2}{2m_e}\frac{d^2}{dr^2}u_{n,\ell}(r) + \left[\frac{\hbar^2\ell(\ell+1)}{2m_e r^2} - \frac{Ze^2}{4\pi}\frac{1}{r}\right]u_{n,\ell}(r) = Eu_{n,\ell}(r)\,. \tag{4.86}$$

It is convenient to define the following parameters to simplify the radial Schrödinger equation.

$$\kappa^2 \equiv \frac{-2m_e E}{\hbar^2}\,, \tag{4.87}$$

Here, κ is a positive real number for bound states with $E < 0$, and it has the dimension of inverse length. The dimensionless radius is $\rho \equiv \kappa r$. Furthermore, a dimensionless parameter ξ is defined as follows.

$$\xi \equiv \frac{2m_e}{\hbar^2} \times \frac{2e^2}{4\pi\kappa}\,, \tag{4.88}$$

Here, ξ is a dimensionless parameter that characterizes the strength of the Coulomb potential relative to the kinetic energy term and the centrifugal potential term in the radial Schrödinger equation. The differential equation is simplified by introducing the dimensionless variables ρ and ξ.

$$-\frac{d^2u_{n,\ell}}{d\rho^2} + \left[-\frac{\xi}{\rho} + \frac{l(l+1)}{\rho^2}\right] u_{n,\ell} = -u_{n,\ell}\,. \tag{4.89}$$

Then, let us find the asymptotic behavior of the solution at small and large ρ. At small ρ, i.e., $\rho \to 0$, the centrifugal term dominates, and the radial wave function should behave as $u_{n,\ell}(\rho) \propto \rho^{\ell+1}$. Meanwhile, at large ρ, i.e., $\rho \to \infty$, both the Coulomb potential and the centrifugal potential are negligible and the exponential decay dominates, so the radial wave function should behave as $u_{n,\ell}(\rho) \propto e^{-\rho}$. There would be another solution at $\rho \to \infty$, e^{ρ}, but it is exponentially growing and not normalizable, so it is not physically acceptable. Then, we can write the radial wave function as a product of the asymptotic behaviors at small and large ρ and an unknown function $F(\rho)$ that is regular everywhere.

$$u_{n,\ell} = \rho^{\ell+1}e^{-\rho}F_{n,\ell}(\rho)\,. \tag{4.90}$$

The function $F_{n,\ell}(\rho)$ satisfies the following differential equation.

$$\frac{d^2F_{n,\ell}}{d\rho^2} - 2\left(1 - \frac{\ell+1}{\rho}\right)\frac{dF_{n,\ell}}{d\rho} + \frac{\xi - 2(\ell+1)}{\rho}F_{n,\ell} = 0\,. \tag{4.91}$$

The differential equation for $F_{n,\ell}(\rho)$ can be solved using a series expansion around $\rho = 0$.

$$F_{n,\ell}(\rho) = \sum_{s=0}^{\infty} a_s^{n,\ell}\rho^s\,. \tag{4.92}$$

We note that $a_0^{n,\ell} \neq 0$, this is because we have already know that the radial function behaves as $u_{n,\ell} \propto \rho^{\ell+1}$ for $\rho \to 0$. This ensures that the series expansion for $F_{n,\ell}(\rho)$ starts with a non-zero constant term around $\rho = 0$. Once substituting the series expansion for $F_{n,\ell}(\rho)$ into the differential equation, we obtain a recurrence relation for the coefficients $a_s^{n,\ell}$.

$$\sum_{s=0}^{\infty}\rho^{s-1}\left\{(s+1)[s+2(\ell+1)]a_{s+1}^{n,\ell} + [\xi - 2(\ell+1) - 2s]a_s^{n,\ell}\right\} = 0\,. \tag{4.93}$$

This recurrence relation should hold for any $\rho > 0$, which implies that the coefficient of each power of ρ must vanish independently.

$$(s+1)[s+2(\ell+1)]a_{s+1}^{n,\ell} + [\xi - 2(\ell+1) - 2s]a_s^{n,\ell} = 0\,. \tag{4.94}$$

Now, we consider the asymptotic solutions $F_{n,\ell}(\rho)$ at large ρ. To analyze the asymptotic behavior of the series at large ρ, the series expansion is dominated by the terms with large s. Thus, we can focus on the large s contribution, which is approximately given by

$$\frac{a_{s+1}^{n,\ell}}{a_s^{n,\ell}} \approx \frac{2}{s}\,. \tag{4.95}$$

So, all terms at large s have the same sign, and the asymptotic behavior of the series is dominated by these large s terms. As far as the asymptotic behavior is concerned, the series behaves like an exponential function at large ρ. Indeed, the coefficient $a_s^{n,\ell}$ at large s is approximately given by

$$a_s^{n,\ell} \approx \frac{2}{s-1}\cdot\frac{2}{s-2}\cdot\cdots \approx c\frac{2^s}{(s+b)!}\,. \tag{4.96}$$

Here, parameters b and c reflect that this expression assumes the asymptotic form of the coefficients at large s. The approximation that $a_{s+1}^{n,\ell}/a_s^{n,\ell} \approx 2/s$ holds only for large s, and hence the true coefficient would of course deviate from this expression. When b is not an integer, the factorial $(s+b)!$ should be interpreted using the Gamma function as $(s+b)! \simeq \Gamma(s+b+1)$. The asymptotic behavior of the series at large s is therefore captured by this expression, which leads to the exponential growth of the series at large ρ. Indeed, the function $F_{n,\ell}(\rho)$ with the asymptotic form of the coefficients at large s exhibits,

$$F_{n,\ell}(\rho) \approx \sum_{n=0}^{\infty} c\frac{(2\rho)^s}{(s+b)!} \approx c(2\rho)^{-b}e^{2\rho}\,, \tag{4.97}$$

and combined solution for the radial function $u_{n,\ell}(\rho)$ is asymptotically behave like $u_{n,\ell}(\rho) \approx \rho^{\ell+1}e^{\rho}$ at large ρ. Of course, this is not surprising that the asymptotic behavior of the radial function at large ρ is dominated by the exponential growth, as expected from the form of the differential equation. However, this exponential growth is not physically acceptable for bound state solutions, which must remain normalizable. There is some missing discussion to keep the solution normalizable. To ensure normalizability, the series expansion for $F_{n,\ell}(\rho)$ must terminate at some finite $s = s_0$, leading to a polynomial solution rather than an infinite series. From Eq. (4.94), we can have the termination condition for the series,

$$\xi - 2(\ell+1) - 2s_0 = 0 \tag{4.98}$$

This termination condition imposes quantization on the energy levels, which leads to the discrete spectrum of the hydrogen atom. Let us take the quantization condition

$\xi = 2n$ with n being a positive integer, and the corresponding termination index is given by $s_0 = n - (\ell + 1)$. Then, the polynomial coefficients $a_s^{n,\ell}$ vanish for $s > s_0$. The integer n is called as the *principal quantum number*, which determines the energy level of the hydrogen atom as we will see later. Finally, the function $F_{n,\ell}(\rho)$ can be expressed as a finite polynomial of degree $s_0 = n - (\ell + 1)$:

$$F_{n,\ell}(\rho) = \sum_{s=0}^{n-\ell-1} a_s^{n,\ell} \rho^s \,. \tag{4.99}$$

The differential equation for $F_{n,\ell}(\rho)$, Eq. (4.91), is known as the *associated/generalized Laguerre differential equation*. Indeed, the differential equation for $F_{n,\ell}(\rho)$ can be transformed into the standard form of the associated Laguerre differential equation: replace the variables and parameters with $z = 2\rho\,, p = 2\ell + 1\,,$ and $q = n - \ell - 1$, then

$$z\frac{d^2F_{n,\ell}}{dz^2} + (p + 1 - z)\frac{dF_{n,\ell}}{dz} + qF_{n,\ell} = 0\,. \tag{4.100}$$

It is a conventional notation for the solution of the associated Laguerre differential equation, and the solution is called the *associated/generalized Laguerre polynomials* $L_q^p(z)$.[1] We express the function $F_{n,\ell}(\rho)$ in terms of the associated Laguerre polynomials:

$$F_{n,\ell}(\rho) = L_{n-\ell-1}^{2\ell+1}(2\rho)\,. \tag{4.101}$$

We list the specific forms of the associated Laguerre polynomials for the first few values of n (as taking $a_0^{n,\ell} = 1$):

$$F_{n,\ell}(\rho) = 1\,, \quad (n = \ell + 1) \tag{4.102}$$

$$F_{n,\ell}(\rho) = 1 - \frac{\rho}{\ell + 1}\,, \quad (n = \ell + 2)\,, \tag{4.103}$$

Substituting these polynomial solutions and parameters into the radial wave function, we obtain,

$$u_{n,\ell} = \rho^{\rho+1} e^{-\rho} L_{n-\ell-1}^{2\ell+1}(2\rho)\,, \tag{4.104}$$

and

$$R_{n,\ell}(r) = r u_{n,\ell}(r) = N_{n,\ell}(\kappa_n r)^\ell e^{-\kappa_n r} L_{n-\ell-1}^{2\ell+1}(2\kappa_n r)\,. \tag{4.105}$$

Here, $N_{n,\ell}$ is the normalization constant for the radial wave function. We remark that $\xi = 2n$ for terminating the associated Laguerre polynomials, and accordingly, the principal quantum number n determines the energy levels of the hydrogen-like atom.

$$\kappa_n = \frac{m_e Z e^2}{4\pi\hbar^2 n} \equiv \frac{1}{n a_B}\,, \tag{4.106}$$

[1] This form of the differential equation is also known as the *confluent hypergeometric differential equation*, and its solution is also expressed by $F_{n,\ell} \propto {}_1F_1(-q; p + 1; z)$, where ${}_1F_1$ is the *confluent hypergeometric function of the first kind*. We do not further discuss the confluent hypergeometric function here, as our focus is on the polynomial solutions corresponding to the associated Laguerre polynomials.

where we define the *Bohr radius* a_B, which have the dimension of length. The numerical value of the Bohr radius is estimated by the several fundamental physical constants as follows: $m_e c^2 \simeq 511\,\mathrm{keV}$, $\hbar c \simeq 197.327\,\mathrm{MeV\,fm}$, and $e^2/(4\pi\hbar c) \simeq 1/137$. Then, using these constants with high precision, the numerical value of the Bohr radius is known with high accuracy as follows:

$$a_B = \frac{4\pi\hbar^2}{m_e Z e^2} \cong 0.529177210544(82) \times 10^{-10} Z^{-1}\ \mathrm{m}\,. \tag{4.107}$$

This accurate value is found in the Particle Data Group (2026) reference.

Using this Bohr radius, we can express the radial wave function and energy levels in terms of a_B. The Bohr radius a_B serves as a natural length scale for the hydrogen-like atom.

$$R_{n,\ell}(r) = N_{n,\ell} \left(\frac{r}{n a_B}\right)^{\ell} e^{-r/(n a_B)} L^{2\ell+1}_{n-\ell-1}\left(\frac{2r}{n a_B}\right)\,. \tag{4.108}$$

From this expression, we can see that the radial wave function is localized around the Bohr radius a_B. In particular, for the ground state ($n = 1, \ell = 0$), the probability density is maximized around $r \sim a_B$. In this sense, we can interpret the Bohr radius a_B as the characteristic "size" of the hydrogen-like atom in its ground state. The discretized energy levels of the hydrogen-like atom are given by

$$E_n = -\frac{\hbar^2 \kappa_n^2}{2m_e} = -\frac{\hbar^2}{2m_e a_B^2 n^2} = -\frac{Z^2}{n^2} E_{\mathrm{Ry}}\,. \tag{4.109}$$

Here, E_{Ry} is called the Rydberg energy. The Rydberg energy is also determined by the fundamental physical constants, electron mass $m_e c^2$, and the fine-structure constant $\alpha = e^2/(4\pi\hbar c)$. The numerical value of the Rydberg energy is also obtained from the fundamental physical constants as shown below.

$$E_{\mathrm{Ry}} \equiv \frac{1}{2}(m_e c^2)\alpha^2 = 13.605693122990(15)\,\mathrm{eV}\,. \tag{4.110}$$

We discuss the physical consequences and implications from the hydrogen-like atom. The meaning of "Hydrogen-like atom" is an atom with one electron orbiting around a nucleus with charge Ze. To be more realistic, we have to take into account several effects: we neglect magnetic fields and relativistic corrections. For the smaller values of Z, the system corresponds to hydrogen-like atoms such as hydrogen ($Z = 1$) and singly ionized helium ($Z = 2$). For the larger Z, we have to consider the effects of inner electrons more carefully: in particular the screening of the nuclear charge by the inner electrons.

We now consider the case of hot hydrogen gas. The excitation of hydrogen-like atom from $n = 1$ to $n = 2$ requires energy of the order of 10 eV. This excitation energy corresponds to a temperature of the order of 10^5 K. Hot gases in astrophysical environments would cool by emitting radiation from atoms excited to higher energy levels: and therefore it is difficult to cool a gas of hot hydrogen below the temperature

Table 1: Spectroscopic notation for azimuthal quantum numbers ℓ.

ℓ	0	1	2	3	4	5	6	7	$\cdots$
Letters	s	p	d	f	g	h	i	k	$\cdots$
Name	Sharp	Principle	Diffuse	Fundamental					

corresponding to the excitation energy of the first excited state, which is of the order of 10^5 K. Meanwhile, the outer electrons in multi-electron atoms can be excited at lower energies since they all have relatively larger values of n, allowing these atoms to cool more efficiently than hydrogen.

Let us consider the degeneracy of the energy levels in a hydrogen-like atom. For the fixed n, the energy eigenvalue E_n is the same for all allowed values of ℓ and m. For a given n, the allowed values of ℓ are constrained by the condition that $n-\ell-1 \geq 0$ to terminate the polynomial solutions of the radial equation: namely $0 \leq \ell \leq n-1$. As we saw in the previous discussion, for each value of ℓ, the magnetic quantum number m can take $2\ell+1$ values: $m = 0, \pm 1, \cdots, \pm\ell$. In total, the number of degenerate states for a given n is

$$N_{\text{deg.}} = \sum_{\ell=0}^{n-1}(2\ell+1) = n^2\,, \tag{4.111}$$

These n^2 degenerate states correspond to the different combinations of the azimuthal quantum number ℓ and the magnetic quantum number m for a given principal quantum number n. Once we include several effects, such as external magnetic fields or relativistic corrections, these degenerate states can be split. Some of the resolution of these degeneracies can be discussed in the section for "perturbation theory".

Special letters are assigned to the spectra from a given ionization state of an atom. This is known as the *spectroscopic notation*, where each letter corresponds to a specific azimuthal quantum number ℓ. The same notation is frequently used even for the processes which are not strictly related to atomic systems. The table 1 summarizes the spectroscopic notation for the azimuthal quantum numbers in atomic spectra. The first-four letters s, p, d, and f have specific names: sharp, principle, diffuse, and fundamental, respectively. These names originate from early spectroscopic observations of atomic spectra, where the lines corresponding to these transitions exhibited distinct characteristics. It is important to note that for azimuthal quantum numbers $\ell \geq 4$, there are no commonly used names, and the letters g, h, i, etc., are used in alphabetical order.[2]

The hydrogen atom is a fundamental example of a central potential problem in quantum mechanics. In this subsection, we have considered the electron wave function under the Coulomb potential created by the (point-like) nucleus. However, the hydrogen atom is actually a two-body system consisting of the nucleus and the electron, and the Coulomb potential is an approximation that treats the nucleus as a fixed point.

[2]Only j is traditionally omitted from the spectroscopic notation. There seems to be several historical reasons: the distinction between "i" and "j" was a recent phenomenon, some language used "j" differently, and to avoid confusion with other labels (such as total angular momentum j).

In this subsection, we have treated the nucleus mass as effectively infinite compared to the electron mass, allowing us to consider it as a fixed point. This approximation simplifies the problem to a one-body problem with a central potential, which is much easier to solve analytically. For more precise treatment, we can think of the hydrogen atom as a true two-body system, where both the nucleus and the electron are allowed to move, and the Coulomb potential depends on their relative position.

4.6 Harmonic oscillators

Last but not least in this section, we consider the quantum harmonic oscillator as another example for the bound state system. The potential energy of the quantum harmonic oscillator is given by

$$V(r) = \frac{1}{2} M\omega^2 r^2 \tag{4.112}$$

where M denotes the mass of a particle, and ω is the angular frequency of the oscillator. There are several physical motivations for studying the harmonic oscillator. In historical context, the harmonic oscillator played a crucial role in the development of quantum mechanics, which is used as a toy model for atomic spectra (algebraic methods for hydrogen energy levels and radiative transition in the matrix mechanics) and as a model of atomic nuclei. The harmonic oscillator is useful for dealing with energy levels of electrons in magnetic fields. In the quantum field theory, the harmonic oscillator plays a fundamental role in the second quantization of fields in the flat space time.

The Schrödinger equation for the harmonic oscillator is given by

$$\left[-\frac{\hbar^2}{2M} \nabla^2 + \frac{1}{2} M\omega^2 r^2 \right] \psi(x) = E\psi(x) \,. \tag{4.113}$$

In this section, we do not solve the Schrödinger equation in spherical coordinates, but rather in Cartesian coordinates. Since $r^2 = \boldsymbol{x}^2$, then the Schrödinger equation is secondarily separable and can be written as a sum of three one-dimensional harmonic oscillator equations:

$$\sum_{i=1}^{3} \left[-\frac{\hbar^2}{2M} \frac{\partial^2}{\partial x_i^2} + \frac{1}{2} M\omega^2 x_i^2 \right] \psi(x) = E\psi(x) \,. \tag{4.114}$$

The solution for this differential equation can be written as a product of three one-dimensional wavefunctions:

$$\psi(x) = \psi_{n_1}(x_1)\psi_{n_2}(x_2)\psi_{n_3}(x_3) \,, \tag{4.115}$$

where $\psi_{n_i}(x_i)$ denotes a solution of the one-dimensional Schrödinger equation, and n_i labels the energy eigenvalues of the i-th one-dimensional harmonic oscillator.

$$\left[-\frac{\hbar^2}{2M} \frac{d^2}{dx_i^2} + \frac{1}{2} M\omega^2 x_i^2 \right] \psi_{n_i}(x_i) = E_{n_i} \psi_{n_i}(x_i) \,. \tag{4.116}$$

The total energy of the system is given by the summation of each one-dimensional energy eigenvalue:

$$E = \sum_i E_{n_i} \,. \tag{4.117}$$

Now, we solve the one-dimensional harmonic oscillator using the *operator method*. For this purpose, we introduce the *lowering and raising operators* as combinations of the position and momentum operators:

$$a_i \equiv \frac{1}{\sqrt{2M\hbar\omega}} \left(-i\hbar \frac{d}{dx_i} - iM\omega x_i \right) , \tag{4.118}$$

$$a_i^\dagger \equiv \frac{1}{\sqrt{2M\hbar\omega}} \left(-i\hbar \frac{d}{dx_i} + iM\omega x_i \right) . \tag{4.119}$$

Here, a_i and $a_i^\dagger$ are the lowering and raising operators, respectively. By the use of the commutator of position and momentum operators, $[x_i, p_j] = i\hbar\delta_{ij}$, we can find the commutation relations of the lowering and raising operators:

$$[a_i, a_j^\dagger] = \delta_{ij} \,, \tag{4.120}$$

$$[a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0 \,. \tag{4.121}$$

The one-dimensional Hamiltonian is rewritten in terms of the lowering and raising operators as follows.

$$H_i \equiv -\frac{\hbar^2}{2M}\frac{d^2}{dx_i^2} + \frac{1}{2}M\omega^2 x_i^2 = \hbar\omega(a_i^\dagger a_i + \frac{1}{2}) \,. \tag{4.122}$$

Using the commutation relation among the lowering and raising operators, the commutation relations between the Hamiltonian H_i and a_i and $a_i^\dagger$ are given by:

$$[H_i, a_i] = -\hbar\omega a_i \,, \quad \left[H_i, a_i^\dagger\right] = \hbar\omega a_i^\dagger \,. \tag{4.123}$$

These commutation relations imply that the lowering operator a_i decreases the energy of a state by $\hbar\omega$, while the raising operator $a_i^\dagger$ increases the energy by $\hbar\omega$. Indeed, for given energy state ψ_{n_i} with energy E_{n_i}, we can consider the action of the lowering and raising operators: $a_i\psi_{n_i}$ and $a_i^\dagger\psi_{n_i}$. By the using of the commutation relations with the Hamiltonian, we can prove that $a_i\psi_{n_i}$ is an eigenstate of H_i with energy $E_{n_i} - \hbar\omega$, and $a_i^\dagger\psi_{n_i}$ is an eigenstate with energy $E_{n_i} + \hbar\omega$.

$$H_i(a_i\psi_{n_i}) = (E_{n_i} - \hbar\omega)(a_i\psi_{n_i}) \,, \tag{4.124}$$

$$H_i(a_i^\dagger\psi_{n_i}) = (E_{n_i} + \hbar\omega)(a_i^\dagger\psi_{n_i}) \,. \tag{4.125}$$

Now, we can construct the bound state solutions for the harmonic oscillator. Since the energy eigenstates with the higher energy can be obtained by applying the raising operator $a_i^\dagger$ to the lower energy states, we first need to find the ground state, which is

annihilated by the lowering operator a_i: $a_i\psi_0(x_i) = 0$. The action of a_i on the wave function reduce the energy eigenvalue in unit of $\hbar\omega$, and hence we call it a "*ground*" state. Thus, the ground state wavefunction $\psi_0(x_i)$ can be found by solving the equation $a_i\psi_0(x_i) = 0$.

$$\frac{1}{\sqrt{2M\hbar\omega}}\left(-i\hbar\frac{d}{dx_i} - iM\omega x_i\right)\psi_0 = 0\,, \tag{4.126}$$

$$\psi_0 \propto \exp\left(-\frac{M\omega}{2\hbar}x_i^2\right)\,. \tag{4.127}$$

Once we impose the normalization condition for the ground state wavefunction, we obtain the properly normalized $\psi_0(x_i)$. It is also questionable if there is a normalizable solution for the equation $a_i^\dagger\psi = 0$. However, the sign in front of the position operator is flipped in this case, and hence the solution for $a_i^\dagger\psi = 0$ is exponentially glowing as $x_i \to \infty$. There is no normalizable solution for $a_i^\dagger\psi = 0$. This implies that there is no highest energy state for the harmonic oscillator, as one can always apply the raising operator to obtain a state with higher energy.

Since the acting of the lowering operator on ψ_0 vanishes the energy eigenstate, the energy eigenvalue for ψ_0 is easily found,

$$H_i\psi_0 = \hbar\omega\left(a_i^\dagger a_i + \frac{1}{2}\right)\psi_0 = \frac{\hbar\omega}{2}\psi_0\,. \tag{4.128}$$

The ψ_0 is the ground state wavefunction of the harmonic oscillator, corresponding to the lowest energy eigenvalue $\hbar\omega/2$. Any solutions with the energy higher than $\hbar\omega/2$ can be constructed by applying $a_i^\dagger$ repeatedly to the ground state ψ_0.

$$\psi_{n_i} \propto \left(a_i^\dagger\right)^{n_i}\psi_0\,. \tag{4.129}$$

The overall normalization constant for ψ_{n_i} can be determined by ensuring that the wavefunction is properly normalized. It is easy to confirm that ψ_{n_i} defined above is indeed an eigenstate of the Hamiltonian with eigenvalue $E_{n_i} = \hbar\omega\left(n_i + \frac{1}{2}\right)$.

$$H_i\psi_{n_i} = C_{n_i}\hbar\omega\left(a_i^\dagger a_i + \frac{1}{2}\right)\left(a_i^\dagger\right)^{n_i}\psi_0 = E_{n_i}\psi_{n_i}\,. \tag{4.130}$$

Here, in the intermediate step, the normalization factor C_{n_i} appears. This equation is easily obtained after moving the lowering operator a_i to the right by using the commutation relation $[a_i, a_i^\dagger] = 1$. In this sense, the operator $N_i \equiv a_i^\dagger a_i$ counts the number of quanta in the i-th mode which carry the energy of $\hbar\omega$ and is called the *number operator*.

As for the normalization constant for the ground state wavefunction, we can find it by requiring that the wavefunction is properly normalized:

$$\int_{-\infty}^{\infty} dx_i\,|\psi_0(x_i)|^2 = 1\,, \tag{4.131}$$

then the explicit form of the ground state wavefunction is

$$\psi_0(x_i) = \left(\frac{M\omega}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{M\omega}{2\hbar}x_i^2\right) . \tag{4.132}$$

Next, we consider the wave function for given any energy eigenvalue E_{n_i}. As a first attempt, we try to solve the Schrödinger equation by factoring out the Gaussian part of the wave function and by expressing the remaining part as a power series, as we did in the case of the hydrogen-like atom. It is convenient to define the dimensionless quantities:

$$\xi_i \equiv \sqrt{\frac{M\omega}{\hbar}}x_i\,, \quad \kappa_{n_i} \equiv \frac{E_{n_i}}{\hbar\omega}\,, \tag{4.133}$$

and make the one-dimensional Schrödinger equation dimensionless. In addition, it is also convenient to factor out the Gaussian part of the wavefunction:

$$\psi_{n_i}(x_i) = h(\xi_i)\exp\left(-\frac{\xi_i^2}{2}\right) . \tag{4.134}$$

The Schrödinger equation for the one-dimensional harmonic oscillator then becomes the differential equation for $h(\xi_i)$ as follows:

$$\frac{d^2}{d\xi_i^2}h(\xi_i) - 2\xi_i\frac{d}{d\xi_i}h(\xi_i) + (\kappa_{n_i} - 1)h(\xi_i) = 0\,. \tag{4.135}$$

As usual way, we try to find the solution in the form of a power series. Let the solution be expressed as a power series:

$$h(\xi_i) = \sum_{n=0}^{\infty}\alpha_n(\xi_i)^n\,. \tag{4.136}$$

When inserting this solution to the differential equation, we can obtain the recurrence relations. Series expansion of $h(\xi_i)$ leads to the following equality,

$$\sum_{n=0}^{\infty}\left[(n+1)(n+2)\alpha_{n+2} - 2n\alpha_n + (\kappa_{n_i} - 1)\alpha_n\right](\xi_i)^n = 0\,, \tag{4.137}$$

This equality holds for any ξ_i, and we obtain the recurrence relation for the coefficients α_n as follows:

$$\alpha_{n+2} = \frac{2n + 1 - \kappa_{n_i}}{(n+1)(n+2)}\alpha_n\,. \tag{4.138}$$

From this expression, we can see that the series would be terminated if the energy is quantized in the unit of $\hbar\omega$ as seen before. In the following, we will again confirm that the energy eigenstate cannot be normalized when the energy eigenvalue is not quantized. It is similar to the case of the hydrogen-like atom, the behavior of the

coefficients at large n determines the asymptotic form of the solution (at $\xi_i \to \infty$). The large n behavior of the coefficients is given by

$$\alpha_{n+2} \simeq \frac{2}{n}\alpha_n \simeq \cdots \simeq \frac{\alpha_0}{(n/2)!}\,, \tag{4.139}$$

and hence the asymptotic form of $h(\xi_i)$ is

$$h(\xi_i) \sim \sum_{n\in\text{even}} \frac{\alpha_0}{(n/2)!}(\xi_i)^n + \sum_{n\in\text{odd}} \frac{\alpha_1}{(n/2)!}(\xi_i)^n \sim e^{\xi_i^2}\,. \tag{4.140}$$

Therefore, the asymptotic form of the wave function exponentially grows, and it is not normalizable unless the series terminates. In the similar way to the hydrogen-like atom, we require that the series terminates to ensure normalizability of the wavefunction, then the energy levels are quantized. At some n, the series must terminate, which gives the condition $2n + 1 - \kappa_{n_i} = 0$.

The terminated polynomials for $h(\xi_i)$ are known as the *Hermite polynomials*. The general solution for the wave function under the quadratic potential is written as

$$\psi_n(x) = \left(\frac{M\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{M\omega}{\hbar}}x\right) \exp\left(-\frac{M\omega}{2\hbar}x^2\right)\,. \tag{4.141}$$

Here, $H_n(\xi)$ denotes the *Hermite polynomials* of order n. Since $H_n(\xi)$ can be obtained by applying multiple raising operators

$$a^\dagger = -\frac{i}{\sqrt{2}}\left(\frac{d}{d\xi} - \xi\right)\,, \tag{4.142}$$

on $H_0(\xi)e^{-\xi^2/2}$. Comparing with Eq. (4.132), we can find $H_0(\xi) = 1$ for the normalized wave function. The raising operator is odd under the inversion of the variable $\xi \to -\xi$, and hence the Hermite polynomials can be odd or even function depending on the order n.

$$H_n(-\xi) = (-1)^n H_n(\xi)\,. \tag{4.143}$$

Combining the one-dimensional solutions, we obtain the three-dimensional wave function as follows:

$$\psi_{n_1n_2n_3}(\boldsymbol{x}) = \left(\frac{M\omega}{\pi\hbar}\right)^{3/4} \prod_{i=1}^{3}\left[\frac{1}{\sqrt{2^{n_i}n_i!}} H_{n_i}\left(\sqrt{\frac{M\omega}{\hbar}}x_i\right)\right] \exp\left(-\frac{M\omega r^2}{2\hbar}\right)\,, \tag{4.144}$$

and the total energy E_N is given by

$$E_N = \hbar\omega\left(N + \frac{3}{2}\right)\,, \qquad N \equiv n_1 + n_2 + n_3\,. \tag{4.145}$$

This solution can also be obtained using the operator method, which provides an alternative and often more convenient way to construct the wave functions for the

harmonic oscillator. We may write the wave functions in terms of multiple raising operators acting $a^\dagger$ on the ground state. The one-dimensional wave function can be written as

$$\psi_0(x) = \left(\frac{M\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{M\omega x^2}{2\hbar}\right), \tag{4.146}$$

and hence the n-th excited state can be obtained by applying the raising operator $a^\dagger$ n times on the ground state:

$$\psi_n(x) = C_n \left(a^\dagger\right)^n \psi_0(x), \tag{4.147}$$

where C_n denotes the normalization constant. This should be $C_0 = 1$ to be consistent for $n = 0$.

To find the normalization constant for n, we need to evaluate the integral of $|\psi_n(x)|^2$. This integration can be simplified using the commutation relation of the raising and lowering operators, which is given by $[a, a^\dagger] = 1$. In particular, we have a useful commutation relation between the lowering operator a and the n-th power of the raising operator $a^\dagger$:

$$[a, (a^\dagger)^n] = n(a^\dagger)^{n-1}. \tag{4.148}$$

Then, the integration of $|(a^\dagger)^n \psi_0(x)|^2$ can be evaluated as follows:

$$\int_{-\infty}^{\infty} \left|\left(a^\dagger\right)^n \psi_0(x)\right|^2 dx = n \int_{-\infty}^{\infty} \left|\left(a^\dagger\right)^{n-1} \psi_0(x)\right|^2 dx = n! \int_{-\infty}^{\infty} |\psi_0(x)|^2 dx. \tag{4.149}$$

Therefore, by taking a proper normalization for $\psi_0(x)$, the normalization constant for $\psi_n(x)$ is given by

$$C_n = \frac{1}{\sqrt{n!}}. \tag{4.150}$$

So, the one-dimensional wave function for the n-th excited state can be expressed as

$$\psi_n(x) = \frac{1}{\sqrt{n!}} \left(a^\dagger\right)^n \psi_0(x) = \left(\frac{M\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{n!}} \left(a^\dagger\right)^n \exp\left(-\frac{M\omega x^2}{2\hbar}\right), \tag{4.151}$$

Comparing with the standard form of the one-dimensional harmonic oscillator wave function given in Eq. (4.141), we can find the form of Hermite polynomials $H_n(\xi)$ in terms of operators.

$$H_n\left(\sqrt{\frac{M\omega}{\hbar}} x\right) = 2^{n/2} \exp\left(\frac{M\omega x^2}{2\hbar}\right) \left(a^\dagger\right)^n \exp\left(-\frac{M\omega x^2}{2\hbar}\right). \tag{4.152}$$

The three-dimensional wave function can be expressed in terms of the raising operators as follows:

$$\psi_{n_1 n_2 n_3}(\boldsymbol{x}) = \frac{1}{\sqrt{n_1! n_2! n_3!}} \left(\frac{M\omega}{\pi\hbar}\right)^{3/4} (a_1^\dagger)^{n_1} (a_2^\dagger)^{n_2} (a_3^\dagger)^{n_3} \exp\left(-\frac{M\omega r^2}{2\hbar}\right). \tag{4.153}$$

Now, we consider the degeneracy of the energy levels. The system has the same energy for fixed $N = n_1 + n_2 + n_3$, and thus there are multiple degenerate states for a given N. The total degeneracy for a given N is obtained by counting all possible combinations of n_1, n_2, and n_3 that satisfy $N = n_1 + n_2 + n_3$, which leads to the following expression:

$$N_N = \sum_{n_1=0}^{N} \sum_{n_2=0}^{N-n_1} 1 = \frac{1}{2}(N+1)(N+2)\,. \tag{4.154}$$

As one of the application of the harmonic oscillator, now we derive Heisenberg's complex amplitude. Focusing on the one-dimensional harmonic oscillator, we can write the position operators in terms of the raising and lowering operators as follows:

$$x = i\sqrt{\frac{\hbar}{2M\omega}}(a - a^\dagger)\,. \tag{4.155}$$

The time dependent wave function is written as

$$\psi_n(x,t) = e^{-i\frac{E_n}{\hbar}t}\psi_n(x)\,, \tag{4.156}$$

where $\psi_n(x)$ is the solution of the Schrödinger equation with the quadratic potential. Applying the raising or lowering operators to the wave function with n gives the wave function with $n \pm 1$. The matrix element of the position operator requires the principal quantum number to change by one, i.e., $\Delta n = \pm 1$. The non-zero matrix element for the position operator is given by the wave function with n and $n \pm 1$, and its expression is

$$\begin{aligned}
[x]_{n+1,n} &= \int_{-\infty}^{\infty} dx \psi^*_{n+1}(x,t) x \psi_n(x,t) \\
&= e^{i\frac{E_{n+1}-E_n}{\hbar}t} \times i\sqrt{\frac{\hbar}{2M\omega}} \int_{-\infty}^{\infty} dx \psi^*_{n+1}(x)(-a^\dagger)\psi_n(x) \\
&= -i\sqrt{\frac{(n+1)\hbar}{2M\omega}} e^{i\frac{E_{n+1}-E_n}{\hbar}t}\,.
\end{aligned} \tag{4.157}$$

This final expression is consistent with the results in Heisenberg's matrix mechanics (up to phase factor).

5 Spins and et cetera

5.1 Rotations

Rotation is one of the fundamental operations to change only the orientation of a vector in three-dimensional space. A rotation transformation of the vector $\boldsymbol{x}$ can be written as

$$x_i \mapsto \sum_{j=1}^{3} R_{ij} x_j \,, \tag{5.1}$$

with R being a transformation matrix. The rotation transformation remains the inner product of vector $\boldsymbol{x} \cdot \boldsymbol{y}$ unchanged.

$$\boldsymbol{x} \cdot \boldsymbol{y} = \sum_i x_i y_i \mapsto \sum_i \left(\sum_j R_{ij} x_j \right) \left(\sum_k R_{ik} y_k \right) = \sum_{jk} \delta_{jk} x_j y_k \,. \tag{5.2}$$

The last equality is imposed to ensure that the rotation transformation preserves the inner product. It leads to the condition for the rotation matrix elements:

$$\sum_i R_{ij} R_{ik} = \delta_{jk} \,, \tag{5.3}$$

and in terms of the matrix notation, $R^T R = 1$ where R^T is the transpose of R. Thus, the transformation matrix R for rotation must be an *orthogonal matrix*.

The determinant of an orthogonal matrix has a special property: by taking the determinant of both sides of $R^T R = 1$, the determinant must be either $+1$ or -1. The non-zero determinant ensures that the rotation matrix is invertible, and the inverse is given by the transpose of the matrix: $R^{-1} = R^T$. There is an important property of orthogonal matrices: for A and B orthogonal matrices, $A^T A = 1$ and $B^T B = 1$, their product AB is also orthogonal. Of course, the identity matrix is also an orthogonal matrix. Thus, a set of orthogonal matrices is closed under matrix multiplication and contains the identity matrix, forming a group under matrix multiplication. This group is known as the *orthogonal group* $O(N)$, where N is the dimension of the space. We remark that not all orthogonal matrices correspond to rotations; those with determinant -1 include reflections: $\boldsymbol{x} \mapsto -\boldsymbol{x}$. So, R with $\det R = 1$ corresponds to a pure rotation without reflection: and the set of such matrices forms the *special orthogonal group* $SO(N)$. In the following, we will focus on the three-dimensional case, $SO(3)$, which is the group of all rotations in three-dimensional space.

The spatial rotation R acting on coordinates $\boldsymbol{x}$ induces a unitary transformation of the physical state in Hilbert space. This unitary transformation is represented by an operator $U(R)$ acting on the state Ψ in Hilbert space:

$$\Psi \to U(R)\Psi \,. \tag{5.4}$$

We may consider the application of two successive rotations: two rotations R_1 and R_2 applied in order. The successive rotation is represented by the product of the two

rotation matrices: R_2R_1. Therefore, the unitary transformation corresponding to the successive rotation is given by the product of the two unitary operators:

$$\Psi \to U(R_2)U(R_1)\Psi\,, \tag{5.5}$$

This corresponds to the unitary transformation of the combined rotation R_2R_1.

$$U(R_2)U(R_1) = U(R_2R_1)\,. \tag{5.6}$$

Let us consider a vector operator $\boldsymbol{V}$ under rotation: this vector operator is related to the spatial coordinates and transforms accordingly under rotations (such as the position operator or the momentum operator). In terms of transformation of operators, the vector operator $\boldsymbol{V}$ transforms following the similarity transformation:

$$(\Psi', \boldsymbol{V}\Psi) \to (U(R)\Psi', \boldsymbol{V}U(R)\Psi) = (\Psi', U^{-1}(R)\boldsymbol{V}U(R)\Psi)\,. \tag{5.7}$$

Meanwhile, the vector operator $\boldsymbol{V}$ itself transforms under the rotation as $\boldsymbol{V} \to R\boldsymbol{V}$. Thus, the similarity transformation of the vector operator must induce the rotation of the vector itself:

$$U^{-1}(R)\boldsymbol{V}U(R) = R\boldsymbol{V}\,, \tag{5.8}$$

or equivalently, component-wise:

$$U^{-1}(R)V_iU(R) = \sum_j R_{ij}V_j\,. \tag{5.9}$$

The rotation transformation, without the inversion, can be continuously connected to the identity transformation. We consider the infinitesimal rotation, where the rotation matrix is close to the identity, and the small parameters ω_{ij} characterize the deviation from the identity:

$$R_{ij} = \delta_{ij} + \omega_{ij} + \mathcal{O}(\omega^2)\,. \tag{5.10}$$

From the orthogonality condition of the rotation matrix,$R^TR = 1$, we obtain the antisymmetric property of the infinitesimal rotation parameters:

$$\omega_{ji} = -\omega_{ij}\,, \tag{5.11}$$

or equivalently, in matrix form: $\omega^T = -\omega$. Under the infinitesimal rotation, the unitary transformation of the state is expanded as:

$$U(R) = U(1 + \omega + \cdots) = 1 + \frac{i}{2\hbar}\sum_{ij}\omega_{ij}J_{ij} + \mathcal{O}(\omega^2)\,. \tag{5.12}$$

Here, $J_{ij} = -J_{ji}$ are a set of Hermitian operators, representing the generators of rotations. For a later convenience, we introduce the normalization factor $\hbar$ to give J the same dimension as angular momentum. The infinitesimal transformation of the

vector operator $\boldsymbol{V}$ can now be analyzed using this expansion. The vector operator $\boldsymbol{V}$ under the infinitesimal rotation in terms of the similarity transformation is given by:

$$U^{-1}(R)V_iU(R) = V_i + \frac{i}{2\hbar}\sum_{k,l}\omega_{kl}(-J_{kl}V_i + V_iJ_{kl}) + \mathcal{O}\left(\omega^2\right) , \tag{5.13}$$

meanwhile, the vector operator $\boldsymbol{V}$ under the infinitesimal rotation in terms of the rotation matrix is given by:

$$\sum_j R_{ij}V_j = V_i + \sum_j \omega_{ij}V_j + \mathcal{O}(\omega^2) . \tag{5.14}$$

These two transformations must be equivalent, $U^{-1}(R)\boldsymbol{V}U(R) = R\boldsymbol{V}$, which leads to the commutation relation between the vector operator $\boldsymbol{V}$ and the generators J_{ij}:

$$\frac{i}{\hbar}[V_i, J_{kl}] = \delta_{ik}V_l - \delta_{il}V_k . \tag{5.15}$$

In a similar way, we can find the transformation properties of the generators J_{ij} themselves under rotations. Let us consider the group property of the rotation operators:

$$U^{-1}\left(R'\right)U(1+\omega)U\left(R'\right) = U\left(1 + R'^{-1}\omega R'\right) . \tag{5.16}$$

The right-hand side of the equation represents the rotation of the infinitesimal rotation parameters by the matrix R'. Thus, in terms of the generators J_{ij}, the right-hand side is written as:

$$\begin{aligned} U\left(1 + R'^{-1}\omega R'\right) &= 1 + \frac{i}{2\hbar}\sum_{ij}\left(R'^{-1}\omega R'\right)_{ij}J_{ij} \\ &= 1 + \frac{i}{2\hbar}\sum_{ijkl}\omega_{ij}R'_{ik}R'_{jl}J_{kl} + \mathcal{O}(\omega^2) . \end{aligned} \tag{5.17}$$

In the last equality, we just use the fact R' is orthogonal $R'^{-1} = R'^T$. Meanwhile, the left-hand side of the equation can be expanded using the infinitesimal form of $U(1+\omega)$:

$$U^{-1}(R')U(1+\omega)U(R') = 1 + \frac{i}{2\hbar}\sum_{ij}\omega_{ij}U^{-1}(R')J_{ij}U(R') . \tag{5.18}$$

Therefore, at the order of ω, we have the relation:

$$U^{-1}(R')J_{ij}U(R') = \sum_{kl}R'_{ik}R'_{jl}J_{kl} . \tag{5.19}$$

In other words, the generators J_{ij} transform as rank-two tensors under the rotation transformation. Considering the infinitesimal rotation $R' = 1 + \omega'$, we can derive the algebraic relations of the generators J_{ij}. The left-hand side of the equation, expanded to first order in ω', gives the commutation relations of the generators J_{ij}:

$$U^{-1}(R')J_{ij}U(R') = J_{ij} + \frac{i}{2\hbar}\sum_{kl}\omega'_{kl}[J_{ij}, J_{kl}] + \mathcal{O}(\omega'^2) . \tag{5.20}$$

On the other hand, the right-hand side of the equation, expanded to first order in ω', gives:

$$\sum_{kl} R'_{ik} R'_{jl} J_{kl} = J_{ij} + \sum_{kl} \left(\omega'_{ik}\delta_{jl} + \delta_{ik}\omega'_{jl}\right) J_{kl} + \mathcal{O}(\omega'^2)\,. \tag{5.21}$$

We remark that the terms linear in ω' on both sides must be equal, which leads to the commutation relations of the generators J_{ij}. We can rewrite the second term on the right-hand side as follows.

$$\begin{aligned}\sum_{kl} \left(\omega'_{ik}\delta_{jl} + \delta_{ik}\omega'_{jl}\right) J_{kl} &= \sum_{k} \left(\omega'_{ik} J_{kj} + \omega'_{jk} J_{ik}\right) \\ &= \frac{1}{2}\sum_{kl} \omega'_{kl}(-\delta_{il}J_{kj} + \delta_{ik}J_{lj} + \delta_{jk}J_{il} - \delta_{jl}J_{ik})\,. \end{aligned} \tag{5.22}$$

Then, the commutation relations of the generators J_{ij} can be directly read off from the comparison of the linear terms in ω'.

$$\frac{i}{\hbar}\left[J_{ij}, J_{kl}\right] = -\delta_{il}J_{kj} + \delta_{ik}J_{lj} + \delta_{jk}J_{il} - \delta_{jl}J_{ik}\,. \tag{5.23}$$

This is the general form of commutation relations for the generators of rotations in three dimensions. This is actually inconvenient for practical calculations, because it involves four indices. It is more convenient to work with the three independent components of J_{ij}, which can be defined using the Levi-Civita symbol. Indeed, the antisymmetric tensor J_{ij} has only three independent components in three dimensions, *corresponding to the three components of angular momentum*. Therefore, we can define the three components of angular momentum J_i as follows:

$$J_{ij} = \sum_{k} \epsilon_{ijk} J_k\,. \tag{5.24}$$

or equivalently, we can express J_i in terms of J_{jk} using the inverse relation:

$$J_i = \frac{1}{2}\sum_{jk} \epsilon_{ijk} J_{jk}\,. \tag{5.25}$$

To prove the inverse relation, the following identity of the Levi-Civita symbol is useful:

$$\sum_{k} \epsilon_{ijm}\epsilon_{klm} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}\,. \tag{5.26}$$

Using the vector notation for the angular momentum components, we can rewrite the unitary operator for the rotation as follows:

$$U(1+\omega) = \mathbf{1} + \frac{i}{\hbar}\boldsymbol{\omega}\cdot\boldsymbol{J} + \mathcal{O}(\omega^2)\,, \qquad \omega_i \equiv \frac{1}{2}\sum_{jk}\epsilon_{ijk}\omega_{jk}\,. \tag{5.27}$$

Using vector notation for the angular momentum, the commutation relation of the vector operator and the angular momentum operator is simplified as follows:

$$[J_i, V_j] = i\hbar \sum_k \epsilon_{ijk} V_k \,. \tag{5.28}$$

It is straightforward to prove this from the commutation relation between the vector operator and the tensorial angular momentum operator, and the use of the identity of the Levi-Civita symbol in Eq. (5.26). In particular, J_i itself is a 3-vector operator, and hence there is the commutation relation for the components of angular momentum:

$$[J_i, J_j] = i\hbar \sum_k \epsilon_{ijk} J_k \,. \tag{5.29}$$

This has the same structure as the commutation relation for the orbital angular momentum of $\boldsymbol{L}$. Remind that $\boldsymbol{L}$ is the generator of rotations in the space, in particular of the $SO(3)$ unitary transformations. The generators of the continuum symmetry transformations would form an algebra, specifically a *Lie algebra*, characterized by commutation relations of the generators.

There is a difference between $\boldsymbol{J}$ and $\boldsymbol{L}$: $\boldsymbol{J}$ is the generator of rotations in quantum mechanics, while $\boldsymbol{L}$ is constructed from the particle's position and momentum operators, specifically as $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$. In other words, $\boldsymbol{J}$ may include all contributions for rotation even beyond the orbital angular momentum, and hence it is generally referred to as the *total angular momentum*. The orbital angular momentum $\boldsymbol{L}$ is a vectorial operator, and it has the commutation relation with the total angular momentum $\boldsymbol{J}$ as well as with itself:

$$[J_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k \,, \qquad [L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k \,. \tag{5.30}$$

We can define the difference between the total angular momentum and the orbital angular momentum, $\boldsymbol{S} \equiv \boldsymbol{J} - \boldsymbol{L}$. The operator $\boldsymbol{S}$ represents the *intrinsic* angular momentum of the particle, in the sense that it has nothing to do with the particle's motion, and is therefore commonly referred to as the *spin* of the particle. Indeed, the commutators of $\boldsymbol{L}$ with $\boldsymbol{J}$ and with itself lead to the conclusion that $\boldsymbol{S}$ commutes with $\boldsymbol{L}$:

$$[S_i, L_j] = 0 \,. \tag{5.31}$$

Hence, the physical states of the system can be simultaneously eigenstates of both the orbital angular momentum $\boldsymbol{L}$ and the spin angular momentum $\boldsymbol{S}$. It is straightforward to show that $\boldsymbol{S}$ itself satisfies the angular momentum commutation relations:

$$[S_i, S_j] = i\hbar \sum_k \epsilon_{ijk} S_k \tag{5.32}$$

As shown in this section and the previous discussion on the spherical harmonics, the total angular momentum and the orbital angular momentum have the commutation

relations with the position and momentum operators as follows.

$$[L_i, x_j] = i\hbar \sum_k \epsilon_{ijk} x_k \,, \qquad [L_i, p_j] = i\hbar \sum_k \epsilon_{ijk} p_k \,, \tag{5.33}$$

$$[J_i, x_j] = i\hbar \sum_k \epsilon_{ijk} x_k \,, \qquad [J_i, p_j] = i\hbar \sum_k \epsilon_{ijk} p_k \,. \tag{5.34}$$

Therefore, the spin angular momentum $\boldsymbol{S}$ commutes with both the position and momentum operators:

$$[S_i, x_j] = 0 \,, \quad [S_i, p_j] = 0 \,. \tag{5.35}$$

This fact is quite natural, since spin operators are intrinsic to the particle and should have nothing to do with its position or momentum.

It is straightforward to generalize the above discussion to a system of N particles. Of course, the total angular momentum for an N-particle system is also uniquely determined by the rotation transformation, and the total angular momentum operator can be expressed as the sum of the orbital and spin angular momentum operators for each particle:

$$\boldsymbol{J} = \sum_n \boldsymbol{L}_n + \sum_n \boldsymbol{S}_n \,, \tag{5.36}$$

where n labels each particle in the system, and $\boldsymbol{L}_n$ and $\boldsymbol{S}_n$ are the orbital and spin angular momentum operators for the n-th particle, respectively.

5.2 Angular Momentum Multiplet

Thanks to the commutation relations for $\boldsymbol{J}$, we can define the raising and lowering operators for the angular momentum. Since the commutation relation with J_3 is given by

$$[J_3, (J_1 \pm iJ_2)] = \pm\hbar(J_1 \pm iJ_2) \,, \tag{5.37}$$

the operators $J_1 \pm iJ_2$ act as raising and lowering operators for the eigenvalues of J_3. Then, we define the raising and lowering operators as

$$J_\pm \equiv J_1 \pm iJ_2 \,. \tag{5.38}$$

Once the state Ψ is an eigenstate of J_3 with eigenvalue $m\hbar$, i.e.,

$$J_3\Psi = m\hbar\Psi \,, \tag{5.39}$$

then the action of the raising and lowering operators on this state is also eigenstates of J_3 with eigenvalues shifted by $\pm\hbar$.

$$J_3(J_\pm\Psi) = (m \pm 1)\hbar(J_\pm\Psi) \,. \tag{5.40}$$

J_3 is not commutative with J_1 and J_2, but it commutes with J^2. Therefore, we can choose simultaneous eigenstates of J^2 and J_3, denoted by Ψ_j^m. Let us denote the eigenvalue of J^2 by a and the eigenvalue of J_3 by b, i.e.,

$$J^2\Psi_j^m = a\Psi_j^m\,, \quad J_3\Psi_j^m = b\Psi_j^m\,. \tag{5.41}$$

Then, the eigenvalue a of J^2 must satisfy $a \geq b^2$ due to the positivity of the operator $J^2 - J_3^2 = J_1^2 + J_2^2$:

$$a - b^2 = (\Psi_j^m, (J^2 - J_3^2)\Psi_j^m) = (\Psi_j^m, (J_1^2 + J_2^2)\Psi_j^m) \geq 0\,, \tag{5.42}$$

We can define the states with the maximum and minimum J_3 values for a given J^2 eigenstate. Let Ψ_j^j be the state with the maximum J_3 value and $\Psi_j^{j'}$ be the state with the minimum J_3 value. Namely, $J_+\Psi_j^j = 0$ and $J_-\Psi_j^{j'} = 0$, and we have the eigenvalues of J_3 for these states as

$$J_3\Psi_j^j = \hbar j\Psi_j^j\,, \quad J_3\Psi_j^{j'} = \hbar j'\Psi_j^{j'}\,. \tag{5.43}$$

The operator J^2 can be expressed in terms of the raising and lowering operators as follows:

$$J^2 = J_+J_- + J_3^2 - \hbar J_3 = J_-J_+ + J_3^2 + \hbar J_3\,. \tag{5.44}$$

The action of order J_+ and J_- is different in the first and second equality, depending on whether we are applying the raising or lowering operator first. The first equality is useful when acting on a state with the maximum J_3 value, while the second equality is useful when acting on a state with the minimum J_3 value. Indeed, the action of J^2 on the states with maximum and minimum J_3 values can be directly computed using these expressions:

$$J^2\Psi_j^j = \left(J_3^2 + \hbar J_3\right)\Psi_j^j = j(j+1)\hbar^2\Psi_j^j\,, \tag{5.45}$$

$$J^2\Psi_j^{j'} = \left(J_3^2 - \hbar J_3\right)\Psi_j^{j'} = j'(j'-1)\hbar^2\Psi_j^{j'}\,. \tag{5.46}$$

But, both are eigenstates of J^2, and the eigenvalues should be same, i.e., $j(j+1) = j'(j'-1) = a$. Here, we may have two solutions for j': $j' = j+1\,, -j$. However, since j is the maximum J_3 value, the solution $j' = j+1$ is not possible, and then we must have $j' = -j$. $j - j' = 2j$ counts the number of the lowering operator application to obtain the state $\Psi_{j'}^j$ from Ψ_j^j, so $2j$ must be an integer. Therefore, j is either an integer or a half-integer. The eigenvalues of J_3 for a given J^2 eigenstate range from $-j$ to j in integer steps. The application of J_3 and J^2 on the states Ψ_j^m is summarized as follows:

$$J_3\Psi_j^m = m\hbar\Psi_j^m\,, \qquad (m = -j\,, -j+1\,, \cdots, j-1\,, j)\,, \tag{5.47}$$

$$J^2\Psi_j^m = j(j+1)\hbar^2\Psi_j^m\,. \tag{5.48}$$

As we expect, states with different m are orthogonal, and hence we can take them to form an orthonormal basis:

$$\begin{aligned}(\Psi_j^{m'}, J_3\Psi_j^m) &= m(\Psi_j^{m'}, \Psi_j^m) = m'(\Psi_j^{m'}, \Psi_j^m) \\ &\Rightarrow (m - m')(\Psi_j^{m'}, \Psi_j^m) = 0\,.\end{aligned} \tag{5.49}$$

Here, we use the hermitian property of J_3, i.e., $J_3^\dagger = J_3$. Choosing the proper normalization for the states, we have

$$(\Psi_j^{m'}, \Psi_j^m) = \delta_{m',m}\,. \tag{5.50}$$

In the following, we focus on the orthonormal basis of states Ψ_j^m.

Next, we will find the relation between the action of the raising and lowering operators $J_\pm$ on the states Ψ_j^m and the normalized states $\Psi_j^{m\pm1}$. Application of the raising and lowering operators on the states Ψ_j^m gives the states with m increased or decreased by one, which can be written as

$$J_\pm \Psi_j^m = \alpha^\pm(j,m)\Psi_j^{m\pm1}\,, \tag{5.51}$$

where $\alpha^\pm(j,m)$ are coefficients that depend on j and m. From the relation $J_\mp J_\pm = J^2 - J_3^2 \mp \hbar J_3$, we can derive the following relation for the coefficients:

$$\alpha^\mp(j,m\pm1)\alpha^\pm(j,m) = \hbar^2[j(j+1) - m^2 \mp m]\,. \tag{5.52}$$

We can determine the explicit form of the coefficients $\alpha^\pm(j,m)$ by considering the norm of the states $J_\pm\Psi_j^m$. The norm of the states $J_\pm\Psi_j^m$ gives the squared magnitude of the coefficients $\alpha^\pm(j,m)$ since all the states Ψ_j^m are normalized.

$$\begin{aligned} |\alpha^\pm(j,m)|^2 &= (J_\pm\Psi_j^m, J_\pm\Psi_j^m) = (\Psi_j^m, J_\mp J_\pm \Psi_j^m) \\ &= [j(j+1) - m^2 \mp m]\hbar^2\,. \end{aligned} \tag{5.53}$$

Here, we use the hermitian property of the raising and lowering operators, i.e., $J_\pm^\dagger = J_\mp$. To obtain the value, we use the relation of the coefficients derived above, $\alpha^\mp(j,m\pm1)\alpha^\pm(j,m)$. The sign and phase of the coefficients $\alpha^\pm(j,m)$ can be adjusted according to a specific convention: indeed it is possible to choose the phase convention for the states Ψ_j^m. In the following, we adopt the convention that the coefficients $\alpha^-(j,m)$ are chosen to be positive real numbers. Once we choose the conventional phase for the states Ψ_j^m such that $\alpha^-(j,m)$ are positive real numbers, the coefficients $\alpha^+(j,m)$ can be complex because it is no longer possible to choose the phase freely for the states Ψ_j^m. However, since we have chosen $\alpha^-(j,m)$ to be positive real, the relation $\alpha^\mp(j,m\pm1)\alpha^\pm(j,m) = \hbar^2[j(j+1) - m^2 \mp m]$ implies that $\alpha^+(j,m)$ is also positive real. The above relations summarize the action of the angular momentum operators on the eigenstates Ψ_j^m.

$$J_3\Psi_j^m = m\hbar\Psi_j^m\,, \qquad (m = -j\,, -j+1\,, \cdots, j-1\,, j)\,, \tag{5.54}$$

$$J_\pm\Psi_j^m = \hbar\sqrt{j(j+1) - m^2 \mp m}\,\Psi_j^{m\pm1}\,, \tag{5.55}$$

$$\alpha^\pm(j,m) \equiv \hbar\sqrt{j(j+1) - m^2 \mp m}\,. \tag{5.56}$$

This provides a complete description of how the angular momentum operators act on the eigenstates Ψ_j^m. In the group theoretical context, the set of applications $J_{3,\pm}$ on the state Ψ_j^m is given by the relations above, which furnishes *representation* of the commutation relations.

As an example, considering the case of $j = 1/2$, this is the situation for spin-1/2 particles with zero orbital angular momentum: the application of the angular momentum operator $J_{3\,,\pm}$ on the states $\Psi^{\pm 1/2}_{1/2}$. The application of operators is given by

$$J_{\pm}\Psi^{\mp 1/2}_{1/2} = \hbar\Psi^{\pm 1/2}_{1/2}\,, \tag{5.57}$$

$$J_{\pm}\Psi^{\pm 1/2}_{1/2} = 0\,, \tag{5.58}$$

$$J_3\Psi^{\pm 1/2}_{1/2} = \pm\frac{\hbar}{2}\Psi^{\pm 1/2}_{1/2}\,. \tag{5.59}$$

These relations can be rewritten in the matrix form as

$$\left(\Psi^{m'}_{1/2}, \boldsymbol{J}\Psi^{m}_{1/2}\right) = \frac{\hbar}{2}\boldsymbol{\sigma}_{m'm}\,, \tag{5.60}$$

where $\boldsymbol{\sigma}$ denotes the vector of 2×2 matrices, $\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$, and their indices $m\,, m'$ run the values $-1/2$ and $1/2$. These matrices are known as the *Pauli matrices*, which play a fundamental role in the description of spin-1/2 particles in quantum mechanics. The explicit matrix forms of the Pauli matrices are given by

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{5.61}$$

The Pauli matrices appear in many situations in quantum mechanics.

5.3 Addition of Angular Momenta

Angular momentum addition is an important concept in quantum mechanics, especially when dealing with composite systems such as atomic nuclei or multi-electron atoms. Suppose J' and J'' are distinct angular momentum operators, and normalized state for each operators are given by $\Psi^{m'}_{j'}$ and $\Psi^{m''}_{j''}$, respectively. We can consider the tensor product of these two states, which forms a basis for the combined system:

$$\Psi^{m'}_{j'} \otimes \Psi^{m''}_{j''} \equiv \Psi^{m',m''}_{j',j''}\,. \tag{5.62}$$

The tensor product $\Psi^{m',m''}_{j',j''}$ are simultaneous eigenstates of J'^2, J''^2, J'_3, and J''_3. Therefore the applications of these operators on the tensor product states yield the following eigenvalue equations:

$$J'^2\Psi^{m',m''}_{j',j''} = \hbar^2 j'(j'+1)\Psi^{m',m''}_{j',j''}\,, \tag{5.63}$$

$$J''^2\Psi^{m',m''}_{j',j''} = \hbar^2 j''(j''+1)\Psi^{m',m''}_{j',j''}\,, \tag{5.64}$$

$$J'_3\Psi^{m',m''}_{j',j''} = \hbar m'\Psi^{m',m''}_{j',j''}\,, \tag{5.65}$$

$$J''_3\Psi^{m',m''}_{j',j''} = \hbar m''\Psi^{m',m''}_{j',j''}\,. \tag{5.66}$$

We may consider the sum of the angular momentum operators, $\boldsymbol{J} = \boldsymbol{J'} + \boldsymbol{J''}$ and the corresponding eigenstates of the total angular momentum, denoted by Ψ^m_j, where j

and m are the quantum numbers associated with $\boldsymbol{J}^2$ and J_3, respectively. The purpose of this subsection is to understand how the total angular momentum states Ψ^m_j can be constructed from the tensor product states $\Psi^{m'm''}_{j'j''}$.

Since we have interested in the state of the total angular momentum constructed from the tensor product states with given j' and j'', we proceed as follows. We can write the normalized state of the total angular momentum as $\Psi^m_j = \Psi^m_{j',j'',j}$, in which the dependence on j' and j'' is explicitly shown. The normalized state $\Psi^m_{j',j'',j}$ satisfies the following eigenvalue equations: first the eigenvalue equations for the total angular momentum operators,

$$J^2\Psi^m_{j',j'',j} = \hbar^2 j(j+1)\Psi^m_{j',j'',j}\,, \tag{5.67}$$

$$J_3\Psi^m_{j',j'',j} = \hbar m\Psi^m_{j',j'',j}\,, \tag{5.68}$$

$$J_\pm\Psi^m_{j',j'',j} = \hbar\sqrt{j(j+1)-m^2\mp m}\Psi^m_{j',j'',j}\,, \tag{5.69}$$

and the state $\Psi^m_{j',j'',j}$ is the direct sum of the tensor product states with given j' and j'', and hence it must also satisfy the eigenvalue equations for J'^2 and J''^2:

$$J'^2\Psi^m_{j',j'',j} = \hbar^2 j'(j'+1)\Psi^m_{j',j'',j}\,, \tag{5.70}$$

$$J''^2\Psi^m_{j',j'',j} = \hbar^2 j''(j''+1)\Psi^m_{j',j'',j}\,. \tag{5.71}$$

The normalized state $\Psi^m_{j',j'',j}$ can be expressed as a linear combination of the tensor product states $\Psi^{m',m''}_{j',j''}$. For given j' and j'', there are $(2j'+1)(2j''+1)$ tensor product states $\Psi^{m',m''}_{j',j''}$, corresponding to all possible values of m' and m''. Some of the states can contribute to the total angular momentum state $\Psi^m_{j',j'',j}$, while others do not.

$$\Psi^m_{j',j'',j} = \sum_{m'm''} C_{j'j''}(jm;m'm'')\Psi^{m',m''}_{j',j''}\,. \tag{5.72}$$

Here, $C_{j'j''}(jm;m'm'')$ are the *Clebsch-Gordan coefficients* (CGC), which determine the contribution of each tensor product state $\Psi^{m',m''}_{j',j''}$ to the total angular momentum state $\Psi^m_{j',j'',j}$. The normalized state $\Psi^m_{j',j'',j}$ is no longer the eigenstates of J'_3 and J''_3 individually, but it remains an eigenstate of the total angular momentum operators J^2 and J_3. In this sense, $\Psi^m_{j',j'',j}$ is called as the *coupled basis*, while the tensor product states $\Psi^{m',m''}_{j',j''}$ are called the *uncoupled basis*. From the summation of vectors, the third component of $\boldsymbol{J}$ is simply given by the sum of the third components of $\boldsymbol{J'}$ and $\boldsymbol{J''}$: this simply implies $m = m' + m''$. Both m' and m'' are constrained by the values of j' and j'', and their sum must equal m.

$$j = |j'-j''|\,, |j'-j''|+1\,, \cdots, j'+j''\,, \tag{5.73}$$

$$m = -j\,, -j+1\,, \cdots, j-1\,, j\,. \tag{5.74}$$

Let us first construct the state for the maximum value of j and m: namely, $j = j'+j''$ and $m = j'+j''$. Since it is the state with the maximum m, it must correspond to the tensor product state with the maximum m' and m''. There is only one uncoupled-basis state that has the maximum m' and m'', which is $\Psi^{j',j''}_{j',j''}$. We choose a phase convention

to be same as the uncoupled-basis state with maximum m' and m''. Then, the coupled-basis state with the maximum j and m is simply the uncoupled-basis state with the maximum m' and m''.

$$\Psi^{j'+j''}_{j',j'',j'+j''} = \Psi^{j',j''}_{j',j''} \,. \tag{5.75}$$

The lower m states are obtained by applying the lowering operator J_-. J_- is decomposed into the sum of the lowering operators for the individual angular momenta: $J_- = J'_- + J''_-$. As an example, the state with $m = j' + j'' - 1$ can be obtained by applying J_- to the state with $m = j' + j''$:

$$\Psi^{j'+j''-1}_{j',j'',j'+j''} = \frac{1}{\sqrt{j'+j''}} \left(\sqrt{j'} \Psi^{j'-1,j''}_{j',j''} + \sqrt{j''} \Psi^{j',j''-1}_{j',j''} \right) , \tag{5.76}$$

The coefficients in front of the uncoupled-basis states are determined by the action of the lowering operator and the requirement of orthonormality. Thanks to the determination of the coefficients from the orthonormality requirement, the state in the left-hand side is automatically normalized. There is another state with the same $m = j' + j'' - 1$, which is orthogonal to the one obtained above. Such a state corresponds to the different total angular momentum value, in this case $j = j' + j'' - 1$.

$$\Psi^{j'+j''-1}_{j',j'',j'+j''-1} = \frac{1}{\sqrt{j'+j''}} \left(\sqrt{j'} \Psi^{j'-1,j''}_{j',j''} - \sqrt{j''} \Psi^{j',j''-1}_{j',j''} \right) , \tag{5.77}$$

By continuing this procedure, we can obtain all coupled-basis states systematically, for all allowed values of the total angular momentum j and its projection m.

Now, let us confirm the number of states (suppose $j' > j''$). For each j, there are $(2j + 1)$ states corresponding to the different values of m. The values of j range from $j' - j''$ to $j' + j''$, and then the total number of states is

$$\sum_{j=j'-j''}^{j'+j''} (2j + 1) = (2j' + 1)(2j'' + 1) \,. \tag{5.78}$$

Thus, the total number of states obtained from the addition of angular momenta matches the number of states in the tensor product of the individual angular momenta. This is consistent with the fact the the coupled-basis states span the same Hilbert space as the tensor product of the uncoupled-basis states. Here, we have assumed $j' > j''$, but the argument is symmetric, and hence we obtain the same result for $j'' > j'$.

Now, let us discuss the properties of Clebsch-Gordan coefficients. Let us label the states in the coupled basis by Φ_n and the states in the uncoupled basis by $\bar{\Phi}_a$. The Clebsch-Gordan coefficients C_{na} relate these two bases:

$$\Phi_n = \sum_a C_{na} \bar{\Phi}_a \,. \tag{5.79}$$

The number of states in each basis is the same, and hence n and a run over the same number of values. Here, each basis Φ_n and $\bar{\Phi}_a$ is an orthonormal basis.

$$(\Phi_n, \Phi_m) = \delta_{nm} \,, \quad (\bar{\Phi}_a, \bar{\Phi}_b) = \delta_{ab} \,. \tag{5.80}$$

The orthonormality of the bases implies the following relation for the Clebsch-Gordan coefficients:

$$(\Phi_n, \Phi_m) = \sum_{a\,,b} C_{na} C^*_{mb} (\bar{\Phi}_a, \bar{\Phi}_b)\,, \quad \rightarrow \quad \delta_{nm} = \sum_a C_{na} C^*_{ma}\,, \tag{5.81}$$

which is called the orthonormality condition for the Clebsch-Gordan coefficients. Once the Clebsch-Gordan coefficients are regarded as the elements of a matrix denoted by C, the orthonormality condition can be written in matrix form, $CC^\dagger = \mathbf{1}$

$$CC^\dagger = \mathbf{1}\,. \tag{5.82}$$

Since both indices n and a run over the same number of values, the matrix C is a square unitary matrix. Therefore, there exists the inverse matrix, and the orthonormality condition can be equivalently written as $C^\dagger C = \mathbf{1}$. This can be written in component form as

$$\delta_{ab} = \sum_n C^*_{na} C_{nb}\,. \tag{5.83}$$

Therefore, the uncoupled-basis state is given by the linear combination of the coupled-basis states

$$\bar{\Phi}_a = \sum_n C^*_{na} \Phi_n\,. \tag{5.84}$$

In terms of the Clebsch-Gordan coefficients, the relations between the coupled and uncoupled bases can be expressed as follows.

$$\sum_{jm} C_{jj''}(jm; m'm'') C_{jj''}(jm; \tilde{m}'\tilde{m}'') = \delta_{m'\tilde{m}'}\delta_{m''\tilde{m}''} \tag{5.85}$$

$$\sum_{m'm''} C_{j'j''}(jm; m'm'') C_{j'j''}(\tilde{j}\tilde{m}; m'm'') = \delta_{j\tilde{j}}\delta_{m\tilde{m}}\,. \tag{5.86}$$

There are several important applications of Clebsch-Gordan coefficients in physics. One of the example is the hydrogen atom. The spin of the electron is $s = 1/2$, and total angular momentum of the electron is given by the sum of total angular momentum of the orbital motion and the spin:

$$j = \ell \pm \frac{1}{2}\,. \tag{5.87}$$

There are possible states and are labeled as $n\ell_j$, such as $1s_{1/2}, 2p_{3/2}, 2p_{1/2}, 2s_{1/2}, \ldots$, with letters ℓ denoting spectroscopic notation, see Table 1. The energy levels are further split due to the spin-orbit interaction.

$$\boldsymbol{L}\cdot\boldsymbol{S} = \frac{1}{2}(\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2)\,. \tag{5.88}$$

The states with the same principal quantum number n have the same energy as discussed in Section 4.5. The presence of the spin-orbit interaction gives rise to the *fine structure* of the hydrogen-like atom, which leads to the energy split of states with the same n and ℓ but different j values: for example, the $2p_{3/2}$ and $2p_{1/2}$ states. The energy difference between them is about $\sim 4.5283 \times 10^{-5}$eV, and this kind of energy splitting is called as "*Fine structure*" (in particular, of the spectra). There are several types of energy shift in atomic physics: such as the Lamb shift, Zeeman/Stark effects, and hyperfine splitting. The Lamb shift is a small energy difference between the $2s_{1/2}$ and $2p_{1/2}$ levels in hydrogen, arising from quantum electrodynamics effects. The Zeeman effect refers to the splitting of atomic energy levels in the presence of an external magnetic field, while the Stark effect refers to the splitting due to an external electric field. The hyperfine splitting arises from the interaction between the magnetic moments of the nucleus and the electron, leading to further splitting of atomic energy levels.

The another strong advantage of understanding addition of angular momentum is that one can systematically calculate the matrix elements of operators acting on the combined angular momentum states. Let us think a set of operators O_j^m acting on the combined angular momentum states, where $m = j, j-1, \ldots, -j$, which is called as spin $\hat{j}$ operators. The matrix element of the operator with the different angular momentum states is decomposed into a product of a Clebsch-Gordan coefficient and a reduced matrix element:

$$(\Phi_{j''}^{m''}, O_j^m \Psi_{j'}^{m'}) = C_{jj'}\,(j''m''; mm')\,(\Phi||O||\Psi)\,. \tag{5.89}$$

Here, $(\Phi||O||\Psi)$ is the reduced matrix element, which does not depend on the magnetic quantum numbers m, m', m''. This decomposition is ensured by the "*Wigner-Eckart theorem*". This theorem greatly simplifies the calculation of matrix elements for operators acting on angular momentum states, and hence it is practically very important and useful: in particular the Clebsch-Gordan coefficients determine the selection rules for non-zero matrix elements. In this note, we will not go further into the proof or deeper aspects of the Wigner-Eckart theorem.

5.4 Bosons and Fermions

As far as we know, every electron in the Universe is identical except for the values of their positions (or momenta) and spin 3-components. This aspect is same for any other types of particles. We call particles that obey this indistinguishability principle as "*identical particles*". We can represent the state of a system with multiple identical particles using a state vector that includes all the particles' positions (or momenta) and spin components. $\Phi_{\boldsymbol{x}_1,m_1;\boldsymbol{x}_2,m_2;\ldots}$ represents the state of multiple identical particles, including all their positions and spin components. This notation assumes a single species of particles. The order of the particles in the state vector is arbitrary, reflecting the indistinguishability of identical particles. For example, in the notation of $\Phi_{\boldsymbol{x}_1,m_1;\boldsymbol{x}_2,m_2;\ldots}$, there is one particle with position $\boldsymbol{x}_1$ and spin component m_1, and another particle with position $\boldsymbol{x}_2$ and spin component m_2, and so on. Here, we do not specify the order of particles, such as "the first particle" or "the second particle". Therefore, exchanging

the labels of two particles should not change the physical state represented by the state vector. So, another state vector $\Phi_{\boldsymbol{x}_2,m_2;\boldsymbol{x}_1,m_1;\dots}$ (just changing labels) also represents the same state. Therefore, they are identical up to phase (since the states are given by a ray of Hilbert space).

$$\Phi_{\boldsymbol{x}_2,m_2;\boldsymbol{x}_1,m_1;\dots} = \alpha\Phi_{\boldsymbol{x}_1,m_1;\boldsymbol{x}_2,m_2;\dots}\,, \tag{5.90}$$

where α represents the phase factor associated with exchanging two identical particles. This phase factor does not depend on which momentum and spin components are exchanged. As a result, exchanging the same two particles twice should return the state to its original state, implying

$$\Phi_{\boldsymbol{x}_1,m_1;\boldsymbol{x}_2,m_2;\dots} = \alpha^2\Phi_{\boldsymbol{x}_1,m_1;\boldsymbol{x}_2,m_2;\dots}\,. \tag{5.91}$$

The states in both sides are same, and hence the factor should satisfy $\alpha^2 = 1$. α can only take the values ± 1: $\alpha = +1$ corresponds to *Bosons* (symmetric under exchanging of labels), and $\alpha = -1$ corresponds to *Fermions* (antisymmetric under exchange of labels). There is a fact that, in quantum field theory, there is a theorem connecting the particle spins and its statistics, named as the *spin-statistics theorem*. In the three-dimensional space (with one time dimension), the spin-statistics theorem states that particles with integer spins are Bosons, and particles with half-integer spins are Fermions.

Let us denote $\Phi_{\xi_1\xi_2,\dots}$ the multi-identical particle state, where ξ_i collectively denote momentum and spin z-component of each particle. The distinction of bosons and fermions is important for the system where the total Hamiltonian approximately acts separately on each particle, such as

$$H\Phi_{\xi_1\xi_2\dots} \approx \int d\xi_1' H_{\xi_1'\xi_1}\Phi_{\xi_1'\xi_2\dots} + \int d\xi_2' H_{\xi_2'\xi_2}\Phi_{\xi_1\xi_2'\dots} + \cdots\,, \tag{5.92}$$

where the one-particle complete set (determined by the one-particle states Φ_ξ) is inserted. The integration over ξ implies the integration over the continuum parameters and the summation over the discrete parameters. $H_{\xi'\xi}$ denotes the effective Hamiltonian of the one-particle state defined by

$$H_{\xi'\xi} = (\Phi_{\xi'}, H^{\text{eff}}\Phi_\xi)\,, \tag{5.93}$$

Here, H^{eff} denotes the one-particle part of total Hamiltonian H. For example, the Hamiltonian is in general written as

$$H = \sum_i \frac{\boldsymbol{p}_i^2}{2m_i} + \sum_i U(\boldsymbol{x}_i) + \sum_{ij} V(|\boldsymbol{x}_i - \boldsymbol{x}_j|)\cdots \tag{5.94}$$

Here, U denotes the one-particle potential, and V denotes the two-particle interaction potential. The one-particle part of the Hamiltonian corresponds to the kinetic energy and the one-particle potential terms.

If Hamiltonian can be decomposed into a sum of one-particle Hamiltonians, the state Ψ is an eigenstate of Hamiltonian when its wave function representation is given by

$$(\Phi_{\xi_1\xi_2\cdots}, \Psi) = \psi_1(\xi_1)\psi_2(\xi_2)\cdots\,. \tag{5.95}$$

Here, $\psi_i(\xi_i)$ denotes the eigenfunction of the one-particle Hamiltonian: for given the one-particle state Φ_{ξ_i}, the one-particle eigenfunction is written as

$$\psi_i(\xi_i) \equiv (\Phi_{\xi_i}, \Psi)\,, \tag{5.96}$$

The product of one-particle eigenfunctions is called the *Hartree Product*. Indeed, when $\psi_a(\xi_a)$ is an eigenfunction of one particle Hamiltonian, the action of the one-particle Hamiltonian on $\psi_a(\xi_a)$ yields its corresponding eigenvalue.

$$\begin{aligned}\int d\xi_a' H_{\xi_a'\xi_a}\psi_a(\xi_a') &= \int d\xi_a'(\Phi_{\xi_a}, H^{\text{eff}}\Phi_{\xi_a'})(\Phi_{\xi_a'}, \Psi) \\ &= (\Phi_{\xi_a}, H^{\text{eff}}\Psi) = E_a\psi_a(\xi_a)\,.\end{aligned} \tag{5.97}$$

In this case, we have the inner product of the multi-identical particle state and the Hamiltonian acting on the state Ψ:

$$\begin{aligned}(\Phi_{\xi_1\xi_2\cdots}, H\Psi) &= (H\Phi_{\xi_1\xi_2\cdots}, \Phi) \\ &= \int d\xi_1' H^*_{\xi_1'\xi_1}(\Phi_{\xi_1'\xi_2\cdots}, \Psi) + \int d\xi_2' H^*_{\xi_2'\xi_2}(\Phi_{\xi_1\xi_2'\cdots}, \Psi) + \cdots \\ &= \int d\xi_1' H^*_{\xi_1'\xi_1}\psi_1(\xi_1')\psi_2(\xi_2)\cdots + \int d\xi_2' H^*_{\xi_2'\xi_2}\psi_1(\xi_1)\psi_2(\xi_2')\cdots + \cdots\,.\end{aligned} \tag{5.98}$$

Using Hermiticity of the one-particle Hamiltonian, each term gives the energy eigenvalue: for instance,

$$\begin{aligned}\int d\xi_1' H^*_{\xi_1'\xi_1}\psi_1(\xi_1')\psi_2(\xi_2)\cdots &= \int d\xi_1' H_{\xi_1\xi_1'}\psi_1(\xi_1')\psi_2(\xi_2)\cdots \\ &= E_1\psi_1(\xi_1)\psi_2(\xi_2)\cdots\end{aligned} \tag{5.99}$$

Then, it leads to $(\Phi_{\xi_1\xi_2\cdots}, H\Psi) = (E_1+E_2+\cdots)(\Phi_{\xi_1\xi_2\cdots}, \Psi)$, and hence Ψ is an eigenstate of H.

$$H\Psi = (E_1 + E_2 + \cdots)\Psi\,. \tag{5.100}$$

However, the wave function representation with Hartree Product does not respect the required symmetry or antisymmetry of the multi-particle wave function. Instead, we need to construct the wave function with the proper symmetry for identical particles.

$$(\Phi_{\xi_1,\xi_2,\cdots}, \Psi) = \sum_{\{\sigma\}} \text{sign}(\sigma)\psi_1(\xi_{\sigma_1})\psi_2(\xi_{\sigma_2})\cdots\,, \tag{5.101}$$

where $\{\sigma\}$ represents permutation of particle indices, and σ denotes a map of the sequence of particle indices.

$$\sigma : \{1, 2, \cdots, N\} \to \{\sigma_1, \sigma_2, \cdots, \sigma_N\}\,, \tag{5.102}$$

sign(σ) denotes the parity of the permutation: +1 for even permutations and -1 for odd permutations.

Let us consider the two-particle system for an example. For the two-particle system, the properly symmetrized (for bosons) or antisymmetrized (for fermions) wave function is given by

$$(\Phi_{\xi_1\xi_2}, \Psi) = \psi_1(\xi_1)\psi_2(\xi_2) \pm \psi_1(\xi_2)\psi_2(\xi_1)\,. \tag{5.103}$$

The plus sign corresponds to bosons (symmetric wave function), and the minus sign corresponds to fermions (antisymmetric wave function). In the general case, the properly antisymmetrized wave function for fermions can be written as a *Slater determinant*.

$$(\Phi_{\xi_1\xi_2\cdots}, \Psi) = \begin{vmatrix} \psi_1(\xi_1) & \psi_1(\xi_2) & \cdots \\ \psi_2(\xi_1) & \psi_2(\xi_2) & \cdots \\ \vdots & & \ddots \end{vmatrix}\,. \tag{5.104}$$

There are several remarks on the identical particles. For fermionic identical particles, the wave function is written by using the Slater determinant, which ensures the antisymmetry with respect to the exchange of any two particles. In other words, the wave function vanishes if any two fermions occupy the same single-particle state, which is a manifestation of the *Pauli exclusion principle*. In terms of the Slater determinant, this property is evident because the determinant vanishes if any two rows (corresponding to identical single-particle states) are identical.

$$\begin{vmatrix} \psi_1(\xi_1) & \psi_1(\xi_2) & \cdots \\ \psi_1(\xi_1) & \psi_1(\xi_2) & \cdots \\ \psi_3(\xi_1) & & \end{vmatrix} = 0 \tag{5.105}$$

For bosonic identical particles, the wave function is symmetric with respect to the exchange of any two particles. It allows two or more bosons to occupy the same single-particle state, which is a manifestation of the phenomenon known as *Bose-Einstein condensation*.

5.5 Algebraic Derivation of Hydrogen Spectrum

There is another solvable system in Heisenberg's Matrix Mechanics, which is given by Wolfgang Pauli. One of outstanding examples of the use of "*Dynamical symmetry*". The hydrogen atom is a prime example where dynamical symmetry plays a crucial role in determining its spectrum. Dynamical symmetry refers to the presence of conserved quantities beyond those associated with obvious geometric symmetries, which can simplify the analysis of a system and lead to exact solutions for its spectrum. The generators of the symmetry commute with Hamiltonian (namely, they are conserved) and commutators of generators themselves depend on Hamiltonian. One of the most famous examples is the *Laplace-Runge-Lenz vector*. It appears in the two-body system with a $1/r$ potential, such as in the Kepler problem, which describes the motion of planets around the Sun under the influence of the inverse-square law of gravitation. The Laplace-Runge-Lenz vector $\boldsymbol{R}$ is defined as

$$\boldsymbol{R} = -\frac{Ze^2\boldsymbol{x}}{4\pi r} + \frac{1}{2m}(\boldsymbol{p}\times\boldsymbol{L} - \boldsymbol{L}\times\boldsymbol{p})\,, \tag{5.106}$$

$$\boldsymbol{L} = \boldsymbol{x}\times\boldsymbol{p}\,, \tag{5.107}$$

where $r = |\boldsymbol{x}|$, and $\boldsymbol{x}$ and $\boldsymbol{p}$ are the position and momentum operators, respectively. In classical mechanics, the second term of the Laplace-Runge-Lenz vector is defined composed only of $\boldsymbol{p} \times \boldsymbol{L}$, without the need for symmetrization as in the quantum case. It is easy to show that $\boldsymbol{R}$ is a conserved vector in classical mechanics: (i) using the Poisson bracket with the Hamiltonian, $\{\boldsymbol{R}, H\} = \boldsymbol{0}$, and (ii) using the time derivative, $\dot{\boldsymbol{R}} = 0$.

It is corresponding to the quantum mechanical case, where the Laplace-Runge-Lenz vector $\boldsymbol{R}$ commutes with the Hamiltonian, $[H, \boldsymbol{R}] = 0$, with the Hamiltonian given by

$$H = \frac{\boldsymbol{p}^2}{2m} - \frac{Ze^2}{4\pi r}\,. \tag{5.108}$$

We now proceed to verify the commutation relations involving the Laplace-Runge-Lenz vector and the Hamiltonian. Using the commutation relations of the angular momentum and momentum operators, we have using $[L_i, p_j] = i\hbar \sum_k \epsilon_{ijk} p_k$. The third term in the expression of the Laplace-Runge-Lenz vector, $-\boldsymbol{L} \times \boldsymbol{p}$, arises due to the symmetrization required in the quantum case, is rewritten as follows.

$$(\boldsymbol{L} \times \boldsymbol{p})_i = -(\boldsymbol{p} \times \boldsymbol{L})_i + 2i\hbar \boldsymbol{p}_i\,. \tag{5.109}$$

Hence, the Laplace-Runge-Lenz vector can be rewritten as

$$\boldsymbol{R} = -\frac{Ze^2\boldsymbol{x}}{4\pi r} + \frac{1}{m}(\boldsymbol{p} \times \boldsymbol{L}) - \frac{i\hbar}{m}\boldsymbol{p}\,. \tag{5.110}$$

From the expression, it is apparent that the the Laplace-Runge-Lenz vector $\boldsymbol{R}$ is orthogonal to the angular momentum vector $\boldsymbol{L}$, i.e., $\boldsymbol{L} \cdot \boldsymbol{R} = \boldsymbol{R} \cdot \boldsymbol{L} = 0$. This is because the orbital angular momentum $\boldsymbol{L} = \boldsymbol{x} \times \boldsymbol{p}$ is perpendicular to the position vector $\boldsymbol{x}$, the momentum vector $\boldsymbol{p}$, and of course $\boldsymbol{p} \times \boldsymbol{L}$. Once we rewrite the Laplace-Runge-Lenz vector in the form given above, it becomes more straightforward to show the $[H, R_i] = 0$ after some algebraic manipulations involving the commutation relations of the position, momentum, and angular momentum operators.

Now, we find the spectrum of $\boldsymbol{R}^2$. It is straightforward, but it is lengthy and tediously algebraic calculations, so for brevity, we can skip the detailed steps and present the final result.

$$\begin{aligned}\boldsymbol{R}^2 &= \left(-\frac{Ze^2\boldsymbol{x}}{4\pi r} + \frac{1}{m}(\boldsymbol{p} \times \boldsymbol{L}) - \frac{i\hbar}{m}\boldsymbol{p}\right)^2 \\ &= \frac{Z^2e^4}{(4\pi)^2} + \frac{2}{m}\left(\frac{\boldsymbol{p}^2}{2m} - \frac{Ze^2}{4\pi r}\right)\left(\boldsymbol{L}^2 + \hbar^2\right) = \frac{Z^2e^4}{(4\pi)^2} + \frac{2}{m}H\left(\boldsymbol{L}^2 + \hbar^2\right)\end{aligned} \tag{5.111}$$

From the expression, we can find the energy spectrum once we know the spectrum of $\boldsymbol{R}^2$ and $\boldsymbol{L}^2$. Since the Laplace-Runge-Lenz vector $\boldsymbol{R}$ is composed of the position vector $\boldsymbol{x}$ and the momentum vector $\boldsymbol{p}$, the commutation relations of $\boldsymbol{R}$ with $\boldsymbol{L}$ can be derived.

$$[L_i, R_j] = i\hbar \sum_k \epsilon_{ijk} R_k\,. \tag{5.112}$$

In addition, we can consider the commutator of $\boldsymbol{R}$ with itself, i.e., $[R_i, R_j]$. After the long and tedious algebraic manipulations, one can show that the commutator of the Laplace-Runge-Lenz vector with itself is proportional to the angular momentum vector:

$$[R_i, R_j] = -i\hbar \frac{2}{m} H \sum_k \epsilon_{ijk} L_k \,. \tag{5.113}$$

As seen from the expression, the commutator of $\boldsymbol{R}$ itself contains the Hamiltonian, which is known as the *dynamical symmetry*. Then, we have prepared all ingredients to consider the algebraic structure of the Laplace-Runge-Lenz vector and the angular momentum vector. Once we take linear combinations of these vectors,

$$\boldsymbol{A}_\pm \equiv \frac{1}{2}\left(\boldsymbol{L} \pm \sqrt{\frac{m}{-2H}}\boldsymbol{R}\right) , \tag{5.114}$$

we can decompose the algebra of the Laplace-Runge-Lenz vector and the angular momentum vector into two independent $SO(3)$ algebras.

$$[A_{\pm i}\,, A_{\pm j}] = i\hbar \sum_k \epsilon_{ijk} A_{\pm k}\,, \qquad [A_{\pm i}\,, A_{\mp j}] = 0\,. \tag{5.115}$$

From the expression, we can conclude that $\boldsymbol{A}_\pm$ independently form the same structure as $SO(3)$, and there is no mixing between the two algebras. Therefore, we can represent the algebraic structure as a direct product of two independent $SO(3)$ algebras. As with the usual angular momentum algebra, the eigenvalues of $\boldsymbol{A}_\pm^2$ are of the form $a_\pm(a_\pm + 1)\hbar^2$. Since $\boldsymbol{L}\cdot\boldsymbol{R} = \boldsymbol{R}\cdot\boldsymbol{L} = 0$, there is no difference between the eigenvalues of $\boldsymbol{A}_+^2$ and $\boldsymbol{A}_-^2$:

$$\boldsymbol{A}_\pm^2 = \frac{1}{4}\left[\boldsymbol{L}^2 + \left(\frac{m}{-2H}\right)\boldsymbol{R}^2\right] . \tag{5.116}$$

Therefore, we can set $a_+ = a_- \equiv a$, and the common eigenvalue of $\boldsymbol{A}_\pm^2$ is given by

$$\boldsymbol{A}_\pm^2 = a(a+1)\hbar^2\,. \tag{5.117}$$

Using the expression for $\boldsymbol{A}_\pm^2$ in terms of $\boldsymbol{L}^2$ and $\boldsymbol{R}^2$ mentioned above, and the expression for $\boldsymbol{R}^2$, we can find, for the energy eigenvalue $H = E$,

$$\begin{aligned}\boldsymbol{A}_\pm^2 &= \frac{1}{4}\left[\boldsymbol{L}^2 + \frac{m}{-2E}\left(\frac{Z^2e^4}{(4\pi)^2} + \frac{2E}{m}\left(\boldsymbol{L}^2 + \hbar^2\right)\right)\right] \\ &= \frac{1}{4}\left[\frac{m}{-2E}\frac{Z^2e^4}{(4\pi)^2} - \hbar^2\right] .\end{aligned} \tag{5.118}$$

Connecting this expression for $\boldsymbol{A}_\pm^2$ with the eigenvalue $a(a+1)\hbar^2$, we get the energy eigenvalues as the function of the quantum number a. If we define the principal quantum number by $n = 2a + 1$, then we can find the energy eigenvalue as

$$E_n = -\frac{m}{2\hbar^2 n^2}\left(\frac{Ze^2}{4\pi}\right)^2 . \tag{5.119}$$

This is apparently same as the well-known result, precisely the Coulomb spectrum of the hydrogen atom. Here, we have derived the energy spectrum of the hydrogen atom without solving the Schrödinger equation explicitly.

We should notice that only negative energy appears in this calculation. This is expected from the definition of $\boldsymbol{A}_\pm$, Eq. (5.114). It requires the square root to be real so that $\boldsymbol{A}_\pm$ is Hermitian. If $E > 0$, then $m/(-2E) < 0$ and $\sqrt{m/(-2E)}$ is purely imaginary. In that case, $\boldsymbol{A}_\pm$ is no longer Hermitian, and the eigenvalues of $\boldsymbol{A}_\pm^2$ cannot be written in the usual form $a_\pm(a_\pm+1)\hbar^2$ with integer or half-integer $a_\pm$. Therefore the derivation applies only to the negative-energy bound states.

Mathematically, the algebra generated by $\boldsymbol{L}$ and $\boldsymbol{R}$ is non-compact. For positive energy, it is associated with scattering states, whose spectrum is continuous; for negative energy, it gives the discrete bound-state spectrum. The key point is that the Kepler problem has a hidden dynamical symmetry that is realized only for bound states, and the resulting algebraic structure determines the bound-state spectrum.

The hydrogen bound states are labeled by the integer $n = 2a+1$, and the degeneracy is determined by the independent eigenvalues of A_{+3} and A_{-3}. Since each component can take $2a+1 = n$ values, the total degeneracy is

$$(2a+1)^2 = n^2\,. \tag{5.120}$$

This is the familiar degeneracy of the hydrogen energy levels.

Now, we consider the degeneracy from the perspective of the underlying symmetry. The underlying symmetry responsible for this degeneracy is $SO(3) \otimes SO(3)$, which contains two independent sets of three generators, so altogether there are six generators. This matches the number of antisymmetric generators of $SO(4)$: for the antisymmetric generators, there are ${}_4\mathrm{C}_2 = 6$ independent components. In general, in d dimensions, the rotation group $SO(d)$ has generators

$$J_{\alpha\beta} = -J_{\beta\alpha}, \qquad \alpha,\beta = 1,\ldots,d, \tag{5.121}$$

with $d(d-1)/2$ independent components. As also mentioned in the first part of this section, the commutation relation for the $SO(d)$ generators is given by

$$[J_{\alpha\beta}, J_{\gamma\delta}] = i\hbar\big(\delta_{\alpha\gamma}J_{\beta\delta} - \delta_{\alpha\delta}J_{\beta\gamma} + \delta_{\beta\delta}J_{\alpha\gamma} - \delta_{\beta\gamma}J_{\alpha\delta}\big)\,. \tag{5.122}$$

For $d = 4$, we can decompose the index $\alpha = (i,4)$ with $i = 1,2,3$. For the components J_{ij}, we identify them with the orbital angular momentum generators L_i via

$$J_{ij} = \sum_k \epsilon_{ijk} L_k\,. \tag{5.123}$$

Choosing $(\alpha,\beta) = (i,4)$ and $(\gamma,\delta) = (j,4)$ yields

$$[J_{i4}, J_{j4}] = i\hbar J_{ij} = i\hbar \sum_k \epsilon_{ijk} L_k\,. \tag{5.124}$$

Thus, once we identify

$$J_{i4} \equiv \sqrt{\frac{m}{-2H}}\, R_i, \tag{5.125}$$

we recover the commutator structure of the Kepler problem. In particular, the fact that H and $\boldsymbol{R}$ commute implies that the commutator of J_{i4} with itself gives Eq. (5.113). The remaining $SO(4)$ commutators reproduce the other relations among L_i and R_i.

In terms of the linear combination operators, Eq. (5.114), the $SO(4)$ generators can be written as

$$J_{ij} = \sum_k \epsilon_{ijk}(A_{+k} + A_{-k}), \qquad J_{i4} = A_{+i} - A_{-i}\,. \tag{5.126}$$

Hence, the hydrogen states are classified according to irreducible representations of $SO(4)$. Because $\boldsymbol{R}\cdot\boldsymbol{L} = \boldsymbol{L}\cdot\boldsymbol{R} = 0$ and the two Casimirs are equal, the relevant representations are the traceless symmetric tensors of rank r in four dimensions. The number of independent components is

$$\binom{r+3}{3} - \binom{r+1}{3} = (r+1)^2\,. \tag{5.127}$$

The first term is the number of independent components of a symmetric rank-r tensor in four dimensions. While, the second term is the number of independent components of a symmetric rank-$(r-2)$ tensor in four dimensions, which are subtracted to account for the traceless condition. Identifying $r = n-1$, we obtain the degeneracy

$$(r+1)^2 = n^2\,. \tag{5.128}$$

This matches the familiar degeneracy of the hydrogen spectrum.

$n = 1$. $r = 0$, the number of independent components is 1, corresponding to the scalar representation of $SO(4)$;

- only $1s$ state

$n = 2$. $r = 1$, the number of independent components is 4, corresponding to a 4-vector representation of $SO(4)$, v_α;

- the spatial components v_i describe the $2p$ states
- the time-like component v_4 gives the $2s$ state

$n = 3$. $r = 2$, the number of independent components is 9, corresponding to a traceless symmetric rank-2 tensor representation of $SO(4)$, $t_{\alpha\beta}$;

- the traceless spatial components t_{ij} give the $3d$ states
- the mixed components t_{i4} give the $3p$ states
- $t_{44} = -\sum_i t_{ii}$ gives the $3s$ state

This is understood as the energy levels of the hydrogen atom being classified according to the irreducible representations of the $SO(4)$ symmetry group. Meanwhile, the same energy level contains the states corresponding to the different ℓ values. This classification corresponds to the decomposition of the $SO(4)$ representations into $SO(3)$ representations, which are labeled by the orbital angular momentum quantum number ℓ.

6 Supersymmetric Quantum Mechanics

A useful way to generalize ordinary quantum mechanics is to construct a system in which the Hamiltonian can be factorized into two first-order operators, often called the supercharges. The resulting structure is called supersymmetric quantum mechanics (SUSY QM). In one dimension, the basic idea is to write a Hamiltonian as the product of two adjoint differential operators and then regard the two partner Hamiltonians as a supersymmetric pair. The terminology "*supersymmetry*" is widely known as a symmetry relating bosons and fermions.

6.1 Supersymmetric Hamiltonians

Let us consider a one-dimensional particle with momentum operator p and a real potential generated by a function $W(x)$, called the superpotential. The superpotential is assumed to be Hermite, $W^\dagger(x) = W(x)$. Define

$$A \equiv \frac{1}{\sqrt{2m}} \left[p - iW'(x) \right] , \quad A^\dagger = \frac{1}{\sqrt{2m}} \left[p + iW'(x) \right] . \tag{6.1}$$

Here, $W'(x) = dW/dx$ is the derivative of the superpotential. From the definition, we can find that the function W has the same dimension as $\hbar$. In a similar way to the ladder operators for the harmonic oscillators, we can construct the Hamiltonians from the operators A and $A^\dagger$. Since $p = -i\hbar\, d/dx$, one readily finds

$$A^\dagger A = \frac{p^2}{2m} + \frac{1}{2m} \left[W'(x) \right]^2 - \frac{\hbar}{2m} W''(x) , \tag{6.2}$$

$$AA^\dagger = \frac{p^2}{2m} + \frac{1}{2m} \left[W'(x) \right]^2 + \frac{\hbar}{2m} W''(x) . \tag{6.3}$$

The third terms in both expressions arise from the standard commutator relation between the derivative of the superpotential and the momentum operator.

$$[p, W'(x)] = -i\hbar W''(x) . \tag{6.4}$$

Both $A^\dagger A$ and $AA^\dagger$ contain the kinetic energy part, $p^2/(2m)$. We can define two different Hamiltonians using a single function $W(x)$.

$$H_+ = A^\dagger A = \frac{p^2}{2m} + V_+(x) , \tag{6.5}$$

$$H_- = AA^\dagger = \frac{p^2}{2m} + V_-(x) , \tag{6.6}$$

with

$$V_\pm(x) = \frac{1}{2m} \left[W'(x) \right]^2 \mp \frac{\hbar}{2m} W''(x) . \tag{6.7}$$

These are usually called the *supersymmetric partner potentials*, or the *Witten model*. The two Hamiltonians differ by a potential term but are related by the intertwining operators A and $A^\dagger$.

6.2 Supersymmetric Spectra

Let $\psi_n^{(\pm)}$ be an eigenstate of $H_\pm$ with eigenvalue $E_n^{(\pm)}$:

$$H_\pm \psi_n^{(\pm)} = E_n^{(\pm)} \psi_n^{(\pm)} \,. \tag{6.8}$$

The orthonormal eigenstates $\psi_n^{(+)}$ and $\psi_n^{(-)}$ form complete sets and span the Hilbert spaces $\mathcal{H}_+$ and $\mathcal{H}_-$ for H_+ and H_-, respectively. Then, we may consider that the state $A\psi_n^{(+)}$ is an eigenstate of H_- with the same energy as $\psi_n^{(+)}$:

$$H_-(A\psi_n^{(+)}) = AH_+\psi_n^{(+)} = E_n^{(+)}(A\psi_n^{(+)}) \,. \tag{6.9}$$

Similarly, for an eigenstate $\psi_n^{(-)}$ of H_-, we can find

$$H_+(A^\dagger\psi_n^{(-)}) = A^\dagger H_-\psi_n^{(-)} = E_n^{(-)}(A^\dagger\psi_n^{(-)}) \,. \tag{6.10}$$

Therefore, the partner Hamiltonians H_+ and H_- have the same spectrum, except possibly for zero-energy states. In particular, if $E_n^{(+)} \neq 0$, then every positive-energy eigenstate of H_+ is mapped to a corresponding eigenstate of H_- by A, and vice versa by $A^\dagger$.

This observation immediately implies that the energy eigenvalues are non-negative. Indeed, the energy eigenvalues are given by the expectation values of each Hamiltonians. The Hamiltonians are written in terms of the bilinear operators, and hence we have

$$E_n^{(+)} = (\psi_n^{(+)}, A^\dagger A\psi_n^{(+)}) = (A\psi_n^{(+)}, A\psi_n^{(+)}) \geq 0 \,, \tag{6.11}$$

$$E_n^{(-)} = (\psi_n^{(-)}, AA^\dagger\psi_n^{(-)}) = (A^\dagger\psi_n^{(-)}, A^\dagger\psi_n^{(-)}) \geq 0 \,. \tag{6.12}$$

The spectra of supersymmetric partner Hamiltonians are bounded from below by zero.

The whole supersymmetric spectra would distinct Hilbert spaces, $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$, which are spanned by states $\psi_n^{(+)}$ and $\psi_n^{(-)}$, respectively. The operators A and $A^\dagger$ map a state into another state in a different Hilbert space: namely, A maps $\mathcal{H}_+ \to \mathcal{H}_-$, while $A^\dagger$ maps $\mathcal{H}_- \to \mathcal{H}_+$. Therefore, the states $A\psi_n^{(+)}$ and $A^\dagger\psi_n^{(-)}$ are expanded by the complete sets of states in the opposite Hilbert spaces, namely $\{\psi_n^{(-)}\}$ and $\{\psi_n^{(+)}\}$, respectively. Meanwhile, both transformed states are also energy eigenstates: therefore the mapping induced by A and $A^\dagger$ preserves the energy spectrum, except possibly for zero-energy states. Once we write $A\psi_n^{(+)} = c_{nn'}^{(-)}\psi_{n'}^{(-)}$ and $A^\dagger\psi_n^{(-)} = c_{nn'}^{(+)}\psi_{n'}^{(+)}$, we can see that the action of H_- and H_+ on this state gives

$$H_-(A\psi_n^{(+)}) = E_n^{(+)}(A\psi_n^{(+)}) = E_{n'}^{(-)}c_{nn'}^{(-)}\psi_{n'}^{(-)} \,, \tag{6.13}$$

$$H_+(A^\dagger\psi_n^{(-)}) = E_n^{(-)}(A^\dagger\psi_n^{(-)}) = E_{n'}^{(+)}c_{nn'}^{(+)}\psi_{n'}^{(+)} \,, \tag{6.14}$$

Therefore, we obtain the relation between the energy eigenvalues of the supersymmetric partner Hamiltonians: $E_n^{(+)} = E_{n'}^{(-)}$ and $E_n^{(-)} = E_{n'}^{(+)}$. We can always reorganize the energy eigenstates in $\mathcal{H}_-$: for instance, in the following we consider $E_n^{(+)} = E_n^{(-)}$ for

all n, except possibly for zero-energy states which may not have a partner in the opposite Hilbert space. This reorganization of spectra depends on a specific choice of the superpotential: for example, depending on which Hilbert space contains the zero-energy mode.

It is trivial to find the normalization constants from the energy eigenvalues: once we rearrange the eigenstates as $E_n^{(+)} = E_n^{(-)} \equiv E_n$, we have

$$c_{nn'}^{(-)} = \sqrt{E_n}\,\delta_{n,n'}\,, \quad c_{nn'}^{(+)} = \sqrt{E_n}\,\delta_{n,n'}\,. \tag{6.15}$$

It is also easy to check the consistency of the mapping: for instance,

$$A^\dagger(A\psi_n^{(+)}) = A^\dagger(\sqrt{E_n}\psi_n^{(-)}) = \sqrt{E_n}A^\dagger\psi_n^{(-)} = E_n\psi_n^{(+)}\,, \tag{6.16}$$

which reproduces the original energy eigenvalue. For the positive energy state, *there always exists a corresponding partner state in the opposite Hilbert space*, related by the action of A or $A^\dagger$.

$$A\psi_n^{(+)} = \sqrt{E_n}\,\psi_n^{(-)}\,, \quad A^\dagger\psi_n^{(-)} = \sqrt{E_n}\,\psi_n^{(+)}\,. \tag{6.17}$$

Next, we consider the case of zero-energy states. For a zero-energy state $\psi_0^{(+)}$ in $\mathcal{H}_+$, we have $H_+\psi_0^{(+)} = 0$, which implies $A^\dagger A\psi_0^{(+)} = 0$. Since $A^\dagger A$ is a positive semi-definite operator, this further implies $A\psi_0^{(+)} = 0$. Similarly, for a zero-energy state $\psi_0^{(-)}$ in $\mathcal{H}_-$, we have $AA^\dagger\psi_0^{(-)} = 0$, which implies $A^\dagger\psi_0^{(-)} = 0$. Therefore, zero-energy states are annihilated by either A or $A^\dagger$ and *need not have a partner in the opposite Hilbert space*. It is straightforward to find these relations explicitly also from the relation above. Since the zero-energy states satisfy $A\psi_0^{(+)} = 0$ or $A^\dagger\psi_0^{(-)} = 0$, it is easy to find a formal solution:

$$\psi_0^{(\pm)} = C_\pm \exp\left[\mp\frac{1}{\hbar}W(x)\right]\,. \tag{6.18}$$

The wavefunctions for the zero-energy states are solely determined by the superpotential $W(x)$. There is one caveat: the zero-energy wavefunction must be normalizable to represent a physical state. Depending on the form of $W(x)$, one of $\psi_0^{(+)}$ or $\psi_0^{(-)}$ may be normalizable while the other is not, which determines in which Hilbert space the zero-energy state resides.[3] When the zero-energy state is found (denoted by ψ_0), we can write the superpotential in terms of the zero-energy wavefunction as

$$W(x) = \hbar\frac{d}{dx}\ln\psi_0(x)\,. \tag{6.19}$$

[3]This discussion is available only when the wavefunction is defined on the whole real line. If the space is compact, the normalizability condition may be satisfied for both $\psi_0^{(+)}$ and $\psi_0^{(-)}$, and the analysis of zero-energy states needs to take the boundary conditions into account.

Witten index

Related to the zero-energy states is the concept of the Witten index, which counts the difference between the number of zero-energy states in the two Hilbert spaces:

$$\Delta_W = \mathcal{N}_{E=0}^{(+)} - \mathcal{N}_{E=0}^{(-)}\,, \tag{6.20}$$

where $\mathcal{N}_{E=0}^{(\pm)}$ denotes the number of normalizable zero-energy states in $\mathcal{H}_\pm$. The Witten index is a topological invariant and remains unchanged under continuous deformations of the superpotential $W(x)$ that do not close the energy gap.

Consider the polynomial superpotential

$$W(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0\,, \tag{6.21}$$

where n is a positive integer. The normalizability of the zero-energy states depends on the asymptotic behavior of $W(x)$ as $x \to \pm\infty$. For instance, if n is even, $\psi_0^{(+)}$ is normalizable for $a_n > 0$ while $\psi_0^{(-)}$ is for $a_n < 0$, leading to a nonzero Witten index. If n is odd, neither $\psi_0^{(+)}$ nor $\psi_0^{(-)}$ is normalizable, resulting in a vanishing Witten index. The above discussion has nothing to do with the detail form of $W(x)$. As far as the asymptotic behavior is unchanged, the Witten index must be the same: indeed, the terms that dominate the asymptotic behavior determine the normalizability of the zero-energy states, and this is the feature of topological invariance.

Meaning of zero-energy states

As shown, the operators A and $A^\dagger$ convert a positive-energy state in one Hilbert space to a positive-energy state in the other Hilbert space, preserving the energy. However, zero-energy states are annihilated by either A or $A^\dagger$ and do not necessarily have a partner, which is why they can exist unpaired and contribute to the Witten index.

The presence of the zero-energy states indicates that the states in the corresponding Hilbert space are annihilated by the operators A or $A^\dagger$ and thus remain invariant under a certain class of transformations. That is to say, the vacuum state (namely, the lowest energy state) in the Hilbert space corresponding to the normalizable zero-energy state is invariant under the action of the operator that annihilates it. This kind of symmetries is referred to as *supersymmetry*.

6.3 Supersymmetric Relations

The structure of the theory becomes clearer if we arrange the two partner Hamiltonians into a single matrix Hamiltonian. Let us assume that the energy eigenvalues for the two partner Hamiltonians are related by $E^{(+)} = E^{(-)} \equiv E$. In the previous subsection, we write the labels for the energy eigenvalues n explicitly for the eigenstates of each partner Hamiltonian. Instead of labeling the energy eigenvalues separately for each partner Hamiltonian, we now use the energy itself as the label. Define the two-component state as follows

$$\Psi_E(x) = \begin{pmatrix} \psi_E^{(+)}(x) \\ \psi_E^{(-)}(x) \end{pmatrix}\,, \tag{6.22}$$

and the matrix Hamiltonian

$$H = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix} . \tag{6.23}$$

The eigenvalue equation for the matrix Hamiltonian reads $H\Psi_E(x) = E\Psi_E(x)$. The supersymmetry generator Q and the operator $(-1)^F$ are defined as follows:

$$Q = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix} , \qquad (-1)^F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} . \tag{6.24}$$

The operator Q is also called the *supercharge*, and $(-1)^F$ is the operator that distinguishes the different Hilbert subspaces corresponding to the two partner Hamiltonians. In the following, we refer $(-1)^F$ as the parity operator, which distinguishes the two partner Hilbert subspaces. Applying these operators on the two-component state $\Psi_E(x)$, we find the following relations among the operators:

$$Q^2 = H\,, \qquad Q(-1)^F = -(-1)^F Q\,, \qquad \left((-1)^F\right)^2 = 1\,. \tag{6.25}$$

These relations are known as the minimal supersymmetry relations. They imply that Q and $(-1)^F$ also commute with the Hamiltonian:

$$[H, Q] = 0\,, \tag{6.26}$$

$$[H, (-1)^F] = 0\,. \tag{6.27}$$

Thus, energy eigenstates can be simultaneously diagonalized and labeled by both the energy and the eigenvalue of Q or $(-1)^F$. Meanwhile, Q and $(-1)^F$ do not commute with each other, we cannot take the simultaneous eigenstates of both operators.

Properties of supersymmetric states

The supersymmetry algebra leads to several important consequences.

(i) Positive-energy states appear in pairs with opposite fermion parity. If $E > 0$, then the states $\psi_E^{(+)}$ and $\psi_E^{(-)}$ are degenerate in energy and are related by the action of Q.

(ii) Zero-energy states need not form a pair. If a normalizable state satisfies $A\psi_0 = 0$ or $A^\dagger\psi_0 = 0$, then it is a zero mode and gives an eigenstate of H with $E = 0$.

(iii) The Witten index

$$\Delta_W = \mathcal{N}_{E=0}^{(+)} - \mathcal{N}_{E=0}^{(-)}\,, \tag{6.28}$$

is topological. Here, $\mathcal{N}_{E=0}^{(\pm)}$ counts the number of zero-energy states with eigenvalue ± 1 of $(-1)^F$.

The Witten index is invariant under smooth deformations of the potential, and it provides a robust criterion for the presence of zero modes. In particular, supersymmetry is broken if there are no normalizable zero-energy states, while it remains unbroken when such states exist. On the other hand, $\Delta_W = 0$ does not indicate whether supersymmetry is broken or unbroken; it only means that the number of zero-energy states with positive and negative fermion parity are equal.

The basic lesson is that supersymmetry allows a Hamiltonian to be built out of two partner Hamiltonians whose spectra are essentially the same, apart from possible zero modes. This is the essential structure behind supersymmetric quantum mechanics and the Witten model.

Supersymmetric Algebra

The matrix form of the supercharge makes the supersymmetry algebra especially transparent. In this representation, the Hilbert space is doubled, and the operator Q acts as an off-diagonal map between the two partner sectors. The parity operator $(-1)^F$ is diagonal and distinguishes the two components of the state. As defined above, the supercharge Q and the parity operator $(-1)^F$ take the following matrix forms:

$$Q = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix} , \quad (-1)^F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} . \tag{6.29}$$

As we saw above, the basic supersymmetry relations are obtained from this form immediately:

$$Q^2 = H , \quad Q(-1)^F = -(-1)^F Q , \quad \left((-1)^F\right)^2 = 1 , \tag{6.30}$$

and the commutation relations with the Hamiltonian are

$$[H, Q] = 0 , \quad [H, (-1)^F] = 0 . \tag{6.31}$$

These operator relations are also called the supersymmetry algebra. The most important feature is that the Hamiltonian operator consists of a certain combination of the supercharges, specifically $H = Q^2$. The energy eigenstates can be chosen to have definite parity, and the supercharge Q maps one partner sector into the other. This is straightforwardly represented by the states

$$\Psi_E^+(x) = \begin{pmatrix} \psi_E^{(+)}(x) \\ 0 \end{pmatrix} , \quad \Psi_E^-(x) = \begin{pmatrix} 0 \\ \psi_E^{(-)}(x) \end{pmatrix} . \tag{6.32}$$

Then, the action of the parity operator $(-1)^F$ on these states is $(-1)^F \Psi_E^\pm(x) = \pm \Psi_E^\pm(x)$, and the action of the supercharge Q is $Q\Psi_E^\pm(x) = \sqrt{E}\Psi_E^\mp(x)$.

As we saw, there is a nontrivial structure in the supersymmetry algebra: the Hamiltonian is constructed from the supercharge Q, and the supercharge Q does not commute with the parity operator $(-1)^F$. The non-commutativity with the parity operator can be rewritten as $Q(-1)^F + (-1)^F Q = 0$: this is referred to as the anticommutation

relation between Q and $(-1)^F$. This motivates us to consider the introduction of multiple real supercharges, leading to the $\mathcal{N} = 2$ supersymmetry algebra. Here, $\mathcal{N}$ counts the number of independent real supercharges. We can define two real supercharges as follows:

$$Q_1 \equiv Q = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}, \quad Q_2 \equiv -iQ(-1)^F = \begin{pmatrix} 0 & iA^\dagger \\ -iA & 0 \end{pmatrix}. \tag{6.33}$$

Instead of the anticommutation relation for the supercharge and the parity operator, we now have the anticommutation relation for the real supercharges.

$$\{Q_i, Q_j\} = 2H\,\delta_{ij}\,. \tag{6.34}$$

Since the Hamiltonian originally commutes with Q and $(-1)^F$, it also commutes with the real supercharges Q_1 and Q_2.

$$[H, Q_i] = 0\,, \quad i = 1\,, 2\,. \tag{6.35}$$

Thus, we remark that the supersymmetry is preserved under time evolution.

It is often convenient to define complex supercharges by linear combinations of the real ones:

$$\mathcal{Q} = \frac{1}{\sqrt{2}}(Q_1 + iQ_2) = \begin{pmatrix} 0 & 0 \\ \sqrt{2}A & 0 \end{pmatrix}, \tag{6.36}$$

$$\mathcal{Q}^\dagger = \frac{1}{\sqrt{2}}(Q_1 - iQ_2) = \begin{pmatrix} 0 & \sqrt{2}A^\dagger \\ 0 & 0 \end{pmatrix}. \tag{6.37}$$

The supersymmetric algebra is rewritten in terms of the complex supercharges as follows

$$\{\mathcal{Q}^\dagger, \mathcal{Q}\} = 2H\,, \tag{6.38}$$

$$\{\mathcal{Q}, \mathcal{Q}\} = \{\mathcal{Q}^\dagger, \mathcal{Q}^\dagger\} = 0\,, \tag{6.39}$$

which implies

$$[H, \mathcal{Q}] = [H, \mathcal{Q}^\dagger] = 0\,. \tag{6.40}$$

This is the familiar complex form of the $N = 2$ supersymmetry algebra. In terms of the complex supercharges, we can remark that

$$\mathcal{Q}^2 = \mathcal{Q}^{\dagger 2} = 0\,. \tag{6.41}$$

This is called as the *nilpotency condition* for the operators $\mathcal{Q}$ and $\mathcal{Q}^\dagger$, or we say that the operators are *nilpotent*.

The nilpotency appears in the spin system. As seen in the matrix form of the supercharges, this structure is naturally realized using the raising and lowering operators for spin, namely the Pauli matrices σ_+ and σ_-.

$$\sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \tag{6.42}$$

Applying the rasing and lowering operators for spin twice, the single spin state should vanish automatically, and this features are encoded in the Pauli matrices σ_+ and σ_-. Therefore, we can write the complex supercharges by utilizing the Pauli matrices

$$\mathcal{Q} = \sqrt{2} A \, \sigma_- \,, \qquad \mathcal{Q}^\dagger = \sqrt{2} A^\dagger \, \sigma_+ \,. \tag{6.43}$$

Because $\sigma_-^2 = \sigma_+^2 = 0$, we immediately get $\mathcal{Q}^2 = (\mathcal{Q}^\dagger)^2 = 0$, which is the nilpotency condition.

Understood from the nilpotency, the complex supercharges $\mathcal{Q}$ and $\mathcal{Q}^\dagger$ can have a property similar to fermions: exchanging the operators leads to the extra minus sign. In the quantum field theories, supersymmetry relates bosonic degrees of freedom (such as spin 1 particle, photon) and fermionic degrees of freedom (such as spin 1/2 particles, electron), and the supercharges act as generators that transform bosonic states into fermionic states and vice versa. On the other hand, in the quantum mechanics, supersymmetry does not mean that the presence of particles with different spins is required. In the supersymmetric quantum mechanics, the supercharges relate different components of the wavefunction, which can be interpreted as bosonic and fermionic degrees of freedom, *without requiring actual particles with different spins*. In the other words, in the supersymmetric quantum mechanics, the fermionic charges $\mathcal{Q}$ and $\mathcal{Q}^\dagger$ are introduced to exchange *the internal degrees of freedom of the wavefunction, which has nothing to do with the actual spin of particles*.

6.4 Examples of Witten Models

The general construction above can be applied to many one-dimensional models and allows us to systematically construct their supersymmetric partners. Thanks to the supersymmetric construction of quantum mechanics, many non-trivial potentials have been extensively studied and understood through their supersymmetric partners. We remind that, for a given superpotential $W(x)$, the partner potentials are defined by

$$V_\pm(x) = \frac{1}{2m} \left[W'(x) \right]^2 \mp \frac{\hbar}{2m} W''(x) \,. \tag{6.44}$$

These examples illustrate how supersymmetry organizes seemingly different potentials into partner pairs.

Harmonic Superpotential

Now, we consider the superpotential which leads to the harmonic oscillator as the partner potentials. The superpotential for the harmonic oscillator is typically chosen as a quadratic function of the position coordinate.

$$W(x) = \frac{1}{2} m \omega x^2 \,. \tag{6.45}$$

Here, m is the mass of the particle and ω is the angular frequency of the harmonic oscillator. The corresponding Hamiltonians for the partner potentials are then given

by

$$H_{\pm} = \frac{p^2}{2m} + \frac{m\omega^2}{2}x^2 \mp \frac{\hbar\omega}{2}\,. \tag{6.46}$$

It is not always true that both partner Hamiltonians possess the same potential up to a constant shift. As learnt from Section 4.6, the energy eigenvalues of the harmonic oscillator are known. Thanks to the constant shift of Hamiltonians, the energy eigenvalues of the partner Hamiltonians are simply shifted versions of the standard harmonic oscillator energy levels.

$$E_n^+ = n\hbar\omega\,, \quad n = 0, 1, 2, \dots\,, \tag{6.47}$$
$$E_n^- = n\hbar\omega\,, \quad n = 1, 2, \dots\,. \tag{6.48}$$

It is apparent that the partner Hamiltonians share the same energy spectrum except for the ground state of H_+. Indeed, the zero-energy eigenstate is given by

$$\psi_0^+(x) = N_{E=0,+}\exp\left(-\frac{m\omega}{2\hbar}x^2\right)\,. \tag{6.49}$$

The ground state of H_+ has zero energy, while H_- does not have a corresponding zero-energy state: this structure is also expected from the general properties of supersymmetric quantum mechanics, where one of the partner Hamiltonians has a zero-energy ground state while the other does not.

Free Hamiltonian and Rosen-Morse Potential

Consider the following superpotential:

$$W(x) = \hbar \ln\cosh\lambda x\,, \tag{6.50}$$

where the parameter λ has dimensions of inverse length and controls the steepness of the potential. The derivatives of the superpotential are then given by

$$W'(x) = \hbar\lambda\tanh\lambda x\,, \qquad W''(x) = \hbar\lambda^2\left[1 - \tanh^2\lambda x\right]\,. \tag{6.51}$$

It is straightforward to find the partner potentials using the standard SUSY-QM formulae.

$$V_+(x) = \frac{\hbar^2\lambda^2}{2m}\left(1 - \frac{2}{\cosh^2\lambda x}\right)\,, \qquad V_-(x) = \frac{\hbar^2\lambda^2}{2m}\,. \tag{6.52}$$

It is quite interesting that the potential $V_-(x)$ is constant and describes the motion of a free particle. There is a partner potential even for free potential: $V_+(x)$ is known as "*Rosen-Morse Potential*". This partner potential has the zero-energy eigenstate, whose wavefunction is given by

$$\psi_0^+(x) = N_{E=0,+}\frac{1}{\cosh\lambda x}\,. \tag{6.53}$$

Delta-function Potential

The delta-function potential is another classic example in supersymmetric quantum mechanics. It can be realized by choosing a superpotential whose derivative has a discontinuity at a point, leading to a delta-function in the partner potentials. The derivative of the superpotential is given by

$$W'(x) = \alpha\hbar\big(\Theta(x) - \Theta(-x)\big)\,, \qquad \Theta(x) := \begin{cases} 1 & (x > 0), \\ 0 & (x < 0), \end{cases} \,. \tag{6.54}$$

Here, α is a constant that determines the strength of the delta-function potential and has the dimensions of inverse length. The Heaviside step function $\Theta(x)$ shows discontinuity of the superpotential, and its derivative gives the delta function:

$$\frac{d}{dx}\Theta(x) = \delta(x)\,. \tag{6.55}$$

Computing the derivatives, we can find the partner potentials as

$$V_\pm(x) \equiv \frac{\hbar^2}{2m}\left\{\alpha^2\left[\Theta(x) - \Theta(-x)\right]^2 \mp 2\alpha\delta(x)\right\}\,. \tag{6.56}$$

The delta function dominates at $x = 0$, then we can ignore the first term in the curly brackets. In addition, $\left[\Theta(x) - \Theta(-x)\right]^2 = 1$ for $x \neq 0$, so it can be treated as a constant away from the origin.[4] Therefore, the partner potentials can be simplified as

$$V_\pm(x) = \frac{\hbar^2}{2m}\left(\alpha^2 \mp 2\alpha\delta(x)\right)\,. \tag{6.58}$$

In the case of the delta-function potential, both partner potentials exhibit an attractive or repulsive singular interaction at the origin, while the sign of the delta-function term differs between them. There is a zero-energy eigenstate for the attractive delta-function potential, which is localized at the origin.

$$\psi_0^+(x) = N_{E=0,+}e^{-\alpha|x|}\,, \tag{6.59}$$

Coulomb potential

The Coulomb potential can be realized even in the one-dimensional quantum mechanics by adding the term corresponding to the centrifugal potential. Once we choose the derivative of the superpotential as

$$W'(x) = \frac{me^2}{4\pi\hbar\ell} - \frac{\hbar\ell}{x}\,. \tag{6.60}$$

[4]There are several identities for the Heaviside step function and the delta function that are useful in this context.

$$\Theta(x)^2 = \Theta(x)\,, \quad \Theta(x)\Theta(-x) = 0\,, \quad \Theta(x) + \Theta(-x) = 1\,. \tag{6.57}$$

Here, ℓ is a positive integer that corresponds to the angular momentum quantum number in the three-dimensional Coulomb problem. The parameters m and e denote the mass and charge of the particle, respectively. Please note that we choose the Lorentz-Heaviside unit for electromagnetism. It is straightforward to find the partner potentials:

$$V_{\pm}(x) = -\frac{e^2}{4\pi x} + \frac{\hbar^2 \ell(\ell \mp 1)}{2mx^2} + \frac{me^4}{2(4\pi)^2\hbar^2\ell^2}\,. \tag{6.61}$$

As for the Coulomb potential, the partner potentials consist of the attractive Coulomb term, the centrifugal term, and a constant shift. We can ignore the constant shift when considering the qualitative behavior of the potential. One of interesting features is that the partner potentials differ only in the centrifugal term: therefore, the potentials with the quantum numbers ℓ and $\ell - 1$ form a supersymmetric pair. There exists a zero-energy eigenstate for the Coulomb potential, which is localized near the origin and decays exponentially at large distances.

$$\psi_0^+(x) = N_{E=0,+} x^{\ell} e^{-\frac{me^2}{4\pi\hbar^2\ell}x}\,. \tag{6.62}$$

Infinite well and Pöschl-Teller potential

One can also consider the superpotential defined in the limited domain. We consider the superpotential whose derivative is given by the tangent function within the limited domain.

$$W'(x) = \hbar\lambda \tan(\lambda x)\,, \tag{6.63}$$

which is defined only within the interval $-\frac{\pi}{2} < \lambda x < \frac{\pi}{2}$. Here, λ is a positive constant and has the dimension of inverse length. Given superpotential derivative, we can construct the partner potentials using the standard SUSYQM formulas.

$$V_+(x) = -\frac{\hbar^2\lambda^2}{2m}\,, \quad V_-(x) = \frac{\hbar^2\lambda^2}{2m}\left(\frac{2}{\cos^2(\lambda x)} - 1\right)\,. \tag{6.64}$$

The potential $V_+(x)$ corresponds to the free Hamiltonian, while $V_-(x)$ represents the *Pöschl-Teller potential* within the limited domain $-\frac{\pi}{2} < \lambda x < \frac{\pi}{2}$. Outside this interval, the potentials are considered to be infinitely large, effectively confining the particle within the domain. There is the zero-energy eigenstate whose wavefunction is given by

$$\psi_0^+(x) = N_{E=0,+} \cos(\lambda x)\,. \tag{6.65}$$